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(12) **United States Patent**
Oh et al.(10) **Patent No.:** **US 12,415,842 B2**(45) **Date of Patent:** ***Sep. 16, 2025**(54) **IMMUNOSTIMULATING IL-2 ANALOGS**(71) Applicant: **HANMI PHARM. CO., LTD.**,
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Hwaseong-si (KR)(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 118 days.

This patent is subject to a terminal disclaimer.(21) Appl. No.: **18/349,656**(22) Filed: **Jul. 10, 2023**(65) **Prior Publication Data**

US 2023/0348550 A1 Nov. 2, 2023

Related U.S. Application Data

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Mar. 31, 2020 (KR) 10-2020-0039476

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CPC **C07K 14/55** (2013.01); **G01N 33/6869** (2013.01)(58) **Field of Classification Search**
CPC C07K 14/55
See application file for complete search history.(56) **References Cited****U.S. PATENT DOCUMENTS**9,428,567 B2 8/2016 Garcia
11,746,137 B2 * 9/2023 Oh C07K 14/55
530/351
2005/0142106 A1 6/2005 Witttrup et al.2006/0234205 A1 10/2006 Cao et al.
2014/0046026 A1 2/2014 Garcia et al.
2019/0321446 A1 10/2019 Higginson-Scott et al.
2021/0101953 A1 4/2021 Sun et al.**FOREIGN PATENT DOCUMENTS**AU 2007201663 A1 10/2008
CA 2019714 A1 12/1990
CN 1309705 A 8/2001
CN 101426916 A 5/2009
CN 107266553 A 10/2017
JP 2016-533167 A 10/2016
KR 10-2015-0087130 A 7/2015
KR 10-2016-0134989 A 11/2016
KR 10-2017-0070091 A 6/2017
KR 10-1989201 B1 6/2019
KR 10-2413691 B1 6/2022
WO 89/04665 A2 6/1989
WO 90/00565 A1 1/1990
WO 2004/022593 A3 3/2004
WO 2005/007121 A2 1/2005
WO 2005/007121 A3 1/2005
WO 2005/086798 A2 9/2005
WO 2015/164815 A1 10/2015
WO 2018/234862 A1 12/2018
WO 2019/147837 A2 8/2019
WO 2019/196815 A1 10/2019
WO 2019/246404 A1 12/2019
WO 2020/020783 A1 1/2020
WO 2020/057646 A1 3/2020
WO 2020/130300 A1 6/2020
WO 2020/228791 A1 11/2020
WO 2021/120350 A1 6/2021
WO 2021/178833 A2 9/2021**OTHER PUBLICATIONS**International Search Report for PCT/KR2021/004028 dated, Jul. 26, 2021 (PCT/ISA/210).
Australian Office Action for AU Application No. 2021202825 dated Jun. 3, 2021.
Australian Office Action for AU Application No. 2021202825 dated May 31, 2022.
Australian Office Action (Acceptance) for AU Application No. 2021202825 dated Jun. 16, 2022.
Extended European Search Report dated Jul. 15, 2024 in Application No. 21781220.5.

* cited by examiner

Primary Examiner — Prema M Mertz(74) *Attorney, Agent, or Firm* — Sughrue Mion, PLLC(57) **ABSTRACT**

Modifications to interleukin-2 alpha receptors are disclosed. Interleukin-2 analogs with increased binding affinity for interleukin-2 beta receptors are disclosed.

5 Claims, 3 Drawing Sheets**Specification includes a Sequence Listing.**

FIG. 1(A)

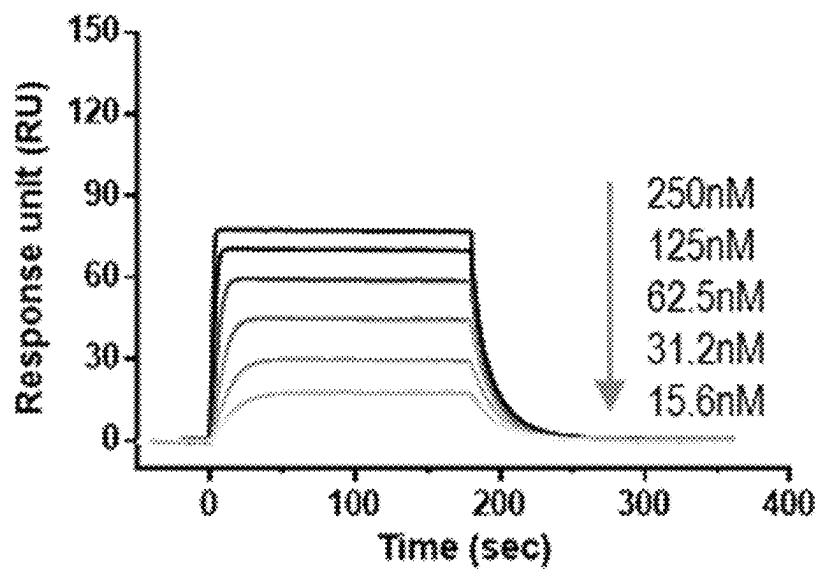


FIG. 1(B)

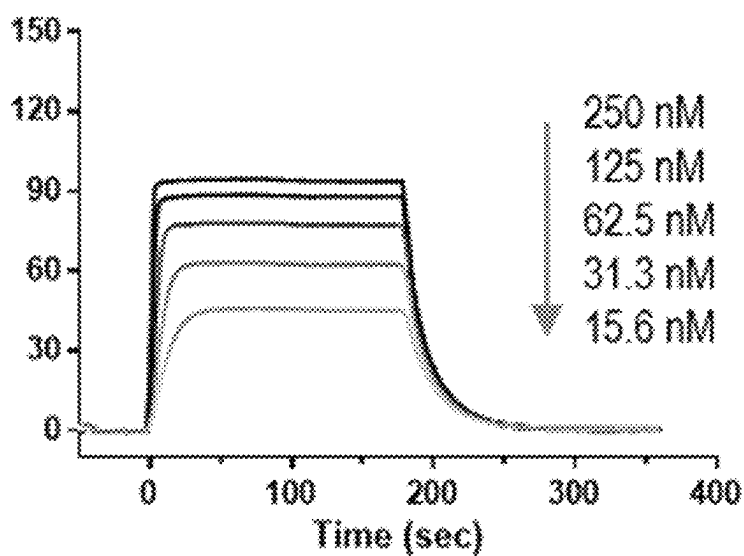


FIG. 1(C)

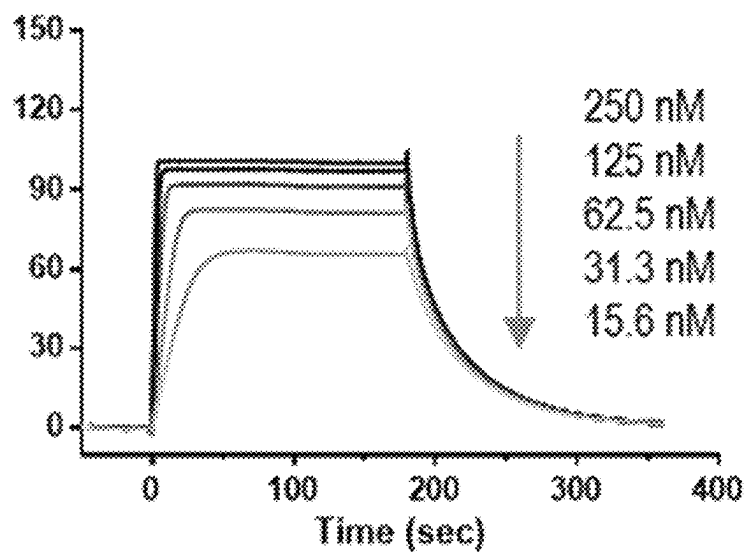


FIG. 2(A)

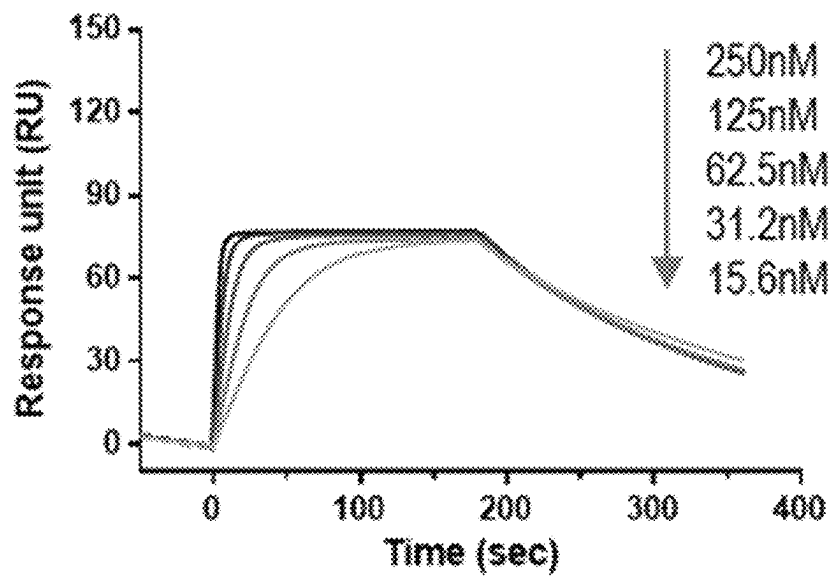


FIG. 2(B)

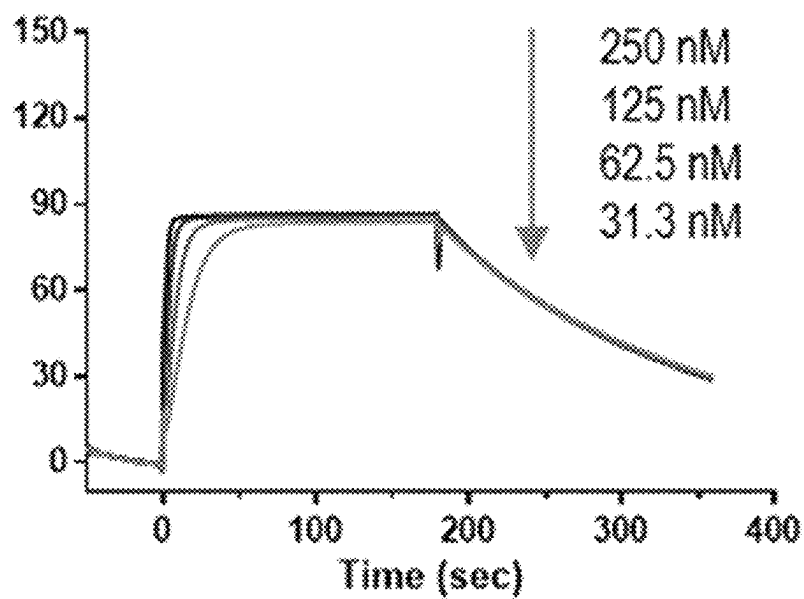
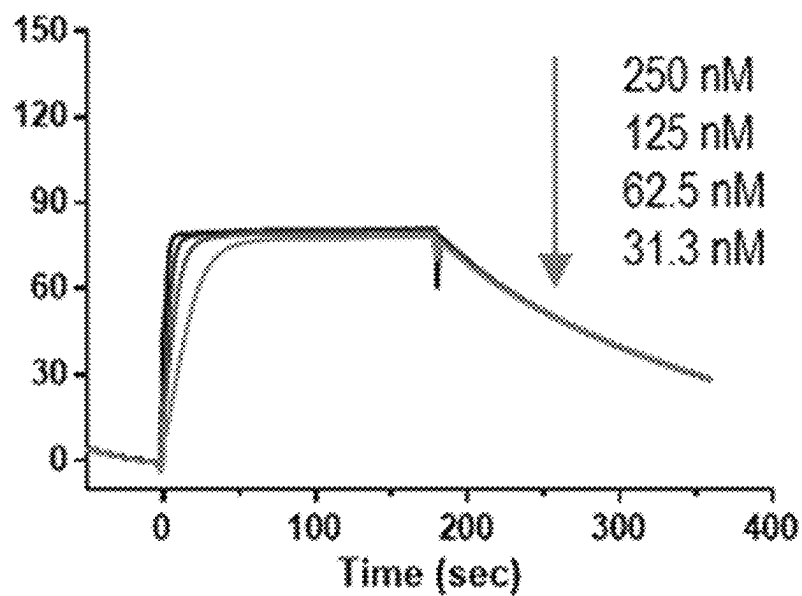


FIG. 2(C)



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IMMUNOSTIMULATING IL-2 ANALOGS**CROSS REFERENCE TO RELATED APPLICATIONS**

This application is a Divisional of U.S. application Ser. No. 17/915,808 filed Sep. 29, 2022, which is National Stage of International Application No. PCT/KR2021/004028 filed Mar. 31, 2021, claiming priority based on Korean Patent Application No. 10-2020-0039476 filed Mar. 31, 2020.

INCORPORATION BY REFERENCE OF SEQUENCE LISTING

The content of the electronically submitted sequence listing, file name: Q288594_SEQ_LIS_ST26_AS_FILED.XML; size: 595,082 bytes; and date of creation: Jul. 7, 2023, filed herewith, is incorporated herein by reference in its entirety.

TECHNICAL FIELD

The present invention relates to a novel interleukin-2 analog.

BACKGROUND ART

Interleukin-2 is an important immunostimulator with a molecular weight of about 15 kDa, which consists of a total of 133 amino acid residues, and activates various cells of the immune system including T cells and B cells. The high efficacy of interleukin-2 as an immune stimulant can be used for the treatment of various immune-related conditions including cancer and AIDS (Korean Patent Application Publication No. 10-2017-0070091). Currently, interleukin-2 (trademark name: Proleukin) is an FDA-approved drug for the treatment of metastatic renal cell carcinoma and metastatic melanoma. However, due to the severe toxicity associated with high-dose interleukin-2 therapy, the applicable patients are limited. In fact, this therapy is applied to only a small number of eligible patients. The toxicity associated with interleukin-2 includes severe fever, nausea, vomiting, vascular leak, severe hypotension, pulmonary edema, and vascular leak syndrome, which causes liver damage.

The interleukin-2 receptor has three kinds of subunit receptors. The subunit consists of an alpha chain (IL-2R α , CD25), a beta chain (IL-2R β or CD122), and a gamma chain (IL-2R γ or CD132). Interleukin-2 can exhibit various functions by binding to receptor subunits of various combinations. A single interleukin-2 alpha receptor is called a low-affinity interleukin-2 receptor, and it is not involved in signaling. A complex of interleukin-2 beta and gamma receptors binds to interleukin-2 with intermediate affinity. A complex of interleukin-2 alpha, beta, and gamma receptors binds to interleukin-2 with high affinity. The complex of interleukin-2 beta and gamma receptors is required for effective signal conversion through kinase activation in multiple signaling pathways. In particular, interleukin-2 beta- and gamma-binding receptors are prominent in CD8⁺ cells and natural killer (NK) cells. In addition, complexes of high-affinity interleukin-2 alpha, beta, and gamma receptors are usually found in CD4⁺ T regulatory cells (Treg), and recently they were also found in activated T cells. Since interleukin-2 beta receptors are distributed in CD8⁺ T cells or natural killer cells (NK cells) and are involved in the immune response in the body, studies have been conducted

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to develop therapeutic agents by increasing the activity of beta receptors for immune activation.

Meanwhile, despite the potential of interleukin-2 as a therapeutic agent for various immune-related conditions, there are still not many drugs which can reduce their doses while reducing toxicity and side effects, and thus there is an increasing demand for studies on new and improved drugs.

DISCLOSURE**Technical Problem**

An object of the present invention is to provide an interleukin-2 analog.

Another object of the present invention is to provide an isolated nucleic acid encoding the interleukin-2 analog; a recombinant expression vector including the nucleic acid; and a transformant including the vector.

Still another object of the present invention is to provide a method for preparing the interleukin-2 analog.

Still another object of the present invention is to provide a method for increasing the binding affinity for interleukin-2 beta receptors, which includes modifying one or more amino acids in native interleukin-2.

Technical Solution

An aspect of the present invention provides a novel interleukin-2 analog (or IL-2 analog). The interleukin-2 analog is an interleukin-2 analog which has an increased binding affinity for interleukin-2 beta receptor compared to native interleukin-2 or aldesleukin (i.e., an interleukin-2 analog). In a specific embodiment, the interleukin-2 analog may include a sequence in which one or more amino acids in native interleukin-2 are modified.

In another specific embodiment, the interleukin-2 analog is characterized in that the interleukin-2 analog includes a sequence in which one or more amino acids corresponding to those at positions 1, 12, 18, 19, 20, 22, 32, 35, 38, 42, 43, 45, 48, 49, 61, 68, 69, 74, 76, 80, 81, 82, 84, 85, 86, 87, 88, 89, 91, 92, 94, 95, 96, 125, 126, and 133 in native interleukin-2 are modified.

The interleukin-2 analog according to any one of the previous specific embodiments is characterized in that it has an altered binding affinity for interleukin-2 alpha receptors and an increased binding affinity for interleukin-2 beta receptors compared to native interleukin-2 or aldesleukin.

The interleukin-2 analog according to any one of the previous specific embodiments is characterized in that one or more amino acids are added to the amino acid corresponding to position 133.

The interleukin-2 analog according to any one of the previous specific embodiments is characterized in that it is native interleukin-2, in which the amino acid at position 1 is removed and the amino acid at position 125 is substituted with a different amino acid.

The interleukin-2 analog according to any one of the previous specific embodiments is characterized in that it further includes 1 to 10 amino acid substitutions. The interleukin-2 analog according to any one of the previous specific embodiments is characterized in that one or more amino acids at positions 18, 19, 20, 22, 38, 42, 43, 45, 61, 68, 69, 74, 80, 81, 84, 85, 86, 88, 89, 91, 92, 94, and 96 are further substituted with different amino acids.

The interleukin-2 analog according to any one of the previous specific embodiments is characterized in that one or more amino acids at positions 18, 19, 22, 38, 42, 43, 45,

The interleukin-2 analog according to any one of the previous specific embodiments is characterized in that it is any one of the following analogs:

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- at position 1 is removed and the amino acids at positions 125, 69, 74, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (ba) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 84, 85, 86, 91, and 92 are substituted with different amino acids;
- (bb) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 85, 86, 92, 94, and 96 are substituted with different amino acids;
- (bc) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 18, 19, 22, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (bd) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 18, 22, 38, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (be) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 18, 22, 61, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (bf) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 18, 22, 68, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (bg) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 35, 38, 42, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (bh) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 45, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (bi) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 85, 86, 91, and 92 are substituted with different amino acids;
- (bj) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 85, 86, 92, and 95 are substituted with different amino acids;
- (bk) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 35, 38, 42, 74, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (bl) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 43, 61, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (bm) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at posi-

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- tions 125, 38, 42, 80, 81, 85, 86, 91, 92, and 95 are substituted with different amino acids;
- (bn) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 35, 38, 42, 74, 80, 81, 82, 85, 86, and 92 are substituted with different amino acids;
- (bo) an interleukin-2 analog in which the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 86, and 92 are substituted with different amino acids in native interleukin-2; and
- (bp) an interleukin-2 analog in which the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 85, and 86 are substituted with different amino acids in native interleukin-2.

The interleukin-2 analog according to any one of the previous specific embodiments is characterized in that it includes any one amino acid substitution selected from the group consisting of the following amino acid substitutions:

- (a) a substitution in which the amino acid at position 12 is substituted with valine or phenylalanine;
- (b) a substitution in which the amino acid at position 18 is substituted with arginine;
- (c) a substitution in which the amino acid at position 19 is substituted with tyrosine, valine, phenylalanine, or arginine;
- (d) a substitution in which the amino acid at position 20 is substituted with valine or phenylalanine;
- (e) a substitution in which the amino acid at position 22 is substituted with glutamic acid;
- (f) a substitution in which the amino acid at position 32 is substituted with cysteine;
- (g) a substitution in which the amino acid at position 35 is substituted with cysteine or glutamic acid;
- (h) a substitution in which the amino acid at position 38 is substituted with alanine or aspartic acid;
- (i) a substitution in which the amino acid at position 42 is substituted with lysine, alanine, or tryptophan;
- (j) a substitution in which the amino acid at position 43 is substituted with cysteine, glutamic acid, or glutamine;
- (k) a substitution in which the amino acid at position 45 is substituted with alanine;
- (l) a substitution in which the amino acid at position 48 is substituted with cysteine;
- (m) a substitution in which the amino acid at position 49 is substituted with cysteine;
- (n) a substitution in which the amino acid at position 61 is substituted with glutamine, arginine, or aspartic acid;
- (o) a substitution in which the amino acid at position 68 is substituted with aspartic acid or glutamine;
- (p) a substitution in which the amino acid at position 69 is substituted with glycine;
- (q) a substitution in which the amino acid at position 74 is substituted with histidine or alanine;
- (r) a substitution in which the amino acid at position 76 is substituted with cysteine;
- (s) a substitution in which the amino acid at position 80 is substituted with phenylalanine, tyrosine, valine, aspartic acid, or tryptophan;
- (t) a substitution in which the amino acid at position 81 is substituted with aspartic acid, glutamic acid, or asparagine;
- (u) a substitution in which the amino acid at position 82 is substituted with glycine or valine;
- (v) a substitution in which the amino acid at position 84 is substituted with glutamic acid, valine, or phenylalanine;

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- (w) a substitution in which the amino acid at position 85 is substituted with valine, alanine, glycine, tryptophan, tyrosine, threonine, isoleucine, glutamic acid, or phenylalanine;
- (x) a substitution in which the amino acid at position 86 is substituted with valine, alanine, glycine, or leucine;
- (y) a substitution in which the amino acid at position 87 is substituted with cysteine;
- (z) a substitution in which the amino acid at position 88 is substituted with glutamine, valine, or phenylalanine;
- (aa) a substitution in which the amino acid at position 89 is substituted with phenylalanine;
- (ab) a substitution in which the amino acid at position 91 is substituted with threonine, phenylalanine, or glutamic acid;
- (ac) a substitution in which the amino acid at position 92 is substituted with phenylalanine, leucine, tyrosine, or tryptophan;
- (ad) a substitution in which the amino acid at position 95 is substituted with aspartic acid;
- (ae) a substitution in which the amino acid at position 96 is substituted with phenylalanine, valine, or isoleucine; and
- (af) a substitution in which the amino acid at position 126 is substituted with threonine.

The interleukin-2 analog according to any one of the previous specific embodiments is characterized in that it is selected from the group consisting of SEQ ID NOS: 3 to 106.

Another aspect to implement the present invention provides an isolated nucleic acid encoding the interleukin-2 analog; a recombinant expression vector which includes the nucleic acid; and a transformant which includes the vector.

Still another aspect to implement the present invention provides a method for preparing the interleukin-2 analog.

Still another aspect to implement the present invention provides a method for increasing the binding affinity for interleukin-2 beta receptors, which includes modifying one or more amino acids in native interleukin-2, in which the modification may be modifications in one or more amino acids selected from the group consisting of amino acids corresponding to positions at 1, 12, 18, 19, 20, 22, 32, 35, 38, 42, 43, 45, 48, 49, 61, 68, 69, 74, 76, 80, 81, 82, 84, 85, 86, 87, 88, 89, 91, 92, 94, 95, 96, 125, 126, and 133.

Still another aspect to implement the present invention provides an interleukin-2 analog which includes any one sequence selected from the group consisting of amino acid sequences of SEQ ID NOS: 3 to 106.

Still another aspect to implement the present invention provides an interleukin-2 analog which includes an amino acid sequence represented by General Formula 1 below:

[General Formula 1]

(General Formula 1, SEQ ID NO: 212)

X1-P-T-S-S-S-T-K-K-T-Q-L-Q-L-E-H-L-X18-X19-D-L-

X22-M-I-L-N-G-I-N-N-Y-K-N-P-K-L-T-X38-M-L-T-X42-

X43-F-X45-M-P-K-K-A-T-E-L-K-H-L-Q-C-L-E-X61-E-L-K-

P-L-E-X68-V-L-N-L-A-X74-S-K-N-F-H-X80-X81-P-R-X84-

X85-X86-S-N-I-N-X91-X92-V-X94-E-X96-K-G-S-E-T-T-F-

M-C-E-Y-A-D-E-T-A-T-I-V-E-F-L-N-R-W-I-T-F-S-Q-S-I-

I-S-T-L-T

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wherein in General Formula 1 above,

- X1 is a deletion;
- X18 is leucine (L) or arginine (R);
- X19 is leucine (L) or tyrosine (Y);
- X22 is glutamic acid (E) or glutamine (Q);
- X38 is alanine (A), aspartic acid (D), or arginine (R);
- X42 is alanine (A), phenylalanine (F), lysine (K), or tryptophan (W);
- X43 is glutamic acid (E), lysine (K), or glutamine (Q);
- X45 is alanine (A) or tyrosine (Y);
- X61 is aspartic acid (D), glutamic acid (E), glutamine (Q), or arginine (R);
- X68 is aspartic acid (D) or glutamic acid (E);
- X74 is histidine (H) or glutamine (Q);
- X80 is phenylalanine (F), leucine (L), valine (V), or tyrosine (Y);
- X81 is aspartic acid (D), glutamic acid (E), or arginine (R);
- X84 is aspartic acid (D) or glutamic acid (E);
- X85 is alanine (A), glutamic acid (E), glycine (G), leucine (L), valine (V), tryptophan (W), or tyrosine (Y);
- X86 is alanine (A), glycine (G), isoleucine (I), or valine (V);
- X91 is threonine (T) or valine (V);
- X92 is phenylalanine (F), isoleucine (I), or tyrosine (Y);
- X94 is phenylalanine (F) or leucine (L); and
- X96 is phenylalanine (F) or leucine (L).

In a specific embodiment, the interleukin-2 analog is characterized in that it includes any one sequence selected from the group consisting of amino acid sequences of SEQ ID NOS: 10, 13, 14, 15, 16, 17, 20, 21, 22, 32, 35, 36, 42, 53, 54, 56, 58, 59, 60, 62, 71, 72, 74, 75, 76, 77, 78, 85, 87, 89, 91, 92, 93, 94, 95, 98, 99, 100, 101, 103, 104, 105, and 106.

In another specific embodiment, the interleukin-2 analog is characterized in that in General Formula 1 above,

- X43 is lysine (K);
- X45 is tyrosine (Y);
- X61 is aspartic acid (D), glutamic acid (E), or glutamine (Q);
- X68 is glutamic acid (E);
- X74 is glutamine (Q);
- X80 is phenylalanine (F) or leucine (L);
- X85 is leucine (L), valine (V), or tyrosine (Y);
- X86 is isoleucine (I) or valine (V); and
- X92 is phenylalanine (F) or isoleucine (I).

The interleukin-2 analog according to any one of the previous specific embodiments is characterized in that it includes any one sequence selected from the group consisting of amino acid sequences of SEQ ID NOS: 10, 13, 14, 16, 17, 20, 21, 22, 32, 35, 36, 42, 53, 54, 87, 89, 91, 92, 93, 94, 98, 99, 100, 101, 103, 104, and 105.

The interleukin-2 analog according to any one of the previous specific embodiments is characterized in that it further includes one or more amino acids at the C-terminus thereof.

Still another aspect to implement the present invention provides an interleukin-2 analog which includes an amino acid sequence represented by General Formula 2 below:

[General Formula 2]

(General Formula 2, SEQ ID NO: 213)

X1-P-T-S-S-S-T-K-K-T-Q-L-Q-L-E-H-L-X18-L-D-L-X22-

M-I-L-N-G-I-N-N-Y-K-N-P-K-L-T-X38-M-L-T-X42-K-F-Y-

M-P-K-K-A-T-E-L-K-H-L-Q-C-L-E-X61-E-L-K-P-L-E-X68-

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-continued

V-L-N-L-A-Q-S-K-N-F-H-F-X81-P-R-D-X85-X86-S-N-I-N-

V-F-V-L-E-L-K-G-S-E-T-T-F-M-C-E-Y-A-D-E-T-A-T-I-V-

E-F-L-N-R-W-I-T-F-S-Q-S-I-I-S-T-L-T

wherein in General Formula 2 above,
X1 is a deletion,

X18 is leucine (L) or arginine (R);
X22 is glutamic acid (E) or glutamine (Q);
X38 is alanine (A) or arginine (R);
X42 is phenylalanine (F) or lysine (K);
X61 is aspartic acid (D) or glutamic acid (E);
X68 is aspartic acid (D) or glutamic acid (E);
X81 is aspartic acid (D) or glutamic acid (E);
X85 is leucine (L) or valine (V); and
X86 is isoleucine (I) or valine (V).

In a specific embodiment, the interleukin-2 analog is characterized in that it includes any one sequence selected from the group consisting of amino acid sequences of SEQ ID NOS: 22, 42, 53, 87, 105, and 106.

In another specific embodiment, the interleukin-2 analog is characterized in that it further includes one or more amino acids at the C-terminus thereof.

Advantageous Effects

The interleukin-2 analog according to the present invention is an analog which has increased binding affinity for interleukin-2 beta receptors in vivo and can be for various purposes.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1(A)-FIG. 1(C) show the results confirming the binding affinity of an interleukin-2 analog for interleukin-2 alpha receptors, in which FIG. 1(A) represents interleukin-2 analog #86, FIG. 1(B) represents interleukin-2 analog #104, and FIG. 1(C) represents interleukin-2 analog #105.

FIG. 2(A)-FIG. 2(C) show the results confirming the binding affinity of an interleukin-2 analog for interleukin-2 beta receptors, in which FIG. 2(A) represents interleukin-2 analog #86, FIG. 2(B) represents interleukin-2 analog #104, and FIG. 2(C) represents interleukin-2 analog #105.

DETAILED DESCRIPTION FOR CARRYING OUT THE INVENTION

The details for carrying out the present invention will be described as follows. Meanwhile, respective descriptions and embodiments disclosed in the present invention may also be applied to other descriptions and embodiments. That is, all combinations of various elements disclosed in the present invention fall within the scope of the present invention. Further, the scope of the present invention cannot be considered to be limited by the specific description below.

Over the entire specification of the present invention, the conventional one-letter and three-letter codes for amino acids are used. Additionally, the amino acids mentioned herein are abbreviated according to the nomenclature rules of the IUPAC-IUB as follows:

alanine	A	arginine	R
asparagine	N	aspartic acid	D
cysteine	C	glutamic acid	E

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-continued

glutamine	Q	glycine	G
histidine	H	isoleucine	I
leucine	L	lysine	K
methionine	M	phenylalanine	F
proline	P	serine	S
threonine	T	tryptophan	W
tyrosine	Y	valine	V

Still another aspect of the present invention provides an interleukin-2 analog. The interleukin-2 analog of the present invention is characterized in that its binding affinity for interleukin-2 receptors is altered, and in particular in that it has increased binding affinity for interleukin-2 beta receptors. Specifically, the interleukin-2 analog of the present invention may be one which has increased binding affinity for interleukin-2 beta receptors, and more specifically one which has altered (increased or decreased) binding affinity for interleukin-2 alpha receptors compared to native interleukin-2 or known aldesleukin.

As used herein, the term "interleukin-2 (IL-2)" refers to a type of cytokine which transmits signals in the immune system in vivo. The interleukin-2 is generally known as an important immunostimulator with a size of about 15 kDa.

As used herein, the term "interleukin-2 analog" refers to native interleukin-2 in which one or more amino acids in the sequence thereof are modified. Particularly in the present invention, the interleukin-2 analog may be one which has reduced or increased binding affinity for interleukin-2 receptors compared to native interleukin-2, in which amino acid(s) in native interleukin-2 is(are) modified. The interleukin-2 analog of the present invention may be one which is not naturally occurring.

The native interleukin-2 may be a human interleukin-2, and its sequence may be obtained from known databases, etc. Specifically, the native interleukin-2 may have an amino acid sequence of SEQ ID NO: 1, but is not limited thereto.

In the present invention, what is meant by "native interleukin-2" may have an amino acid sequence of SEQ ID NO: 1" is that not only the sequence which is the same as SEQ ID NO: 1, but also sequences which have a homology of 80% or higher, 85% or higher, 90% or higher, 91% or higher, 92% or higher, 93% or higher, 94% or higher, 95% or higher, 96% or higher, 97% or higher, 98% or higher, and 99% or higher to SEQ ID NO: 1 belong to the scope of native interleukin-2 of the present invention; and that the corresponding position(s) of amino acid modification is(are) altered on the amino acid sequence of SEQ ID NO: 1 when sequences having a homology of 80% or higher, 85% or higher, 90% or higher, 91% or higher, 92% or higher, 93% or higher, 94% or higher, 95% or higher, 96% or higher, 97% or higher, 98% or higher, and 99% or higher are aligned based on SEQ ID NO: 1.

In the present invention, what is meant by "one or more amino acids in the native sequence are altered" may be that a modification selected from the group consisting of substitution, addition, deletion, modification, and a combination thereof has occurred in at least one amino acid in the native sequence.

Specifically, the interleukin-2 analog of the present invention may include a sequence in which one or more amino acids corresponding to positions 1, 12, 18, 19, 20, 22, 32, 35, 38, 42, 43, 45, 48, 49, 61, 68, 69, 74, 76, 80, 81, 82, 84, 85, 86, 87, 88, 89, 91, 92, 94, 95, 96, 125, 126, and 133 in native interleukin-2 are modified. Specifically, the interleukin-2 analog of the present invention may be native interleukin-2

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in which the amino acid at position 1 is removed and the amino acid at position 125 is substituted with a different amino acid; and which further includes 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more amino acid substitutions. Although not limited thereto, the amino acid at position 125 (i.e., cysteine) may be substituted with serine, and the amino acid(s) at which a further substitution occurs may be amino acids corresponding to positions 12, 18, 19, 20, 22, 32, 35, 38, 42, 43, 45, 48, 49, 61, 68, 69, 74, 76, 80, 81, 82, 84, 85, 86, 87, 88, 89, 91, 92, 94, 95, 96, 126, and 133.

Additionally, interleukin-2 analogs which include substitution, addition, deletion, modification, etc. of amino acid residues in addition to the positions for modification above to the extent that can be performed for the stability and increase of half-life of a peptide known in the art are also included within the scope of the present invention.

As used herein, the term “aldesleukin” or “interleukin-2 analog (aldesleukin)”, which is a commercially available interleukin-2 analog, may be aldesleukin (trademark name: Proleukin®), and specifically may be one which has the amino acid sequence of SEQ ID NO: 2. In the present invention, these terms are used interchangeably with “interleukin-2 analog 1”. The interleukin analog according to the present invention may have altered binding affinity for interleukin-2 alpha receptors and/or increased binding affinity for interleukin-2 beta receptors compared to the interleukin-2 analog 1.

Although interleukin-2 alpha receptors are not known to be involved in the signaling system of interleukin-2, they increase the binding affinity of interleukin-2 for other interleukin-2 receptors (beta or gamma receptors) by 10 to 100 times and are expressed in CD4⁺ regulatory T cells, etc.

Since interleukin-2 beta receptors are mainly distributed in CD8⁺ T cells or natural killer cells (NK cells) and have an important role of activating immune responses and macrophages, it is expected that tumor cell death and activation of the body's immune responses can be promoted through the activation of interleukin-2 beta receptors.

Accordingly, the interleukin-2 analog of the present invention which has increased binding affinity for interleukin-2 beta receptors can have a therapeutic effect where the suppression and death of tumors is increased while side effects are reduced.

In the present invention, the interleukin-2 analog may include a sequence of native interleukin-2 in which the amino acid at position 1 is removed and the amino acid at position 125 is substituted with a different amino acid, and which further includes 1 to 10 amino acid modifications. For example, the interleukin-2 analog may include a sequence of native interleukin-2 in which the amino acid at position 125 is substituted with serine and one or more amino acids at positions 12, 18, 19, 20, 22, 32, 35, 38, 42, 43, 45, 48, 49, 61, 68, 69, 74, 76, 80, 81, 82, 84, 85, 86, 87, 88, 89, 91, 92, 94, 95, 96, and 126 are substituted with different amino acids and/or one or more amino acids are added on the amino acid at position 133, but the sequence is not limited thereto, and any interleukin-2 analog which has altered binding affinity for interleukin-2 alpha receptors and increased binding affinity for interleukin-2 beta receptors compared to native interleukin-2 and/or aldesleukin is included without limitation.

In an embodiment, the interleukin-2 analog may be one in which one or more amino acids are added to the amino acid corresponding to position 133, but the interleukin-2 analog is not limited thereto. For the purpose of the present invention, the amino acids to be added are not limited with regard to the type or length thereof as long as the interleukin-2

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analog has altered binding affinity for interleukin-2 alpha receptors and increased binding affinity for interleukin-2 beta receptors compared to native interleukin-2 and/or aldesleukin, and amino acids which are not naturally occurring and amino acids with a chemical modification can also be added in addition to natural amino acids.

In another embodiment, the interleukin-2 analog may be native interleukin-2 in which the amino acid at position 1 is removed; the amino acid at position 125 is substituted with a different amino acid; and 1, 2, 3, 4, 5, 6, 7, 8, 9, or more amino acids among the amino acids at positions 18, 19, 20, 22, 38, 42, 43, 45, 61, 68, 69, 74, 80, 81, 84, 85, 86, 88, 89, 91, 92, 94, and 96 are substituted with different amino acids, but the interleukin-2 analog is not limited thereto.

In still another embodiment, the interleukin-2 analog may be native interleukin-2 in which the amino acid at position 1 is removed; the amino acid at position 125 is substituted with a different amino acid; and 1, 2, 3, 4, 5, 6, 7, 8, 9, or more amino acids among the amino acids at positions 18, 19, 22, 38, 42, 43, 45, 61, 68, 74, 80, 81, 84, 85, 86, 88, 91, 92, 94, and 96 are substituted with different amino acids, but the interleukin-2 analog is not limited thereto.

In still another embodiment, the interleukin-2 analog may be any one selected from the group consisting of the following analogs:

- (a) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125 and 32 are substituted with different amino acids;
- (b) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125 and 35 are substituted with different amino acids;
- (c) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125 and 38 are substituted with different amino acids;
- (d) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125 and 42 are substituted with different amino acids;
- (e) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125 and 43 are substituted with different amino acids;
- (f) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125 and 48 are substituted with different amino acids;
- (g) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125 and 49 are substituted with different amino acids;
- (h) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125 and 76 are substituted with different amino acids;
- (i) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid

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at position 1 is removed and the amino acids at positions 125, 38, 42, 81, 92, 94, and 96 are substituted with different amino acids;

- [illegible]

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at position 1 is removed and the amino acids at positions 125, 38, 42, and 126 are substituted with different amino acids;

- (w) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 80, and 84 are substituted with different amino acids;
- (x) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 94, and 96 are substituted with different amino acids;
- (y) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 81, and 92 are substituted with different amino acids;
- (z) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 61, 81, and 92 are substituted with different amino acids;
- (aa) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 81, and 92 are substituted with different amino acids;
- (ab) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 80, 81, and 92 are substituted with different amino acids;
- (ac) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 81, 84, and 92 are substituted with different amino acids,
- (ad) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 20, 38, 42, 81, and 92 are substituted with different amino acids;
- (ae) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, and 92 are substituted with different amino acids,
- (af) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 74, 81, and 92 are substituted with different amino acids,
- (ag) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 81, 84, and 92 are substituted with different amino acids;
- (ah) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 81, 88, and 92 are substituted with different amino acids,
- (ai) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid

- tions 125, 38, 42, 80, 81, 85, 86, 91, and 92 are substituted with different amino acids;
- (bj) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 85, 86, 92, and 95 are substituted with different amino acids;
- (bk) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 35, 38, 42, 74, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (bl) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 43, 61, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (bm) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 85, 86, 91, 92, and 95 are substituted with different amino acids;
- (bn) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 35, 38, 42, 74, 80, 81, 82, 85, 86, and 92 are substituted with different amino acids;
- (bo) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 86, and 92 are substituted with different amino acids; and
- (bp) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 85, and 86 are substituted with different amino acids.

In particular, the amino acid substitutions included in the interleukin-2 analog may be any one or more selected from the group consisting of the following amino acid substitutions:

- (a) a substitution in which the amino acid at position 12 is substituted with valine or phenylalanine;
- (b) a substitution in which the amino acid at position 18 is substituted with arginine;
- (c) a substitution in which the amino acid at position 19 is substituted with tyrosine, valine, phenylalanine, or arginine;
- (d) a substitution in which the amino acid at position 20 is substituted with valine or phenylalanine;
- (e) a substitution in which the amino acid at position 22 is substituted with glutamic acid;
- (f) a substitution in which the amino acid at position 32 is substituted with cysteine;
- (g) a substitution in which the amino acid at position 35 is substituted with cysteine or glutamic acid;
- (h) a substitution in which the amino acid at position 38 is substituted with alanine or aspartic acid;
- (i) a substitution in which the amino acid at position 42 is substituted with lysine, alanine, or tryptophan;
- (j) a substitution in which the amino acid at position 43 is substituted with cysteine, glutamic acid, or glutamine;
- (k) a substitution in which the amino acid at position 45 is substituted with alanine;
- (l) a substitution in which the amino acid at position 48 is substituted with cysteine;

- (m) a substitution in which the amino acid at position 49 is substituted with cysteine;
- (n) a substitution in which the amino acid at position 61 is substituted with glutamine, arginine, or aspartic acid;
- (o) a substitution in which the amino acid at position 68 is substituted with aspartic acid or glutamine;
- (p) a substitution in which the amino acid at position 69 is substituted with glycine;
- (q) a substitution in which the amino acid at position 74 is substituted with histidine or alanine;
- (r) a substitution in which the amino acid at position 76 is substituted with cysteine;
- (s) a substitution in which the amino acid at position 80 is substituted with phenylalanine, tyrosine, valine, aspartic acid, or tryptophan;
- (t) a substitution in which the amino acid at position 81 is substituted with aspartic acid, glutamic acid, or asparagine;
- (u) a substitution in which the amino acid at position 82 is substituted with glycine or valine;
- (v) a substitution in which the amino acid at position 84 is substituted with glutamic acid, valine, or phenylalanine;
- (w) a substitution in which the amino acid at position 85 is substituted with valine, alanine, glycine, tryptophan, tyrosine, threonine, isoleucine, glutamic acid, or phenylalanine;
- (x) a substitution in which the amino acid at position 86 is substituted with valine, alanine, glycine, or leucine;
- (y) a substitution in which the amino acid at position 87 is substituted with cysteine;
- (z) a substitution in which the amino acid at position 88 is substituted with glutamine, valine, or phenylalanine;
- (aa) a substitution in which the amino acid at position 89 is substituted with phenylalanine;
- (ab) a substitution in which the amino acid at position 91 is substituted with threonine, phenylalanine, or glutamic acid;
- (ac) a substitution in which the amino acid at position 92 is substituted with phenylalanine, leucine, tyrosine, or tryptophan;
- (ad) a substitution in which the amino acid at position 95 is substituted with aspartic acid;
- (ae) a substitution in which the amino acid at position 96 is substituted with phenylalanine, valine, or isoleucine; and
- (af) a substitution in which the amino acid at position 126 is substituted with threonine.

As used herein, the term "corresponding to" refers to an amino acid residue at a position listed in a peptide, or an amino acid residue which is similar, identical, or homologous to a residue listed in a peptide. Confirmation of the amino acid at the corresponding position may be determining a specific amino acid in a sequence that refers to a specific sequence.

In an embodiment, each amino acid residue in the amino acid sequence can be numbered by aligning any amino acid sequence with SEQ ID NO: 1, and based on the same, referring to the numerical position of the amino acid residue corresponding to the amino acid residue of SEQ ID NO: 1.

As such an alignment, for example, the Needleman-Wunsch algorithm (Needleman and Wunsch, 1970, *J. Mol. Biol.* 48:443-453), the Needle program of the EMBOS package (EMBOSS: The European Molecular Biology Open Software Suite, Rice et al., 2000), (*Trends Genet.* 16:276-277), etc. may be used, but the available programs are not

limited thereto, and a sequence alignment program known in the art, a pairwise sequence comparison algorithm, etc. may be used appropriately.

In the present invention, even if expressed as a specific position of an amino acid in a peptide, such expression may refer to a corresponding position in a reference sequence.

In another embodiment, the interleukin-2 analog may include, consist of or essentially consist of an amino acid sequence which is selected from the group consisting of SEQ ID NOS: 3 to 106, but the interleukin-2 analog is not limited thereto.

Additionally, even if the interleukin-2 analog is expressed as "an interleukin-2 analog consisting of a particular SEQ ID NO" in the present invention, it does not exclude a mutation that may occur by the addition of a meaningless sequence upstream or downstream of the amino acid sequence of the corresponding SEQ ID NO, or a mutation that may occur naturally, or a silent mutation thereof, as long as the interleukin-2 analog has an activity identical or equivalent to the interleukin-2 analog consisting of the amino acid sequence of the corresponding SEQ ID NO, and even if the sequence addition or mutation is present, the interleukin-2 analog apparently belongs to the scope of the present invention.

The interleukin-2 analog of the present invention may include an amino acid sequence which has a homology of 40% or higher, 45% or higher, 50% or higher, 55% or higher, 60% or higher, 65% or higher, 70% or higher, 75% or higher, 80% or higher, 81% or higher, 82% or higher, 83% or higher, 84% or higher, 85% or higher, 86% or higher, 87% or higher, 88% or higher, 89% or higher, 90% or higher, 91% or higher, 92% or higher, 93% or higher, 94% or higher, 95% or higher, 96% or higher, 97% or higher, 98% or higher, or 99% or higher to the amino acid sequences of SEQ ID NOS: 3 to 106, but the interleukin-2 analog is not limited thereto.

In the present invention, the terms "homology" and "identity" refer to a degree of relatedness between two given amino acid sequences or nucleotide sequences and may be expressed as a percentage.

Sequence homology or identity of a conserved polynucleotide or polypeptide may be determined by a standard alignment algorithm, and default gap penalties established by a program to be used may be used in combination. Substantially, homologous or identical sequences may generally hybridize with all or part of the sequences under moderately or highly stringent conditions. It is apparent that hybridization also includes hybridization of a polynucleotide with a polynucleotide, which includes a general codon or a codon where codon degeneracy is considered.

The terms homology and identity can frequently be used interchangeably.

Whether any two nucleotide or peptide sequences have homology, similarity, or identity may be determined by, for example, a known computer algorithm (e.g., the "FASTA" program) using default parameters as in Pearson et al. (1988) *Proc. Natl. Acad. Sci. USA* 85:2444). Alternatively, they may be determined using the Needleman-Wunsch algorithm (Needleman and Wunsch, 1970, *J. Mol. Biol.* 48:443-453) as performed in the Needleman program of the EMBOSS package (EMBOSS: The European Molecular Biology Open Software Suite, Rice et al., 2000, *Trends Genet.* 16:276-277) (version 5.0.0 or later) (including GCG program package (Devereux, J. et al., *Nucleic Acids Research* 12:387 (1984)), BLASTP, BLASTN, FASTA (Atschul, S. F. et al., *J. Molec Biol* 215:403 (1990); *Guide to Huge Computers*, Martin J. Bishop, ed., Academic Press, San Diego, 1994, and CARILLO et al. (1988) *SIAM J Applied Math* 48:1073). For

example, homology, similarity, or identity may be determined using BLAST or ClustalW of the National Center for Biotechnology Information.

Homology, similarity, or identity of nucleotide or peptide sequences may be determined by comparing sequence information using the GAP computer program (e.g., Needleman et al. (1970), *J Mol Biol* 48:443) as disclosed in Smith and Waterman, *Adv. Appl. Math* (1981) 2:482. Briefly, the GAP program defines homology, similarity, or identity as the number of similar aligned symbols (i.e., nucleotides or amino acids), divided by the total number of symbols in the shorter of the two sequences. The default parameters for the GAP program may include: (1) a unary comparison matrix (containing a value of 1 for identities and 0 for non-identities) and the weighted comparison matrix (or EDNA-FULL (EMBOSS version of NCBI NUC4.4) substitution matrix) of Gribskov et al. (1986) *Nucl. Acids Res.* 14:6745, as disclosed by Schwartz and Dayhoff, eds., *Atlas Of Protein Sequence And Structure*, National Biomedical Research Foundation, pp. 353-358 (1979); (2) a penalty of 3.0 for each gap and an additional 0.10 penalty for each symbol in each gap (or gap open penalty 10, gap extension penalty 0.5); and (3) no penalty for end gaps. Therefore, the terms "homology" and "identity", as used herein, represent relevance between sequences.

The above may be applied to other embodiments or other aspects of the present invention, but is not limited thereto.

The interleukin-2 analog of the present invention may be used as a novel interleukin-2 substitute that alters its in vitro activity by weakening or increasing the binding affinity of the interleukin-2 analog for interleukin-2 alpha and/or beta receptors. In particular, the interleukin-2 analog of the present invention can be used as an effective therapeutic agent due to its activities for the two types of receptors because it not only has an increased binding affinity for beta receptors but also has altered (i.e., increased or decreased) binding affinity for alpha receptors.

In the present invention, such modification for preparing analogs of interleukin-2 includes all of the modifications using L-type or D-type amino acids and/or non-natural amino acids; and/or a modification of native sequence, for example, a modification of a side chain functional group, an intramolecular covalent bonding (e.g., a ring formation between side chains), methylation, acylation, ubiquitination, phosphorylation, aminohexanation, biotinylation, etc.

Additionally, the modification includes all of those where one or more amino acids are added to the amino and/or carboxy terminus of native interleukin-2.

As the amino acids to be substituted or added, not only the 20 amino acids commonly observed in human proteins, but also atypical amino acids or those which are not naturally occurring can be used. Commercial sources of atypical amino acids include Sigma-Aldrich, Chem Pep Inc., and Genzyme Pharmaceuticals. The peptides including these amino acids and typical peptide sequences may be synthesized and purchased from commercial suppliers, e.g., American Peptide Company, Bachem (USA), or Anygen (Korea). Amino acid derivatives may be obtained in the same manner, and as one such example, 4-imidazoacetic acid, etc. may be used.

Additionally, the interleukin-2 analog according to the present invention may be in a modified form where the N-terminus and/or C-terminus, etc. of the interleukin-2 is chemically modified or protected by organic groups, or amino acids may be added to the terminus of the peptide, etc. for its protection from proteases in vivo while increasing its stability.

In particular, in the case of a chemically synthesized peptide, its N- and C-termini are electrically charged, and thus the N-terminus of the peptide may be acetylated and/or C-terminus of the peptide may be amidated, but the peptide is not particularly limited thereto.

Additionally, since the interleukin-2 analog according to the present invention is in a peptide form, it may include all of those in the form of the peptide itself, a salt thereof (e.g., a pharmaceutically acceptable salt of the peptide), or a solvate thereof. Additionally, the peptide may be in any pharmaceutically acceptable form.

The kind of the salt is not particularly limited. However, it is desirable that the salt be in a safe and effective form for a subject (e.g., a mammal), but the salt type is not particularly limited thereto.

The term "pharmaceutically acceptable" refers to a material which can be effectively used for a desired purpose without causing excessive toxicity, irritation, allergic reactions, etc. within the scope of medical judgment.

As used herein, the term "pharmaceutically acceptable salt" includes salts which are derived from pharmaceutically acceptable inorganic acids, organic acids, or bases. Examples of suitable acids include hydrochloric acid, bromic acid, sulfuric acid, nitric acid, perchloric acid, fumaric acid, maleic acid, phosphoric acid, glycolic acid, lactic acid, salicylic acid, succinic acid, toluene-p-sulfonic acid, tartaric acid, acetic acid, citric acid, methanesulfonic acid, formic acid, benzoic acid, malonic acid, naphthalene-2-sulfonic acid, benzenesulfonic acid, etc. Salts derived from suitable bases may include alkali metals (e.g., sodium, potassium, etc.), alkaline earth metals (e.g., magnesium, etc.), ammonium, etc.

Additionally, as used herein, the term "solvate" refers to a complex formed between the peptide according to the present invention or a salt thereof and a solvent molecule.

In the present invention, the binding affinity of any interleukin-2 analog for native interleukin-2 receptors can be measured using various known techniques, which are methods for measuring the affinity for the receptors. For example, surface plasmon resonance (SPR) may be used, but the measurement method is not limited thereto.

More specifically, the interleukin-2 analog of the present invention may have reduced or increased binding affinity for interleukin-2 alpha receptors compared to native interleukin-2 or aldesleukin.

Specifically, the interleukin-2 analog of the present invention may have binding affinity for interleukin-2 alpha receptors of about 0.001-fold or greater, about 0.005-fold or greater, about 0.01-fold or greater, about 0.05-fold or greater, about 0.1-fold or greater, about 0.3-fold or greater, about 0.5-fold or greater, about 0.7-fold or greater, about 0.9-fold or greater, about 1.1-fold or greater, about 1.3-fold or greater, about 1.5-fold or greater, or about 1.7-fold or greater compared to the binding affinity of native interleukin-2 or aldesleukin for interleukin-2 alpha receptors, but the numerical value of the binding affinity is not limited, and the value will belong to the scope of the present invention as long as there is a change in the binding affinity compared to that of native interleukin-2 or aldesleukin.

Alternatively, based on the binding affinity of aldesleukin for interleukin-2 alpha receptors (set at 100%), the interleukin-2 analog of the present invention may have no binding affinity for interleukin-2 alpha receptors or have binding affinity for interleukin-2 alpha receptors of about 1% or greater, about 5% or greater, about 7% or greater, about 10% or greater, about 15% or greater, about 20% or greater, about 30% or greater, about 50% or greater, about

70% or greater, about 90% or greater, about 100% or greater, about 150% or greater, or about 200% or greater, but the numerical value of the binding affinity is not limited, and the value will belong to the scope of the present invention as long as there is a change in the binding affinity compared to that of native interleukin-2 or aldesleukin.

Additionally, the interleukin-2 analog of the present invention may specifically have binding affinity for interleukin-2 beta receptors of about 0.1-fold or greater, about 0.3-fold or greater, about 0.5-fold or greater, about 0.7-fold or greater, about 1.0-fold or greater, about 10-fold or greater, about 20-fold or greater, about 30-fold or greater, about 40-fold or greater, about 50-fold or greater, about 60-fold or greater, about 70-fold or greater, about 80-fold or greater, about 90-fold or greater, or about 100-fold or greater compared to the binding affinity of native interleukin-2 or aldesleukin for interleukin-2 beta receptors, but the numerical value of the binding affinity is not limited, and the value will belong to the scope of the present invention as long as there is a change or increase in the binding affinity compared to that of native interleukin-2 or aldesleukin.

Alternatively, based on the binding affinity of aldesleukin for interleukin-2 beta receptors (set at 100%), the interleukin-2 analog of the present invention may have binding affinity for interleukin-2 beta receptors of about 5% or greater, about 9% or greater, about 10% or greater, about 20% or greater, about 30% or greater, about 50% or greater, about 100% or greater, about 200% or greater, about 500% or greater, about 700% or greater, about 1,000% or greater, about 1,500% or greater, about 3,000% or greater, about 5,000% or greater, about 7,000% or greater, about 10,000% or greater, about 12,000% or greater, about 15,000% or greater, about 20,000% or greater, or about 25,000%, but the numerical value of the binding affinity is not limited, and the value will belong to the scope of the present invention as long as there is an increase in the binding affinity compared to that of native interleukin-2 or aldesleukin.

As used herein, the term "about" refers to a range which includes all of ± 0.5 , ± 0.4 , ± 0.3 , ± 0.2 , ± 0.1 , etc. and includes all of the values that are equivalent or similar to those following the values, but the range is not limited thereto.

The interleukin-2 analog of the present invention is characterized in that it has altered binding affinity for interleukin-2 alpha receptors and increased binding affinity for interleukin-2 beta receptors compared to native interleukin-2 or aldesleukin.

In a specific embodiment of the present invention, for the preparation of the interleukin-2 analog of the present invention, an interleukin-2 analog was prepared into which a modification was introduced based on native interleukin-2 (SEQ ID NO: 1). The interleukin-2 analog prepared in the present invention may be one which includes any one amino acid sequence among SEQ ID NOS: 3 to 106, or may be one which is encoded by any one nucleotide sequence among SEQ ID NOS: 108 to 211.

Still another aspect to implement the present invention provides a nucleic acid (polynucleotide) encoding the interleukin-2 analog, a recombinant expression vector including the nucleic acid, and a transformant which includes the nucleic acid or recombinant expression vector.

The nucleic acid encoding the interleukin-2 analog of the present invention may be one which is modified so that a modification (deletion, substitution, and/or addition of an amino acid) can be introduced into an amino acid at a particular position in a native interleukin-2 of SEQ ID NO: 1, and specifically, the interleukin-2 analog of the present invention may include a nucleotide sequence encoding any

one amino acid sequence among SEQ ID NOS: 3 to 106. For example, the nucleic acid of the present invention may have or include a nucleotide sequence of any one among SEQ ID NOS: 108 to 211.

The nucleotide sequence of the present invention may be modified variously in the coding region within a range not altering the amino acid sequence of the interleukin-2 analog of the present invention, considering codon degeneracy or the codons preferred in the organism where the nucleic acid of the present invention is to be expressed. Specifically, the nucleic acid of the present invention may have or include a nucleotide sequence which has a homology or identity of 70% or higher, 75% or higher, 80% or higher, 85% or higher, 90% or higher, 95% or higher, 96% or higher, 97% or higher, 98% or higher, and less than 100% to any one of the sequences of SEQ ID NOS: 108 to 211; or may consist of or essentially consist of a nucleotide sequence which has a homology or identity of 70% or higher, 75% or higher, 80% or higher, 85% or higher, 90% or higher, 95% or higher, 96% or higher, 97% or higher, 98% or higher, and less than 100% to any one of the sequences of SEQ ID NOS: 108 to 211, but the nucleic acid is not limited thereto.

Additionally, the nucleic acid of the present invention can include, without limitation, a probe which can be prepared from a known gene sequence (e.g., a sequence that can hybridize with a sequence complementary to all or part of the nucleic acid sequence of the present invention under stringent conditions). The "stringent conditions" refer to conditions that enable specific hybridization between polynucleotides. Such conditions are described in detail in the literature (see J. Sambrook et al., *Molecular Cloning, A Laboratory Manual*, 2nd Edition, Cold Spring Harbor Laboratory press, Cold Spring Harbor, New York, 1989; F. M. Ausubel et al., *Current Protocols in Molecular Biology*, John Wiley & Sons, Inc., New York, 9.50-9.51, 11.7-11.8).

Hybridization requires that two nucleic acids have complementary sequences, although mismatches between bases may be possible depending on hybridization stringency. The term "complementary" is used to describe the relationship between nucleotide bases that can hybridize to each other. For example, with respect to DNA, adenine is complementary to thymine, and cytosine is complementary to guanine. Accordingly, the nucleic acid of the present invention can include isolated nucleic acid fragments complementary to the entire sequence as well as to substantially similar nucleic acid sequences.

The appropriate stringency for hybridizing polynucleotides depends on the length of the polynucleotides and the degree of complementarity, and variables are well known in the art (e.g., Sambrook et al., *supra*).

The homology and identity are as described above.

The recombination vector according to the present invention may be constructed as a vector for typical cloning or a vector for expression, and may be constructed as a vector for use of eukaryotic or prokaryotic cells as a host cell.

As used herein, the term "vector", which is a recombination vector capable of expressing a target protein in an appropriate host cell, refers to a nucleic acid construct that includes essential control elements operably linked to enable the expression of the nucleic acid insert. In the present invention, it is possible to prepare a recombination vector which includes a nucleic acid encoding an interleukin-2 analog, and the interleukin-2 analog of the present invention can be obtained by transforming or transfecting a host cell with the recombination vector.

As used herein, the term "transformation" refers to a phenomenon in which DNA is introduced into a host cell to

allow DNA to be replicated as a factor of a chromosome or by completion of chromosome integration, and external DNA is introduced into cells to artificially cause genetic changes.

The host suitable for the present invention is not particularly limited as long as it enables the expression of the nucleic acid of the present invention. Specific examples of the host that can be used in the present invention include bacteria of the genus *Escherichia* (e.g., *E. coli*); bacteria of the genus *Bacillus* (e.g., *Bacillus subtilis*); bacteria of the genus *Pseudomonas* (e.g., *Pseudomonas putida*); yeasts (e.g., *Pichia pastoris*, *Saccharomyces cerevisiae*, and *Schizosaccharomyces pombe*); insect cells (e.g., *Spodoptera frugiperda* (SF9)); and animal cells (e.g., CHO, COS, BSC, etc.).

Still another aspect to implement the present invention provides a method for preparing an interleukin-2 analog which includes one or more modifications.

Specifically, the method may include introducing a modification into one or more amino acids selected from the group consisting of amino acids corresponding to those at positions 1, 12, 18, 19, 20, 22, 32, 35, 38, 42, 43, 45, 48, 49, 61, 68, 69, 74, 76, 80, 81, 82, 84, 85, 86, 87, 88, 89, 91, 92, 94, 95, 96, 125, 126, and 133 in native interleukin-2.

More specifically, the method may be:

- (a) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 32 are substituted with different amino acids;
- (b) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 35 are substituted with different amino acids;
- (c) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 38 are substituted with different amino acids;
- (d) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 42 are substituted with different amino acids;
- (e) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 43 are substituted with different amino acids;
- (f) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 48 are substituted with different amino acids;
- (g) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 49 are substituted with different amino acids;
- (h) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 76 are substituted with different amino acids;
- (i) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 81, 92, 94, and 96 are substituted with different amino acids;
- (j) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 87 are substituted with different amino acids;

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- (aq) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (ar) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 42, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (as) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 61, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (at) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 69, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (au) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 85, 86, 91, and 92 are substituted with different amino acids;
- (av) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 18, 22, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (aw) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (ax) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 45, 74, 80, 81, and 92 are substituted with different amino acids;
- (ay) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 68, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (az) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 69, 74, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (ba) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 84, 85, 86, 91, and 92 are substituted with different amino acids;
- (bb) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 85, 86, 92, 94, and 96 are substituted with different amino acids;
- (bc) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 18, 19, 22, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (bd) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 18, 22, 38, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (be) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 18, 22, 61, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (bf) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 18, 22, 68, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (bg) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at

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- positions 125, 35, 38, 42, 80, 81, 85, 86, and 92 are substituted with different amino acids;
 - (bh) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 45, 80, 81, 85, 86, and 92 are substituted with different amino acids;
 - (bi) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 85, 86, 91, and 92 are substituted with different amino acids;
 - (bj) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 85, 86, 92, and 95 are substituted with different amino acids;
 - (bk) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 35, 38, 42, 74, 80, 81, 85, 86, and 92 are substituted with different amino acids;
 - (bl) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 43, 61, 80, 81, 85, 86, and 92 are substituted with different amino acids;
 - (bm) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 85, 86, 91, 92, and 95 are substituted with different amino acids; or
 - (bn) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 35, 38, 42, 74, 80, 81, 82, 85, 86, and 92 are substituted with different amino acids,
 - (bo) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 86, and 92 are substituted with different amino acids; or
 - (bp) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 85, and 86 are substituted with different amino acids;
- but the method is not limited thereto.

The interleukin-2 analog and modification are the same as above.

In another embodiment of a method for preparing the interleukin-2 analog of the present invention, the method for preparing the interleukin-2 analog may include (a) culturing a transformant which includes a nucleic acid encoding the interleukin-2 analog and expressing the interleukin-2 analog; and (b) isolating and purifying the expressed interleukin-2 analog, but the method is not limited to any particular method, and any method known in the art may be used as long as the interleukin-2 analog can be prepared by the same.

In the present invention, the nucleic acid encoding the interleukin-2 analog may include or (essentially) consist of any one nucleotide sequence among SEQ ID NOS: 108 to 211.

The medium used for culturing a transformant in the present invention must meet the requirements for culturing host cells in an appropriate manner. The carbon sources that can be included in the medium for the growth of host cells can be appropriately selected as a decision by those skilled in the art according to the type of transformants being produced, and appropriate culture conditions can be adopted to control the time and amount of culture.

Sugar sources that can be used may include sugars and carbohydrates (e.g., glucose, saccharose, lactose, fructose, maltose, starch, and cellulose); oils and fats (e.g., soybean oil, sunflower oil, castor oil, coconut oil, etc.); fatty acids

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(e.g., palmitic acid, stearic acid, and linoleic acid); alcohols (e.g., glycerol and ethanol); and organic acids (e.g., acetic acid). These materials can be used individually or as a mixture.

Nitrogen sources that can be used may include peptone, yeast extract, gravy, malt extract, corn steep liquor, soybean meal, and urea, or inorganic compounds (e.g., ammonium sulfate, ammonium chloride, ammonium phosphate, ammonium carbonate, and ammonium nitrate). Nitrogen sources can also be used individually or as a mixture.

Phosphorous sources that can be used may include potassium dihydrogen phosphate or dipotassium hydrogen phosphate, or corresponding sodium-containing salts thereof. In addition, the culture medium may contain a metal salt (e.g., magnesium sulfate and iron sulfate) required for growth.

Finally, in addition to these materials, essential growth materials (e.g., amino acids and vitamins) may be used. In addition, suitable precursors for culture media may be used. The above-mentioned raw materials can be added in a batch or continuous mode in a manner appropriate to the culture during the cultivation. Basic compounds (e.g., sodium hydroxide, potassium hydroxide, and ammonia) or acidic compounds (e.g., phosphoric acid and sulfuric acid) can be used in an appropriate manner to adjust the pH of the culture. In addition, antifoaming agents (e.g., fatty acid polyglycol esters) may be used to inhibit bubble generation. In order to maintain aerobic conditions, oxygen or oxygen-containing gas (e.g., air) is injected into the culture.

Culturing of the transformant according to the present invention is usually performed at a temperature of 20° C. to 45° C., specifically 25° C. to 40° C. In addition, the culture is continued until the maximum amount of the desired interleukin-2 analog is obtained, and for this purpose, the culture can usually last for 10 to 160 hours.

As described above, if the appropriate culture conditions are established depending on the host cell, the transformant according to the present invention will produce an interleukin-2 analog, and depending on the composition of the vector and the characteristics of the host cell, the interleukin-2 analog produced can be secreted into the cytoplasm of the host cell, into the periplasmic space, or extracellularly.

Proteins expressed in the host cell or outside thereof can be purified in a conventional manner. Examples of purification methods include salting out (e.g., ammonium sulfate precipitation, sodium phosphate precipitation, etc.), solvent precipitation (e.g., protein fractionation precipitation using acetone, ethanol, etc.), dialysis, gel filtration, ion exchange, chromatography (e.g., reverse-phase column chromatography), ultrafiltration, etc. and can be used alone or in combination.

In a specific embodiment of the present invention, a method for preparing an interleukin-2 analog may include:

- (a) expressing the interleukin-2 analog; and
- (b) isolating the expressed interleukin-2 analog.

In a specific embodiment of the present invention, the following steps may be further included to isolate and purify the interleukin-2 analog expressed in the form of an inclusion body from a transformant:

- (b-1) collecting and disrupting a transformant from the culture medium of the step (a) above;
- (b-2) recovering and refolding an interleukin-2 analog expressed in a disrupted cell lysate; and
- (b-3) purifying the refolded interleukin-2 analog by size-exclusion chromatography.

Still another aspect to implement the present invention provides a method for preparing the interleukin-2 analog by way of a peptide synthesis method. Since the sequences of

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the interleukin-2 analogs of the present invention are already provided, the synthesis of peptides can be performed using a known peptide synthesis method.

The interleukin-2 analog and modification are as described above.

Still another aspect to implement the present invention provides a method for increasing the binding affinity for interleukin-2 beta receptors, which includes modifying one or more amino acids in native interleukin-2.

The method for increasing the binding affinity for interleukin-2 beta receptors according to the present invention may be one which not only increases the binding affinity for interleukin-2 beta receptors but also alters the binding affinity for interleukin-2 alpha receptors compared to native interleukin-2 or aldesleukin.

Specifically, the method may include a step of introducing a modification into one or more amino acids corresponding to those at positions 1, 12, 18, 19, 20, 22, 32, 35, 38, 42, 43, 45, 48, 49, 61, 68, 69, 74, 76, 80, 81, 82, 84, 85, 86, 87, 88, 89, 91, 92, 94, 95, 96, 125, 126, and 133 in native interleukin-2.

More specifically, the method may be:

- (a) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 32 are substituted with different amino acids;
- (b) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 35 are substituted with different amino acids;
- (c) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 38 are substituted with different amino acids;
- (d) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 42 are substituted with different amino acids;
- (e) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 43 are substituted with different amino acids;
- (f) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 48 are substituted with different amino acids;
- (g) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 49 are substituted with different amino acids;
- (h) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 76 are substituted with different amino acids;
- (i) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 81, 92, 94, and 96 are substituted with different amino acids;
- (j) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 87 are substituted with different amino acids;
- (k) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, and 42 are substituted with different amino acids;

- (l) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, and 80 are substituted with different amino acids;
- (m) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, and 84 are substituted with different amino acids;
- (n) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 19, 38, and 42 are substituted with different amino acids;
- (o) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 12, 38, and 42 are substituted with different amino acids;
- (p) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 61 are substituted with different amino acids;
- (q) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 84 are substituted with different amino acids;
- (r) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 88 are substituted with different amino acids;
- (s) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 89 are substituted with different amino acids;
- (t) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 91 are substituted with different amino acids;
- (u) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 94 are substituted with different amino acids;
- (v) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 126 are substituted with different amino acids;
- (w) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 80, and 84 are substituted with different amino acids;
- (x) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 94, and 96 are substituted with different amino acids;
- (y) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 81, and 92 are substituted with different amino acids;
- (z) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 61, 81, and 92 are substituted with different amino acids;
- (aa) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 81, and 92 are substituted with different amino acids;

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- (ar) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 42, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (as) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 61, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (at) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 69, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (au) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 85, 86, 91, and 92 are substituted with different amino acids;
- (av) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 18, 22, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (aw) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (ax) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 45, 74, 80, 81, and 92 are substituted with different amino acids;
- (ay) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 68, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (az) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 69, 74, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (ba) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 84, 85, 86, 91, and 92 are substituted with different amino acids;
- (bb) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 85, 86, 92, 94, and 96 are substituted with different amino acids;
- (bc) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 18, 19, 22, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (bd) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 18, 22, 38, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (be) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 18, 22, 61, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (bf) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 18, 22, 68, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (bg) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 35, 38, 42, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (bh) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at

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- positions 125, 38, 42, 45, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (bi) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 85, 86, 91, and 92 are substituted with different amino acids;
- (bj) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 85, 86, 92, and 95 are substituted with different amino acids;
- (bk) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 35, 38, 42, 74, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (bl) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 43, 61, 80, 81, 85, 86, and 92 are substituted with different amino acids;
- (bm) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 85, 86, 91, 92, and 95 are substituted with different amino acids;
- (bn) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 35, 38, 42, 74, 80, 81, 82, 85, 86, and 92 are substituted with different amino acids;
- (bo) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 86, and 92 are substituted with different amino acids; or
- (bp) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 85, and 86 are substituted with different amino acids, but the method is not limited thereto.

The interleukin-2 analog and modification are the same as above.

Still another aspect to implement the present invention provides an interleukin-2 analog which includes any one sequence selected from the group consisting of amino acid sequences of SEQ ID NOS: 3 to 106.

The definitions of the interleukin-2 analog, modification, and analog represented by a SEQ ID NO are the same as above.

Specifically, the interleukin-2 analog may include, essentially consist of, or consist of any one nucleotide sequence selected from the group consisting of SEQ ID NOS: 3 to 106, but the interleukin-2 analog is not limited thereto.

Still another aspect to implement the present invention provides an interleukin-2 analog which includes an amino acid sequence represented by General Formula 1 below:

[General Formula 1]
(General Formula 1, SEQ ID NO: 212)
X1-P-T-S-S-S-T-K-K-T-Q-L-Q-L-E-H-L-X18-X19-D-L-
X22-M-I-L-N-G-I-N-N-Y-K-N-P-K-L-T-X38-M-L-T-X42-
X43-F-X45-M-P-K-K-A-T-E-L-K-H-L-Q-C-L-E-X61-E-L-K-
P-L-E-X68-V-L-N-L-A-X74-S-K-N-F-H-X80-X81-P-R-X84-
X85-X86-S-N-I-N-X91-X92-V-X94-E-X96-K-G-S-E-T-T-F-
M-C-E-Y-A-D-E-T-A-T-I-V-E-F-L-N-R-W-I-T-F-S-Q-S-I-
I-S-T-L-T

wherein in General Formula 1 above,

X1 is a deletion;

X18 is leucine (L) or arginine (R);

X19 is leucine (L) or tyrosine (Y);

X22 is glutamic acid (E) or glutamine (Q);

X38 is alanine (A), aspartic acid (D), or arginine (R);

X42 is alanine (A), phenylalanine (F), lysine (K), or tryptophan (W);

X43 is glutamic acid (E), lysine (K), or glutamine (Q);

X45 is alanine (A) or tyrosine (Y);

X61 is aspartic acid (D), glutamic acid (E), glutamine (Q), or arginine (R);

X68 is aspartic acid (D) or glutamic acid (E);

X74 is histidine (H) or glutamine (Q);

X80 is phenylalanine (F), leucine (L), valine (V), or tyrosine (Y);

X81 is aspartic acid (D), glutamic acid (E), or arginine (R);

X84 is aspartic acid (D) or glutamic acid (E);

X85 is alanine (A), glutamic acid (E), glycine (G), leucine (L), valine (V), tryptophan (W), or tyrosine (Y);

X86 is alanine (A), glycine (G), isoleucine (I), or valine (V);

X91 is threonine (T) or valine (V);

X92 is phenylalanine (F), isoleucine (I), or tyrosine (Y);

X94 is phenylalanine (F) or leucine (L); and

X96 is phenylalanine (F) or leucine (L).

Additionally, in General Formula 1 above, one or more amino acids may be added to threonine (T), which corresponds to X133, but the sequence is not limited thereto.

Specifically, the interleukin-2 analog may include, essentially consist of, or consist of any one sequence selected from the group consisting of amino acid sequences of SEQ ID NOS: 10, 13, 14, 15, 16, 17, 20, 21, 22, 32, 35, 36, 42, 53, 54, 56, 58, 59, 60, 62, 71, 72, 74, 75, 76, 77, 78, 85, 87, 89, 91, 92, 93, 94, 95, 98, 99, 100, 101, 103, 104, 105, and 106, but the interleukin-2 analog is not limited thereto.

Such an interleukin-2 analog may have increased binding affinity for beta receptors compared to aldesleukin or native interleukin-2, but the binding affinity of the interleukin-2 analog is not limited thereto.

In another embodiment, the interleukin-2 analog of the present invention may be: in General Formula 1 above,

X43 is lysine (K);

X45 is tyrosine (Y);

X61 is aspartic acid (D), glutamic acid (E), or glutamine (Q);

X68 is glutamic acid (E);

X74 is glutamine (Q);

X80 is phenylalanine (F) or leucine (L);

X85 is leucine (L), valine (V), or tyrosine (Y);

X86 is isoleucine (I) or valine (V); and

X92 is phenylalanine (F) or isoleucine (I), but the interleukin-2 analog is not limited thereto.

Specifically, the interleukin-2 analog is characterized in that it includes any one sequence selected from the group consisting of amino acid sequences of SEQ ID NOS: 10, 13, 14, 16, 17, 20, 21, 22, 32, 35, 36, 42, 53, 54, 87, 89, 91, 92, 93, 94, 98, 99, 100, 101, 103, 104, and 105.

In the interleukin-2 analog of the present invention, one or more amino acids may be further added to a C-terminus thereof, but the interleukin-2 analog is not limited thereto.

Still another aspect to implement the present invention provides an interleukin-2 analog which includes an amino acid sequence expressed by General Formula 2 below:

[General Formula 2]

(General Formula 2, SEQ ID NO: 213)

X1-P-T-S-S-S-T-K-K-T-Q-L-Q-L-E-H-L-X18-L-D-L-X22-

5 M-I-L-N-G-I-N-N-Y-K-N-P-K-L-T-X38-M-L-T-X42-K-F-Y-

M-P-K-K-A-T-E-L-K-H-L-Q-C-L-E-X61-E-L-K-P-L-E-X68-

V-L-N-L-A-Q-S-K-N-F-H-F-X81-P-R-D-X85-X86-S-N-I-N-

10 V-F-V-L-E-L-K-G-S-E-T-T-F-M-C-E-Y-A-D-E-T-A-T-I-V-

E-F-L-N-R-W-I-T-F-S-Q-S-I-I-S-T-L-T

wherein in General Formula 2 above,

X1 is a deletion,

X18 is leucine (L) or arginine (R);

X22 is glutamic acid (E) or glutamine (Q);

X38 is alanine (A) or arginine (R);

X42 is phenylalanine (F) or lysine (K);

20 X61 is aspartic acid (D) or glutamic acid (E);

X68 is aspartic acid (D) or glutamic acid (E);

X81 is aspartic acid (D) or glutamic acid (E);

X85 is leucine (L) or valine (V); and

25 X86 is isoleucine (I) or valine (V).

Specifically, the interleukin-2 analog may include any one sequence selected from the group consisting of amino acid sequences of SEQ ID NO: 22, 42, 53, 87, 105, and 106, but the sequence of the interleukin-2 analog is not limited thereto.

Additionally, in General Formula 2 above, one or more amino acids may be added to threonine (T), which corresponds to X133, or alternatively, in the interleukin-2 analog, one or more amino acids may be further added to a C-terminus thereof, but the present invention is not limited thereto.

Meanwhile, unless otherwise required by context in the present specification, expressions such as "include", "including", "containing", etc. mean that they include a specified integer or group of integers, but it should be understood that these expressions do not exclude other integers or a set of integers.

Hereinafter, the present invention will be described in more detail through examples. These Examples are only for describing the present invention in more detail, and the scope of the present invention is not limited by these Examples.

Example 1: Preparation of Expression Vectors for Native Interleukin-2 and Interleukin-2 Analogs

For the preparation of expression vectors for native interleukin-2 encoding 133 amino acids, an interleukin-2 that was synthesized based on the reported interleukin-2 sequence (NM_000586.3; SEQ ID NO: 1) was cloned into the pET-22b vector (Novagen). Additionally, a novel interleukin-2 analog was prepared in which an amino acid(s) of interleukin-2 were modified using the interleukin-2 as a template. The PCR conditions for the amplification of the interleukin-2 analog were 16 cycles of a process consisting of 95° C. for 30 seconds, 55° C. for 60 seconds, and 65° C. for 6.5 minutes. In order to confirm whether the amino acid(s) at the desired position had been correctly substituted, sequence analysis was performed on the mutagenesis product obtained under the conditions above. As a result, it was

confirmed that the modifications shown in Table 1 below were found based on the native type at the desired mutation positions for each interleukin-2 analog. The thus-obtained expression vectors were named pET22b-interleukin-2 analogs 1 to 105.

Table 1 below shows the altered sequences of amino acids and analog names for each. In order to prepare these

interleukin-2 analogs, forward (F) and reverse (R) primers were synthesized, and then PCR was performed to amplify each analog gene.

In Table 1 below, analog 1 represents aldesleukin and primer nos 1 to 203 correspond to SEQ ID NOS: 214 to 417, respectively.

TABLE 1

Types of interleukin-2 analog, positions for modification, and altered sequences thereof		
Analog	Positions for Modification and Altered Sequences Thereof	Primer No.
1	desA1, C125S	197, 198, 201, 202
2	desA1, C125S, S87C	197, 198, 201, 202, 149, 150
3	desA1, C125S, K32C	197, 198, 201, 202, 19, 20
4	desA1, C125S, K35C	197, 198, 201, 202, 21, 22
5	desA1, C125S, K43C	197, 198, 201, 202, 47, 48
6	desA1, C125S, K48C	197, 198, 201, 202, 53, 54
7	desA1, C125S, K49C	197, 198, 201, 202, 55, 56
8	desA1, C125S, K76C	197, 198, 201, 202, 73, 74
9	desA1, C125S, R38A	197, 198, 201, 202, 25, 26
10	desA1, C125S, F42K	197, 198, 201, 202, 35, 36
11	desA1, C125S, F42A	197, 198, 201, 202, 33, 34
12	desA1, C125S, R38A, F42K	197, 198, 201, 202, 25, 26, 35, 36
13	desA1, C125S, R38A, F42A	197, 198, 201, 202, 25, 26, 33, 34
14	desA1, C125S, L19Y, R38A, F42K	197, 198, 201, 202, 13, 14, 25, 26, 35, 36
15	desA1, C125S, R38A, F42K, D84E	197, 198, 201, 202, 25, 26, 35, 36, 109, 110
16	desA1, C125S, R38A, F42K, N88Q	197, 198, 201, 202, 25, 26, 35, 36, 153, 154
17	desA1, C125S, R38A, F42K, V91T	197, 198, 201, 202, 25, 26, 35, 36, 165, 166
18	desA1, C125S, R38A, F42K, E61Q	197, 198, 201, 202, 25, 26, 35, 36, 59, 60
19	desA1, C125S, R38A, F42K, R81D, I92F	197, 198, 201, 202, 25, 26, 35, 36, 97, 98, 167, 168
20	desA1, C125S, R38A, F42K, L85V, I86V, I92F	197, 198, 201, 202, 25, 26, 35, 36, 139, 140, 167, 168
21	desA1, C125S, R38A, F42K, L80F, R81D, L85V, I86V, I92F	197, 198, 201, 202, 25, 26, 35, 36, 79, 80, 167, 168
22	desA1, C125S, L12V, R38A, F42K	197, 198, 201, 202, 3, 4, 25, 26, 35, 36
23	desA1, C125S, L12F, R38A, F42K	197, 198, 201, 202, 1, 2, 25, 26, 35, 36
24	desA1, C125S, L19V, R38A, F42K	197, 198, 201, 202, 11, 12, 25, 26, 35, 36
25	desA1, C125S, L19F, R38A, F42K	197, 198, 201, 202, 9, 10, 25, 26, 35, 36
26	desA1, C125S, R38A, F42K, I89F	197, 198, 201, 202, 25, 26, 35, 36, 157, 158
27	desA1, C125S, R38A, F42K, V91F	197, 198, 201, 202, 25, 26, 35, 36, 163, 164
28	desA1, C125S, R38A, F42K, L94V	197, 198, 201, 202, 25, 26, 35, 36
29	desA1, C125S, R38A, F42K, Q126T	197, 198, 201, 202, 25, 26, 35, 36, 199, 200
30	desA1, C125S, R38A, R81D, I92F	197, 198, 201, 202, 27, 28, 97, 98, 169, 170
31	desA1, C125S, R38A, D84E	197, 198, 201, 202, 27, 28, 109, 110

TABLE 1-continued

Types of interleukin-2 analog, positions for modification, and altered sequences thereof		
Analog	Positions for Modification and Altered Sequences Thereof	Primer No.
32	desA1, C125S, R38A, R81D, D84E, I92F	197, 198, 201, 202, 27, 28, 95, 96, 169, 170
33	desA1, C125S, R38A, L80F	197, 198, 201, 202, 27, 28, 77, 78
34	desA1, C125S, R38A, L80F, D84E	197, 198, 201, 202, 27, 28, 77, 78, 109, 110
35	desA1, C125S, R38A, L94F, L96F	197, 198, 201, 202, 27, 28, 189, 190
36	desA1, C125S, R38A, L94F, L96V	197, 198, 201, 202, 27, 28, 193, 194
37	desA1, C125S, R38A, L94F, L96I	197, 198, 201, 202, 27, 28, 191, 192
38	desA1, C125S, R38A, F42K, R81D, I92F, L94F, L96F	197, 198, 201, 202, 25, 26, 35, 36, 97, 98, 175, 176
39	desA1, C125S, R38A, F42K, R81D, I92F, L94F, L96V	197, 198, 201, 202, 25, 26, 35, 36, 97, 98, 179, 180
40	desA1, C125S, R38A, F42K, R81D, I92F, L94F, L96I	197, 198, 201, 202, 25, 26, 35, 36, 97, 98, 177, 178
41	desA1, C125S, L80F, R81D, L85V, I86V, I92F	197, 198, 201, 202, 25, 26, 29, 30, 35, 36, 41, 42, 79, 80, 167, 168
42	desA1, C125S, R38A, F42K, R81E, I92F	197, 198, 201, 202, 25, 26, 35, 36, 103, 104, 169, 170
43	desA1, C125S, R38A, F42K, R81D, I92L	197, 198, 201, 202, 25, 26, 35, 36, 99, 100, 183, 184
44	desA1, C125S, R38A, F42K, R81D, D84V, I92F	197, 198, 201, 202, 25, 26, 35, 36, 99, 100, 113, 114, 169, 170
45	desA1, C125S, R38A, F42K, R81D, D84F, I92F	197, 198, 201, 202, 25, 26, 35, 36, 99, 100, 111, 112, 169, 170
46	desA1, C125S, D20V, R38A, F42K, R81D, I92F	197, 198, 201, 202, 17, 18, 25, 26, 35, 36, 99, 100, 169, 170
47	desA1, C125S, D20F, R38A, F42K, R81D, I92F	197, 198, 201, 202, 15, 16, 25, 26, 35, 36, 99, 100, 169, 170
48	desA1, C125S, R38A, F42K, R81D, N88V, I92F	197, 198, 201, 202, 25, 26, 35, 36, 99, 100, 155, 156, 169, 170
49	desA1, C125S, R38A, F42K, R81D, N88F, I92F	197, 198, 201, 202, 25, 26, 35, 36, 99, 100, 151, 152, 169, 170
50	desA1, C125S, F42K, L80F, R81D, L85V, I86V, I92F	197, 198, 201, 202, 25, 26, 29, 30, 35, 36, 79, 80, 139, 140, 169, 170
51	desA1, C125S, E61Q, R81D, I92F	197, 198, 201, 202, 59, 60, 99, 100, 169, 170
52	desA1, C125S, R38A, F42K, L80F, R81D, I92F	197, 198, 201, 202, 25, 26, 35, 36, 79, 80, 169, 170
53	desA1, C125S, R38A, F42K, L80F, R81D, D84E, I92F	197, 198, 201, 202, 25, 26, 35, 36, 79, 80, 95, 96, 169, 170
54	desA1, C125S, R38A, F42K, Q74H, R81D, I92F	197, 198, 201, 202, 25, 26, 35, 36, 71, 72, 99, 100, 169, 170
55	desA1, C125S, R38A, F42K, Q74H, L80F, R81D, I92F	197, 198, 201, 202, 25, 26, 35, 36, 71, 72, 79, 80, 169, 170
56	desA1, C125S, R38A, F42K, Y45A, Q74H, R81D, I92F	197, 198, 201, 202, 25, 26, 35, 36, 49, 50, 71, 72, 99, 100, 169, 170
57	desA1, C125S, R38A, F42K, Y45A, Q74H, L80F, R81D, I92F	197, 198, 201, 202, 25, 26, 35, 36, 49, 50, 71, 72, 79, 80, 169, 170
58	desA1, C125S, R38A, F42K, L80F, R81D, L85A, I86A, I92F	197, 198, 201, 202, 25, 26, 35, 36, 79, 80, 115, 116, 167, 168

TABLE 1-continued

Types of interleukin-2 analog, positions for modification, and altered sequences thereof		
Analog	Positions for Modification and Altered Sequences Thereof	Primer No.
59	desA1, C125S, R38A, F42K, L80F, R81D, L85A, I86A, I92Y	197, 198, 201, 202, 25, 26, 35, 36, 79, 80, 115, 116, 187, 188
60	desA1, C125S, R38A, Y45A, L80Y, L85A, I86A, I92Y	197, 198, 201, 202, 27, 28, 35, 36, 41, 42, 51, 52, 93, 94, 115, 116, 187, 188
61	desA1, C125S, R38A, F42K, L80Y, R81D, L85G, I86V, I92Y	197, 198, 201, 202, 25, 26, 35, 36, 87, 88, 125, 126, 187, 188
62	desA1, C125S, R38A, L80W, R81E, L85G, I86A, I92F	197, 198, 201, 202, 27, 28, 35, 36, 41, 42, 85, 86, 123, 124, 171, 172
63	desA1, C125S, R38A, F42K, L80D, R81E, L85T, I86G, I92F	197, 198, 201, 202, 25, 26, 35, 36, 75, 76, 133, 134, 167, 168
64	desA1, C125S, R38A, F42K, L80Y, R81N, L85V, I86V, I92F	197, 198, 201, 202, 25, 26, 35, 36, 91, 92, 139, 140, 167, 168
65	desA1, C125S, R38A, F42K, L80Y, R81E, L85V, I86V, I92F	197, 198, 201, 202, 25, 26, 35, 36, 89, 90, 139, 140, 167, 168
66	desA1, C125S, R38A, F42K, L80F, R81E, L85F, I86V, I92F	197, 198, 201, 202, 25, 26, 35, 36, 81, 82, 121, 122, 167, 168
67	desA1, C125S, R38A, F42K, L80Y, R81D, L85F, I86V, I92W, E95D	197, 198, 201, 202, 25, 26, 35, 36, 87, 88, 121, 122, 185, 186, 195, 196
68	desA1, C125S, R38A, F42K, L80F, R81E, L85I, I86V, V91E, I92F	197, 198, 201, 202, 25, 26, 35, 36, 81, 82, 127, 128, 159, 160, 167, 168
69	desA1, C125S, R38A, F42K, L80Y, R81E, L85F, I86L, V91E, I92W, E95D	197, 198, 201, 202, 25, 26, 35, 36, 89, 90, 119, 120, 161, 162, 185, 186, 195, 196
70	desA1, C125S, R38A, F42K, L80Y, R81D, L85V, I86V, I92F	197, 198, 201, 202, 25, 26, 35, 36, 87, 88, 139, 140, 167, 168
71	desA1, C125S, R38A, F42K, L80F, R81E, L85V, I86V, I92F	197, 198, 201, 202, 25, 26, 35, 36, 81, 82, 139, 140, 167, 168
72	desA1, C125S, R38A, F42K, L80F, R81D, L85V, I86G, I92F	197, 198, 201, 202, 25, 26, 35, 36, 79, 80, 135, 136, 167, 168
73	desA1, C125S, R38A, F42K, L80F, R81D, L85W, I86V, I92F	197, 198, 201, 202, 25, 26, 35, 36, 79, 80, 141, 142, 167, 168
74	desA1, C125S, R38D, F42K, L80Y, R81D, L85V, I86V, I92F	197, 198, 201, 202, 31, 32, 35, 36, 87, 88, 139, 140, 167, 168
75	desA1, C125S, R38A, F42K, Y45A, L80F, R81E, L85V, I86V, I92F	197, 198, 201, 202, 25, 26, 35, 36, 51, 52, 81, 82, 139, 140, 167, 168
76	desA1, C125S, R38A, F42K, K43Q, E61R, L80F, R81D, L85V, I86G, I92F	197, 198, 201, 202, 25, 26, 39, 40, 61, 62, 79, 80, 135, 136, 167, 168
77	desA1, C125S, R38A, F42K, K43E, E61R, L80F, R81D, L85W, I86V, I92F	197, 198, 201, 202, 25, 26, 37, 38, 61, 62, 79, 80, 143, 144, 167, 168
78	desA1, C125S, K35E, R38A, F42K, L80F, R81E, L85V, I86V, I92F	197, 198, 201, 202, 23, 24, 25, 26, 35, 36, 81, 82, 139, 140, 167, 168
79	desA1, C125S, K35E, R38A, F42K, Q74H, L80F, R81E, L85V, I86V, I92F	197, 198, 201, 202, 23, 24, 25, 26, 35, 36, 71, 72, 81, 82, 139, 140, 167, 168
80	desA1, C125S, K35E, R38A, F42K, Q74H, L80F, R81E, P82G, L85V, I86V, I92F	197, 198, 201, 202, 23, 24, 25, 26, 35, 36, 71, 72, 81, 82, 105, 106, 139, 140, 167, 168
81	desA1, C125S, K35E, R38A, F42K, Q74H, L80F, R81E, P82V, L85V, I86V, I92F	197, 198, 201, 202, 23, 24, 25, 26, 35, 36, 71, 72, 81, 82, 107, 108, 139, 140, 167, 168

TABLE 1-continued

Types of interleukin-2 analog, positions for modification, and altered sequences thereof		
Analog	Positions for Modification and Altered Sequences Thereof	Primer No.
82	desA1, C125S, L18R, Q22E, L80F, R81D, L85E, I86V, I92F	197, 198, 201, 202, 7, 8, 25, 26, 29, 30, 35, 36, 43, 44, 79, 80, 117, 118, 167, 168
83	desA1, C125S, L18R, L19R, Q22E, L80F, R81D, L85E, I86V, I92F	197, 198, 201, 202, 5, 6, 25, 26, 29, 30, 35, 36, 43, 44, 79, 80, 117, 118, 167, 168
84	desA1, C125S, L18R, Q22E, L80V, R81D, L85E, I86V, I92F	197, 198, 201, 202, 7, 8, 25, 26, 29, 30, 35, 36, 43, 44, 83, 84, 117, 118, 167, 168
85	desA1, C125S, L80F, R81E, L85V, I86V, I92F	197, 198, 201, 202, 25, 26, 29, 30, 35, 36, 43, 44, 81, 82, 139, 140, 167, 168
86	desA1, C125S, L18R, Q22E, L80F, R81E, L85V, I86V, I92F	197, 198, 201, 202, 7, 8, 25, 26, 29, 30, 35, 36, 43, 44, 81, 82, 139, 140, 167, 168
87	desA1, C125S, L18R, L19R, Q22E, L80F, R81E, L85V, I86V, I92F	197, 198, 201, 202, 5, 6, 25, 26, 29, 30, 35, 36, 43, 44, 81, 82, 139, 140, 167, 168
88	desA1, C125S, E61D, L80F, R81E, L85V, I86V, I92F	197, 198, 201, 202, 25, 26, 29, 30, 35, 36, 43, 44, 57, 58, 81, 82, 139, 140, 167, 168
89	desA1, C125S, R38A, E68Q, L80F, R81E, L85V, I86V, I92F	197, 198, 201, 202, 27, 28, 35, 36, 43, 44, 65, 66, 81, 82, 139, 140, 167, 168
90	desA1, C125S, F42W, L80F, R81E, L85V, I86V, I92F	197, 198, 201, 202, 25, 26, 29, 30, 45, 46, 81, 82, 139, 140, 167, 168
91	desA1, C125S, E61Q, L80F, R81E, L85V, I86V, I92F	197, 198, 201, 202, 25, 26, 29, 30, 35, 36, 43, 44, 59, 60, 81, 82, 139, 140, 167, 168
92	desA1, C125S, L80F, R81E, L85V, I86V, V91T, I92F	197, 198, 201, 202, 25, 26, 29, 30, 35, 36, 43, 44, 81, 82, 139, 140, 147, 148, 167, 168
93	desA1, C125S, L80F, R81E, D84E, L85V, I86V, V91T, I92F	197, 198, 201, 202, 25, 26, 29, 30, 35, 36, 43, 44, 81, 82, 101, 102, 139, 140, 167, 168
94	desA1, C125S, L80F, R81E, L85V, I86V, I92F, L94F, L96F	197, 198, 201, 202, 25, 26, 29, 30, 35, 36, 43, 44, 81, 82, 139, 140, 173, 174
95	desA1, C125S, V69G, L80F, R81E, L85V, I86V, I92F	197, 198, 201, 202, 25, 26, 29, 30, 35, 36, 43, 44, 67, 68, 81, 82, 139, 140, 167, 168
96	desA1, C125S, V69G, Q74A, L80F, R81E, L85V, I86V, I92F	197, 198, 201, 202, 25, 26, 29, 30, 35, 36, 43, 44, 69, 70, 81, 82, 139, 140, 167, 168
97	desA1, C125S, R38A, L80F, R81D, I92F	197, 198, 201, 202, 27, 28, 35, 36, 41, 42, 79, 80, 167, 168
98	desA1, C125S, R38A, L80F, R81E, L85V, I92F	197, 198, 201, 202, 27, 28, 35, 36, 41, 42, 81, 82, 137, 138, 167, 168
99	desA1, C125S, R38A, L80F, R81E, I86V, I92F	197, 198, 201, 202, 27, 28, 35, 36, 41, 42, 81, 82, 131, 132, 167, 168
100	desA1, C125S, L80F, R81E, L85Y, I86V, I92F	197, 198, 201, 202, 25, 26, 29, 30, 35, 36, 41, 42, 81, 82, 145, 146, 167, 168
101	desA1, C125S, L80F, R81E, I86A, I92F	197, 198, 201, 202, 25, 26, 29, 30, 35, 36, 41,

TABLE 1-continued

Types of interleukin-2 analog, positions for modification, and altered sequences thereof	
Analog	Primer No.
Positions for Modification and Altered Sequences Thereof	
102 desA1, C125S, L80F, R81E, L85V, I86V	42, 81, 82, 129, 130, 167, 168 197, 198, 201, 202, 25, 26, 29, 30, 35, 36, 41, 42, 81, 82, 139, 140, 181, 182
103 desA1, C125S, L18R, Q22E, R38A, L80F, R81E, L85V, I86V, I92F	197, 198, 201, 202, 7, 8, 27, 28, 35, 36, 41, 42, 81, 82, 139, 140, 167, 168
104 desA1, C125S, L18R, Q22E, E61D, L80F, R81E, L85V, I86V, I92F	197, 198, 201, 202, 7, 8, 25, 26, 29, 30, 35, 36, 41, 42, 57, 58, 81, 82, 139, 140, 167, 168
105 desA1, C125S, L18R, Q22E, E68D, L80F, R81E, L85V, I86V, I92F	197, 198, 201, 202, 7, 8, 25, 26, 29, 30, 35, 36, 41, 42, 63, 64, 81, 82, 139, 140, 167, 168

In Table 1 above, desA1 represents a deletion of alanine, which is the first amino acid in interleukin-2. ²⁵

Table 2 below shows full-length protein sequences of interleukin-2 analogs. The letters shown in bold in Table 2 represent the positions for modification.

TABLE 2

Sequences of interleukin-2 analogs		
Analog	Protein Sequence	SEQ ID NO of Proteins
1	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	2
2	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLICNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	3
3	PTSSSTKKT QLQLEHLLLD LQMILNGINN YCNPKLTRML TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	4
4	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPCLTRML TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	5
5	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML TFCFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	6
6	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML TFKFYMPCKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	7
7	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML TFKFYMPKCA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	8

TABLE 2-continued

Sequences of interleukin-2 analogs			SEQ ID	SEQ ID
Analog	Protein Sequence		No of Proteins	No of Nucleotides
8	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	9	114	
9	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	10	115	
10	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	11	116	
11	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML TAKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	12	117	
12	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	13	118	
13	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TAKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	14	119	
14	PTSSSTKKT QLQLEHLLYD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	15	120	
15	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	16	121	
16	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISQIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	17	122	
17	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN TIVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	18	123	
18	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE QELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	19	124	
19	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL DPRDLISNIN VFVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	20	125	
20	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDVVSININ VFVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	21	126	
21	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHF DPRDVVSININ VFVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	22	127	

TABLE 2-continued

Sequences of interleukin-2 analogs				
Analog	Protein Sequence	SEQ ID No of Proteins	SEQ ID No of Nucleotides	
22	PTSSSTKKT QVQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	23	128	
23	PTSSSTKKT QFQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	24	129	
24	PTSSSTKKT QLQLEHLLVD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	25	130	
25	PTSSSTKKT QLQLEHLLFD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	26	131	
26	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	27	132	
27	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN FIVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	28	133	
28	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVVELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	29	134	
29	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	30	135	
30	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL DPRDLISNIN VFVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	31	136	
31	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRELISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	32	137	
32	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL DPRELISNIN VFVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	33	138	
33	PTSSSTKKT QLQLEHLLLD LOMILNGINN YKNPKLTAML TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHF RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	34	139	
34	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHF RPRELISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	35	140	
35	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVFEFKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	36	141	
36	PTSSSTKKT QLQLEHLLLD LOMILNGINN YKNPKLTAML TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL	37	142	

TABLE 2-continued

Sequences of interleukin-2 analogs				
Analog	Protein Sequence	SEQ ID No of Proteins	SEQ ID No of Nucleotides	
	RPRDLISNIN VIV FEV KGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT			
37	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT A ML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIV FEI KGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	38	143	
38	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT A ML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL DPRDLISNIN VV FVEF KGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	39	144	
39	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT A ML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL DPRDLISNIN V FVFEV KGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	40	145	
40	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT A ML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL DPRDLISNIN V FVFEI KGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	41	146	
41	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKL T RML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNF H DPRDVSNIN V FVLE KGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	42	147	
42	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT A ML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL EPRDLISNIN V FVLE KGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	43	148	
43	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT A ML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL DPRDLISNIN VLVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	44	149	
44	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT A ML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL DPRVLISNIN V FVLE KGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	45	150	
45	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT A ML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL DPRFLISNIN V FVLE KGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	46	151	
46	PTSSSTKKT QLQLEHLLLV LQMILNGINN YKNPKLT A ML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL DPRDLISNIN V FVLE KGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	47	152	
47	PTSSSTKKT QLQLEHLL L LQMILNGINN YKNPKLT A ML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL DPRDLISNIN V FVLE KGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	48	153	
48	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT A ML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL DPRDLISVIN V FVLE KGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	49	154	
49	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT A ML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL DPRDLISFIN V FVLE KGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	50	155	
50	PTSSSTKKT QLQLEHLLLD LOMILNGINN YKNPKL T RML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNF H DPRDVSNIN V FVLE KGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	51	156	

TABLE 2-continued

Sequences of interleukin-2 analogs			SEQ ID	SEQ ID
Analogue	Protein Sequence		NO of Proteins	NO of Nucleotides
51	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML TKKFYMPKKA TELKHLQCLE QELKPLEEVL NLAQSKNFHL DPRDLISNIN VVLEELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT		52	157
52	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHF DPRDLISNIN VVLEELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT		53	158
53	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHF DPRDLISNIN VVLEELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT		54	159
54	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAHAKNFHL DPRDLISNIN VVLEELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT		55	160
55	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAHAKNFHF DPRDLISNIN VVLEELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT		56	161
56	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFAMPKKA TELKHLQCLE EELKPLEEVL NLAHAKNFHL DPRDLISNIN VVLEELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT		57	162
57	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFAMPKKA TELKHLQCLE EELKPLEEVL NLAHAKNFHF DPRDLISNIN VVLEELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT		58	163
58	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHF DPRDAASNIN VVLEELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT		59	164
59	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHF DPRDAASNIN VVLEELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT		60	165
60	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFAMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHY RPRDAASNIN VVLEELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT		61	166
61	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHY DPRDGVSNIN VVLEELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT		62	167
62	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHW EPRDGASNIN VVLEELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT		63	168
63	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHD EPRDTGSNIN VVLEELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT		64	169
64	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHY NPRDVSNIN VVLEELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT		65	170

TABLE 2-continued

Sequences of interleukin-2 analogs				
Analog	Protein Sequence	SEQ ID No of Proteins	SEQ ID No of Nucleotides	
65	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHY EPRDVSNIN VFLVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	66	171	
66	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHF EPRDFVSNIN VFLVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	67	172	
67	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHY DPRDFVSNIN VWVLDLKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	68	173	
68	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHF EPRDIVSNIN EFVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	69	174	
69	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHY EPRDFLSNIN EWVLDLKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	70	175	
70	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHY DPRDVSNIN VFLVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	71	176	
71	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHF EPRDVSNIN VFLVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	72	177	
72	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHF DPRDVGSNIN VFLVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	73	178	
73	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHF DPRDWVSNIN VFLVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	74	179	
74	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTDML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHY DPRDVSNIN VFLVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	75	180	
75	PTSSSTKKT QLQLEHLLLD LOMILNGINN YKNPKLTAML TKKFAMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHF EPRDVSNIN VFLVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	76	181	
76	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKQFYMPKKA TELKHLQCLE RELKPLEEVL NLAQSKNFHF DPRDVGSNIN VFLVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	77	182	
77	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TKEFYMPKKA TELKHLQCLE RELKPLEEVL NLAQSKNFHF DPRDWVSNIN VFLVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	78	183	
78	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPELTAML TKKFYMPKKA TELKHLOCLE EELKPLEEVL NLAQSKNFHF EPRDVSNIN VFLVLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	79	184	

TABLE 2-continued

Sequences of interleukin-2 analogs			SEQ ID	SEQ ID
Analog	Protein Sequence		No of Proteins	No of Nucleotides
79	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPELTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAH SKNFHF E PRDVVSININ V F VLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT		80	185
80	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPELTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAH SKNFHF E GRDVVSININ V F VLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT		81	186
81	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPELTAML TKKFYMPKKA TELKHLQCLE EELKPLEEVL NLAH SKNFHF E VRDVVSININ V F VLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT		82	187
82	PTSSSTKKT QLQLEHLRLD LEMILNGINN YKNPKLTRML TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH F D PRDEVSNIN V F VLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT		83	188
83	PTSSSTKKT QLQLEHLRRD LEMILNGINN YKNPKLTRML TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH F D PRDEVSNIN V F VLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT		84	189
84	PTSSSTKKT QLQLEHLRLD LEMILNGINN YKNPKLTRML TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH V D PRDEVSNIN V F VLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT		85	190
85	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH F E PRDVVSININ V F VLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT		86	191
86	PTSSSTKKT QLQLEHLRLD LEMILNGINN YKNPKLTRML TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH F E PRDVVSININ V F VLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT		87	192
87	PTSSSTKKT QLQLEHLRRD LEMILNGINN YKNPKLTRML TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH F E PRDVVSININ V F VLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT		88	193
88	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML TFKFYMPKKA TELKHLQCLE D EELKPLEEVL NLAQSKNFH F E PRDVVSININ V F VLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT		89	194
89	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TFKFYMPKKA TELKHLQCLE EELKPLEQVL NLAQSKNFH F E PRDVVSININ V F VLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT		90	195
90	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML TWKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH F E PRDVVSININ V F VLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT		91	196
91	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML TFKFYMPKKA TELKHLQCLE QELKPLEEVL NLAQSKNFH F E PRDVVSININ V F VLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT		92	197
92	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH F E PRDVVSININ T F VLELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT		93	198
93	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH F		94	199

TABLE 2-continued

Sequences of interleukin-2 analogs			
Analog	Protein Sequence	SEQ ID NO of Proteins	SEQ ID NO of Nucleotides
	EPRE VVSNN TFV LELKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT		
94	PTSSSTKKT QLQLEHLLLD LOMILNGINN YKNPKLTRML TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHF EP RDVVSNN VFVFE FKGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	95	200
95	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML TFKFYMPKKA TELKHLQCLE EELKPLEEGL NLAQSKNFHF EP RDVVSNN VFVLE KGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	96	201
96	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML TFKFYMPKKA TELKHLQCLE EELKPLEEGL NLAASKNFHF EP RDVVSNN VFVLE KGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	97	202
97	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHF D PRDLISNN VFVLE KGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	98	203
98	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHF EP RDVISNN VFVLE KGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	99	204
99	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTAML TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHF EP RDVVSNN VFVLE KGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	100	205
100	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHF EP RDVVSNN VFVLE KGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	101	206
101	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHF EP RDLASNN VFVLE KGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	102	207
102	PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHF EP RDVVSNN VIVLE KGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	103	208
103	PTSSSTKKT QLQLEHLRLD L EMILNGINN YKNPKLTAML TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHF EP RDVVSNN VFVLE KGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	104	209
104	PTSSSTKKT QLQLEHLRLD L EMILNGINN YKNPKLTRML TFKFYMPKKA TELKHLQCLE DEL KPLEEVL NLAQSKNFHF EP RDVVSNN VFVLE KGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	105	210
105	PTSSSTKKT QLQLEHLRLD L EMILNGINN YKNPKLTRML TFKFYMPKKA TELKHLQCLE EELKPLEDVL NLAQSKNFHF EP RDVVSNN VFVLE KGSE TTFMCEYADE TATIVEFLNR WITFSQSIIS TLT	106	211

Example 2: Expression of Interleukin-2 Analogs

A recombinant interleukin-2 analog under the control of T7 promoter was expressed using the expression vectors prepared in Example 1.

An expression *E. coli* strain, *E. coli* BL21 DE3 (*E. coli* B F-dcm ompT hsdS(rB-mB-) gal A(DE3); Novagen), was transformed with each recombinant interleukin-2 analog

expression vector. As for the transformation method, a method recommended by Novagen was used. Each single colony, in which each recombinant expression vector was transformed, was collected, inoculated into a 2X Luria Broth medium containing ampicillin (50 µg/mL), and cultured at 37° C. for 15 hours. Each recombinant strain culture solution and the 2x LB medium containing 30% glycerol were mixed at a 1:1 (v/v) ratio, and each 1 mL of the mixture was

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dispensed into a cryo-tube, and stored at -150°C . This was used as a cell stock for the production of a recombinant protein.

For the expression of each recombinant interleukin-2 analog, one vial of each cell stock was dissolved, inoculated into 500 mL of $2\times\text{LB}$, and cultured with shaking at 37°C . for 14 to 16 hours. When the absorbance value at 600 nm reached 4.0 or higher, the culture was terminated, and this was used as a seed culture solution. The seed culture was inoculated into 1.6 L of a fermentation medium, and initial fermentation was started Using a 5 L fermentor (Bioflo-320, NBS, USA). Culture conditions were maintained at a pH of 6.70 using a temperature of 37°C ., an air volume of 2.0 L/min (1 vvm), a stirring speed of 650 rpm, and 30% aqueous ammonia. As for the fermentation process, when nutrients in the culture medium were limited, fed-batch culture was performed by adding an additional medium (feeding solution). The growth of the strain was observed by absorbance, and a final concentration of 500 μM IPTG was introduced at an absorbance value of 70 or higher. The culture was performed further until for about 23 to 25 hours after the introduction of IPTG, and after termination of the culture, and the recombinant strain was recovered using a centrifuge and stored at -80°C . until use.

Example 3: Extraction and Refolding of Interleukin-2 Analogs

In order to convert the interleukin-2 analogs from the interleukin-2 analog expressing *E. coli* obtained in Example 2 in a soluble form, cells were disrupted and refolded. Cell pellets corresponding to 100 mL of the culture were suspended in 1-200 mL of a lysis buffer solution (20 mM Tris-HCl (pH 9.0), 1 mM EDTA (pH 9.0), 0.2 M NaCl, 0.5% Triton X-100), and the recombinant *E. coli* cells were disrupted at 15,000 psi using a microfluidizer. After centrifugation at 13,900 g for 30 minutes, the supernatant was discarded, and the pellet was washed with 400 mL of a first washing buffer solution (50 mM Tris-HCl (pH 8.0), 5 mM EDTA (pH 9.0)). After centrifugation under the same conditions as above, the supernatant was discarded, and the pellet was washed with 400 mL of a second washing buffer solution (50 mM Tris-HCl (pH 8.0), 5 mM EDTA (pH 9.0), 2% Triton X-100). After centrifugation under the same conditions as above, the supernatant was discarded, and the pellet was washed with 400 mL of a third washing buffer solution (50 mM Tris-HCl (pH 8.0), 5 mM EDTA (pH 9.0), 1% sodium deoxycholate). After centrifugation under the same conditions as above, the supernatant was discarded, and the pellet was washed with 400 mL of a fourth washing buffer solution (50 mM Tris-HCl (pH 8.0), 5 mM EDTA (pH 9.0), 1 M NaCl). The resultant was subjected to centrifugation under the same conditions as above and washing, and *E. coli* inclusion bodies were obtained therefrom. The pellet of the washed inclusion bodies was resuspended in 400 mL of soluble/reducing buffer (6 M guanidine, 100 mM Tris (pH 8.0), 2 mM EDTA (pH 9.0), 50 mM DTT) and stirred at 50°C . for 30 minutes. To the soluble/reduced interleukin-2 analogs, 100 mL of distilled water was added to dilute the 6 M guanidine to 4.8 M guanidine, and then the resultant was centrifuged at 13,900 g for 30 minutes and the pellet discarded to obtain only the solution therein. To the diluted solution was additionally added 185.7 mL of distilled water, and the 4.8 M guanidine was diluted to 3.5 M guanidine, and the pH was adjusted to 5.0 using 100% acetic acid. The pH-adjusted solution was stirred at room temperature for one hour. The solution with precipitated impurities was

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centrifuged at 13,900 g for 30 minutes, and the supernatant was discarded and the pellet washed with a final washing buffer solution (3.5 M guanidine, 20 mM sodium acetate (pH 5.0), 5 mM DTT). The resultant was centrifuged under the same conditions as above to obtain a pellet. The washed interleukin-2 analogs were dissolved in 400 mL of a refolding buffer solution (6 mM guanidine, 100 mM Tris (pH 8.0), 0.1 mM CuCl_2). The refolding process was performed by stirring the mixed solution at 4°C . for 15 to 24 hours.

Example 4: Size-Exclusion Column Chromatography

The interleukin-2 analog refolding solution obtained in Example 3 was concentrated to less than 1 mL to be applied to a size-exclusion column for purification. The column was equilibrated with a buffer solution (2 M guanidine, 100 mM Tris (pH 8.0)) before introducing with the refolding solution and was eluted by flowing a buffer solution thereto after the introduction of the refolding solution. Since the eluted sample contained guanidine, it was replaced with a stabilized solution (10 mM sodium acetate (pH 4.5), 5% trehalose), and the purity was measured through RP-HPLC and peptide mapping analysis. The sample was used in the experiment when its measured purity reached 80% or higher.

Example 5: Evaluation of Binding Affinity of Interleukin-2 Analogs for Receptors

In order to measure the binding affinity of the interleukin-2 analogs obtained in Example 4 for each of interleukin-2 alpha receptors and beta receptors, surface plasmon resonance measurement (BIACORE T200, GE Healthcare) was used. The binding affinity of the prepared analogs for the alpha receptors and beta receptors was measured, and the binding affinity of each of the prepared analogs was compared with that of interleukin-2 analog 01 (aldesleukin).

First, an anti-human immunoglobulin antibody (Abcam, #ab97221) was immobilized to CM5 chips (GE Healthcare) by as much as about 5,000 RU (resonance unit) through amine coupling, and then, the immunoglobulin antibody was finally immobilized by allowing the interleukin-2 alpha receptors (SYMANSIS, #4102H) or interleukin-2 beta receptors (SYMANSIS, #4122H), to each of which a human immunoglobulin Fc region was bound, to bind to each immunoglobulin antibody using an antigen-antibody binding reaction. Thereafter, the recombinant interleukin-2 analog prepared above was diluted at various concentrations and was flowed onto the CM5 chips, to which the interleukin-2 receptors were finally immobilized, to measure the binding affinity of each interleukin-2 receptor. The measurement of binding affinity consisted of measurements of an association rate constant (k_a) and a dissociation rate constant (k_d), in which the binding rate was measured by flowing each interleukin-2 analog at a flow rate of 10 L/min for 3 minutes while the dissociation rate was measured from each interleukin-2 receptor by flowing only the experimental buffer for the same period of time and at the same flow rate. After the measurement was completed, the binding affinity for the receptors was evaluated according to the 1:1 binding fitting model in the Biaevaluation program.

$$\text{relative binding affinity}(K_D)(\%) = \frac{\text{binding affinity of analog 01(aldesleukin)}(K_D)}{\text{binding affinity of analog}(K_D)} \times 100$$

In Table 3 below, "cannot be defined" indicates that the corresponding physical quantity cannot be defined for the

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corresponding receptor because no binding to the receptor was observed in the surface plasmon resonance measurement.

TABLE 3

Relative binding affinity of interleukin-2 analogs for interleukin-2 alpha or beta receptors compared to analog 01 (aldesleukin)		
Interleukin-2 Receptor	Test Material	Relative Binding Affinity (%)
Alpha Receptor	analog 01	100.0
	analog 09	74.5
	analog 12	cannot be defined
	analog 13	1.1
	analog 15	cannot be defined
	analog 16	0.2
	analog 17	29.6
	analog 19	cannot be defined
	analog 20	cannot be defined
	analog 21	cannot be defined
	analog 31	5.0
	analog 34	9.4
	analog 35	31.7
	analog 41	121.3
	analog 52	cannot be defined
	analog 53	cannot be defined
	analog 86	71.1
	analog 88	101.5
	analog 90	98.4
	analog 91	7.9
	analog 92	97.3
	analog 93	92.8
	analog 95	10.7
	analog 96	14.9
	analog 97	18.8
	analog 98	7.7
	analog 99	19.9
	analog 100	29.1
	analog 101	24.7
	analog 102	151.4
	analog 103	6.1
Beta Receptor	analog 104	122.4
	analog 105	246.8
	analog 01	100.0
	analog 09	337.4
	analog 12	166.2
	analog 13	148.6
	analog 14	129.7
	analog 15	98.1
	analog 16	1261.8
	analog 17	9.4
	analog 18	35.3
	analog 19	455.0
	analog 20	156.5
	analog 21	14,084.2
	analog 24	37.9
	analog 25	21.7
	analog 31	235.7
	analog 34	321.8
	analog 35	232.7
	analog 41	22,776.2
	analog 52	3,821.1
	analog 53	690.7
	analog 55	3,025.7
	analog 57	2,569.7
	analog 58	7,771.2
	analog 59	1,533.5
	analog 61	1,039.1
	analog 70	10,199.2
	analog 71	17,083.8

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TABLE 3-continued

Relative binding affinity of interleukin-2 analogs for interleukin-2 alpha or beta receptors compared to analog 01 (aldesleukin)		
Interleukin-2 Receptor	Test Material	Relative Binding Affinity (%)
5	analog 73	1,591.8
	analog 74	8,153.4
	analog 75	9,571.2
	analog 76	1,040.4
	analog 77	644.4
	analog 84	710.7
	analog 86	18,745.8
	analog 88	13,856.6
	analog 90	12,776.2
	analog 91	7,361.9
10	analog 92	1,510.3
	analog 93	696.8
	analog 94	35.5
	analog 95	17.1
	analog 96	229.3
15	analog 97	3,019.4
	analog 98	11,084.5
	analog 99	1,509.1
	analog 100	2,534.1
	analog 101	113.1
20	analog 102	4,452.0
	analog 103	13,100.0
	analog 104	25,439.8
	analog 105	26,837.8

As explicitly shown in the test results (FIGS. 1 and 2 and Table 3), it was confirmed that the interleukin-2 analogs of the present invention had no binding affinity, increased/reduced binding affinity for interleukin-2 alpha receptors, etc., thus showing an altered binding affinity for interleukin-2 alpha receptors compared to native interleukin-2 or aldesleukin. In contrast, as for the interleukin-2 beta receptors, the interleukin-2 analogs of the present invention showed a stronger binding affinity of up to 100-fold compared to native interleukin-2 or aldesleukin. From the above results, it was confirmed that the amino acid sequence of the interleukin-2 analog has an effect on its binding to the interleukin-2 alpha or beta receptors. These results suggest that the binding affinity for interleukin-2 receptors can be altered by substituting an amino acid at a specific position.

These experimental results suggest that the interleukin-2 analogs according to the present invention have altered binding affinity for interleukin-2 alpha receptors and interleukin-2 beta receptors and thus can be used in the development of various drugs based on the same.

From the foregoing, one of ordinary skill in the art to which the present invention pertains will be able to understand that the present invention may be embodied in other specific forms without modifying the technical concepts or essential characteristics of the present invention. In this regard, the exemplary embodiments disclosed herein are only for illustrative purposes and should not be construed as limiting the scope of the present invention. On the contrary, the present invention is intended to cover not only the exemplary embodiments but also various alternatives, modifications, equivalents, and other embodiments that may be included within the spirit and scope of the present invention as defined by the appended claims.

SEQUENCE LISTING

Sequence total quantity: 417

SEQ ID NO: 1 moltype = AA length = 133
 FEATURE Location/Qualifiers

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source                1..133
                      mol_type = protein
                      organism = Homo sapiens

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APTSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML TFKFYMPKKA TELKHLQCLE 60
EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TFMCEYADE TATIVEFLNR 120
WITFCQSIIS TLT 133

SEQ ID NO: 2          moltype = AA length = 132
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REGION              1..132
                      note = IL-2 analog 1
source              1..132
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 2
PTSSSTKKTQ LQLEHLLLDL QMILNGINNY KNPCLTRMLT PKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHLR PRDLISNINV IVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 3          moltype = AA length = 132
FEATURE              Location/Qualifiers
REGION              1..132
                      note = IL-2 analog 2
source              1..132
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 3
PTSSSTKKTQ LQLEHLLLDL QMILNGINNY KNPCLTRMLT PKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHLR PRDLICNINV IVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 4          moltype = AA length = 132
FEATURE              Location/Qualifiers
REGION              1..132
                      note = IL-2 analog 3
source              1..132
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 4
PTSSSTKKTQ LQLEHLLLDL QMILNGINNY CNPKLTRMLT PKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHLR PRDLISNINV IVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 5          moltype = AA length = 132
FEATURE              Location/Qualifiers
REGION              1..132
                      note = IL-2 analog 4
source              1..132
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 5
PTSSSTKKTQ LQLEHLLLDL QMILNGINNY KNPCLTRMLT PKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHLR PRDLISNINV IVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 6          moltype = AA length = 132
FEATURE              Location/Qualifiers
REGION              1..132
                      note = IL-2 analog 5
source              1..132
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 6
PTSSSTKKTQ LQLEHLLLDL QMILNGINNY KNPCLTRMLT FCFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHLR PRDLISNINV IVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 7          moltype = AA length = 132
FEATURE              Location/Qualifiers
REGION              1..132
                      note = IL-2 analog 6
source              1..132
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 7
PTSSSTKKTQ LQLEHLLLDL QMILNGINNY KNPCLTRMLT PKFYMPCKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHLR PRDLISNINV IVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

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SEQ ID NO: 8          moltype = AA  length = 132
FEATURE              Location/Qualifiers
REGION              1..132
                    note = IL-2 analog 7
source              1..132
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 8
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTRMLT FKFYMPKCAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNPHLR PRDLISNINV IVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 9          moltype = AA  length = 132
FEATURE              Location/Qualifiers
REGION              1..132
                    note = IL-2 analog 9
source              1..132
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 9
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTRMLT FKFYMPKCAT ELKHLQCLEE 60
ELKPLEEVLN LAQSCNPHLR PRDLISNINV IVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 10         moltype = AA  length = 132
FEATURE              Location/Qualifiers
REGION              1..132
                    note = IL-2 analog 10
source              1..132
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 10
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTRMLT FKFYMPKCAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNPHLR PRDLISNINV IVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 11         moltype = AA  length = 132
FEATURE              Location/Qualifiers
REGION              1..132
                    note = IL-2 analog 10
source              1..132
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 11
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTRMLT KKFYMPKCAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNPHLR PRDLISNINV IVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 12         moltype = AA  length = 132
FEATURE              Location/Qualifiers
REGION              1..132
                    note = IL-2 analog 11
source              1..132
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 12
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTRMLT AKFYMPKCAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNPHLR PRDLISNINV IVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 13         moltype = AA  length = 132
FEATURE              Location/Qualifiers
REGION              1..132
                    note = IL-2 analog 12
source              1..132
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 13
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTRMLT KKFYMPKCAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNPHLR PRDLISNINV IVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

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FEATURE              Location/Qualifiers
REGION              1..132
                    note = IL-2 analog 13
source              1..132
                    mol_type = protein

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                                organism = synthetic construct
SEQUENCE: 14
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ELKPLEEVLN LAQSKNFHLR PRDLISNINV IVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 15      moltype = AA length = 132
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                  note = IL-2 analog 14
source            1..132
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 15
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ELKPLEEVLN LAQSKNFHLR PRDLISNINV IVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 16      moltype = AA length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 15
source            1..132
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 16
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ELKPLEEVLN LAQSKNFHLR PRELISQINV IVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 17      moltype = AA length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 16
source            1..132
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 17
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ELKPLEEVLN LAQSKNFHLR PRDLISQINV IVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 18      moltype = AA length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 17
source            1..132
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 18
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ELKPLEEVLN LAQSKNFHLR PRDLISNINT IVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 19      moltype = AA length = 132
FEATURE           Location/Qualifiers
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                  note = IL-2 analog 18
source            1..132
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 19
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ELKPLEEVLN LAQSKNFHLR PRDLISNINV IVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 20      moltype = AA length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 19
source            1..132
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 20
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ELKPLEEVLN LAQSKNFHLR PRDLISNINV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 21      moltype = AA length = 132

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source	1..132	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 21		
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ELKPLEEVLN LAQSKNFHLR	PRDVSNINV FVLELKGSET TFMCEYADET ATIVEFLNRW	120
ITFSQSIIST LT		132
SEQ ID NO: 22	moltype = AA length = 132	
FEATURE	Location/Qualifiers	
REGION	1..132	
	note = IL-2 analog 21	
source	1..132	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 22		
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ELKPLEEVLN LAQSKNFHFD	PRDVSNINV FVLELKGSET TFMCEYADET ATIVEFLNRW	120
ITFSQSIIST LT		132
SEQ ID NO: 23	moltype = AA length = 132	
FEATURE	Location/Qualifiers	
REGION	1..132	
	note = IL-2 analog 22	
source	1..132	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 23		
PTSSSTKKKTQ VQLEHLLLDL	QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE	60
ELKPLEEVLN LAQSKNFHLR	PRDLISNINV IVLELKGSET TFMCEYADET ATIVEFLNRW	120
ITFSQSIIST LT		132
SEQ ID NO: 24	moltype = AA length = 132	
FEATURE	Location/Qualifiers	
REGION	1..132	
	note = IL-2 analog 23	
source	1..132	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 24		
PTSSSTKKKTQ FQLEHLLLDL	QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE	60
ELKPLEEVLN LAQSKNFHLR	PRDLISNINV IVLELKGSET TFMCEYADET ATIVEFLNRW	120
ITFSQSIIST LT		132
SEQ ID NO: 25	moltype = AA length = 132	
FEATURE	Location/Qualifiers	
REGION	1..132	
	note = IL-2 analog 24	
source	1..132	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 25		
PTSSSTKKKTQ LQLEHLLVDL	QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE	60
ELKPLEEVLN LAQSKNFHLR	PRDLISNINV IVLELKGSET TFMCEYADET ATIVEFLNRW	120
ITFSQSIIST LT		132
SEQ ID NO: 26	moltype = AA length = 132	
FEATURE	Location/Qualifiers	
REGION	1..132	
	note = IL-2 analog 25	
source	1..132	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 26		
PTSSSTKKKTQ LQLEHLLFDL	QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE	60
ELKPLEEVLN LAQSKNFHLR	PRDLISNINV IVLELKGSET TFMCEYADET ATIVEFLNRW	120
ITFSQSIIST LT		132
SEQ ID NO: 27	moltype = AA length = 132	
FEATURE	Location/Qualifiers	
REGION	1..132	
	note = IL-2 analog 26	
source	1..132	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 27		

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PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHLR PRDLISNINV IVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

```

```

SEQ ID NO: 28      moltype = AA  length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 27
source            1..132
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 28
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHLR PRDLISNINV IVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

```

```

SEQ ID NO: 29      moltype = AA  length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 28
source            1..132
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 29
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHLR PRDLISNINV IVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

```

```

SEQ ID NO: 30      moltype = AA  length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 29
source            1..132
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 30
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHLR PRDLISNINV IVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

```

```

SEQ ID NO: 31      moltype = AA  length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 30
source            1..132
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 31
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHLR PRDLISNINV IVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

```

```

SEQ ID NO: 32      moltype = AA  length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 31
source            1..132
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 32
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHLR PRELISNINV IVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

```

```

SEQ ID NO: 33      moltype = AA  length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 32
source            1..132
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 33
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHLR PRELISNINV IVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

```

```

SEQ ID NO: 34      moltype = AA  length = 132
FEATURE           Location/Qualifiers
REGION            1..132

```

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source          note = IL-2 analog 33
                1..132
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 34
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT FKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHFR PRDLISNINV IVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 35    moltype = AA length = 132
FEATURE         Location/Qualifiers
REGION          1..132
                note = IL-2 analog 34
source          1..132
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 35
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT FKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHFR PRELISNINV IVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 36    moltype = AA length = 132
FEATURE         Location/Qualifiers
REGION          1..132
                note = IL-2 analog 35
source          1..132
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 36
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT FKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHLR PRDLISNINV IVFEPKGSSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 37    moltype = AA length = 132
FEATURE         Location/Qualifiers
REGION          1..132
                note = IL-2 analog 36
source          1..132
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 37
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT FKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHLR PRDLISNINV IVFEPKGSSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 38    moltype = AA length = 132
FEATURE         Location/Qualifiers
REGION          1..132
                note = IL-2 analog 37
source          1..132
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 38
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT FKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHLR PRDLISNINV IVFEIKGSSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 39    moltype = AA length = 132
FEATURE         Location/Qualifiers
REGION          1..132
                note = IL-2 analog 38
source          1..132
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 39
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHLD PRDLISNINV FVFEVKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 40    moltype = AA length = 132
FEATURE         Location/Qualifiers
REGION          1..132
                note = IL-2 analog 39
source          1..132
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 40
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHLD PRDLISNINV FVFEVKGSET TFMCEYADET ATIVEFLNRW 120

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ITFSQSIIST LT		132
SEQ ID NO: 41	moltype = AA length = 132	
FEATURE	Location/Qualifiers	
REGION	1..132	
	note = IL-2 analog 40	
source	1..132	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 41		
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAFLT KKFYMPKKAT ELKHLQCLEE		60
ELKPLEEVN LAQSKNFHLD PRDLISNINV FVLELKGSET TFMCEYADET ATIVEFLNRW		120
ITFSQSIIST LT		132
SEQ ID NO: 42	moltype = AA length = 132	
FEATURE	Location/Qualifiers	
REGION	1..132	
	note = IL-2 analog 41	
source	1..132	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 42		
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTRMLT KKFYMPKKAT ELKHLQCLEE		60
ELKPLEEVN LAQSKNFHLD PRDVSNINV FVLELKGSET TFMCEYADET ATIVEFLNRW		120
ITFSQSIIST LT		132
SEQ ID NO: 43	moltype = AA length = 132	
FEATURE	Location/Qualifiers	
REGION	1..132	
	note = IL-2 analog 42	
source	1..132	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 43		
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAFLT KKFYMPKKAT ELKHLQCLEE		60
ELKPLEEVN LAQSKNFHLD PRDLISNINV FVLELKGSET TFMCEYADET ATIVEFLNRW		120
ITFSQSIIST LT		132
SEQ ID NO: 44	moltype = AA length = 132	
FEATURE	Location/Qualifiers	
REGION	1..132	
	note = IL-2 analog 43	
source	1..132	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 44		
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAFLT KKFYMPKKAT ELKHLQCLEE		60
ELKPLEEVN LAQSKNFHLD PRDLISNINV LVLELKGSET TFMCEYADET ATIVEFLNRW		120
ITFSQSIIST LT		132
SEQ ID NO: 45	moltype = AA length = 132	
FEATURE	Location/Qualifiers	
REGION	1..132	
	note = IL-2 analog 44	
source	1..132	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 45		
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAFLT KKFYMPKKAT ELKHLQCLEE		60
ELKPLEEVN LAQSKNFHLD PRVLISNINV FVLELKGSET TFMCEYADET ATIVEFLNRW		120
ITFSQSIIST LT		132
SEQ ID NO: 46	moltype = AA length = 132	
FEATURE	Location/Qualifiers	
REGION	1..132	
	note = IL-2 analog 45	
source	1..132	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 46		
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAFLT KKFYMPKKAT ELKHLQCLEE		60
ELKPLEEVN LAQSKNFHLD PRFLISNINV FVLELKGSET TFMCEYADET ATIVEFLNRW		120
ITFSQSIIST LT		132
SEQ ID NO: 47	moltype = AA length = 132	
FEATURE	Location/Qualifiers	
REGION	1..132	
	note = IL-2 analog 46	
source	1..132	

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mol_type = protein
organism = synthetic construct

SEQUENCE: 47
PTSSSTKKKTQ LQLEHLLLV L QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHLD PRDLISNINV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 48      moltype = AA length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 47
source            1..132
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 48
PTSSSTKKKTQ LQLEHLLLV L QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHLD PRDLISNINV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 49      moltype = AA length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 48
source            1..132
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 49
PTSSSTKKKTQ LQLEHLLLV L QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHLD PRDLISVINV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 50      moltype = AA length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 49
source            1..132
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 50
PTSSSTKKKTQ LQLEHLLLV L QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHLD PRDLISFINV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 51      moltype = AA length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 50
source            1..132
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 51
PTSSSTKKKTQ LQLEHLLLV L QMILNGINNY KNPKLTRMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHFD PRDVVSININV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 52      moltype = AA length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 51
source            1..132
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 52
PTSSSTKKKTQ LQLEHLLLV L QMILNGINNY KNPKLTRMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHLD PRDLISNINV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 53      moltype = AA length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 52
source            1..132
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 53
PTSSSTKKKTQ LQLEHLLLV L QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHFD PRDLISNINV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

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SEQ ID NO: 54      moltype = AA  length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 53
source            1..132
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 54
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHFD PRELISNINV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT                                     132

SEQ ID NO: 55      moltype = AA  length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 54
source            1..132
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 55
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAHKNFHD PRDLISNINV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT                                     132

SEQ ID NO: 56      moltype = AA  length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 55
source            1..132
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 56
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAHKNFHD PRDLISNINV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT                                     132

SEQ ID NO: 57      moltype = AA  length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 56
source            1..132
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 57
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFAMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAHKNFHD PRDLISNINV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT                                     132

SEQ ID NO: 58      moltype = AA  length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 57
source            1..132
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 58
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFAMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAHKNFHD PRDLISNINV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT                                     132

SEQ ID NO: 59      moltype = AA  length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 61
source            1..132
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 59
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHFD PRDAASNINV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT                                     132

SEQ ID NO: 60      moltype = AA  length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 59
source            1..132
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 60
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHFD PRDAASNINV YVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 61      moltype = AA length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 60
source            1..132
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 61
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFAMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHYR PRDAASNINV YVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 62      moltype = AA length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 61
source            1..132
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 62
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHYD PRDGVSNINV YVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 63      moltype = AA length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 62
source            1..132
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 63
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHWE PRDGASNINV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 64      moltype = AA length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 63
source            1..132
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 64
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHDE PRDTGSNINV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 65      moltype = AA length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 64
source            1..132
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 65
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHYN PRDVVSNINV YVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 66      moltype = AA length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 65
source            1..132
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 66
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHYE PRDVVSNINV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 67      moltype = AA length = 132
FEATURE           Location/Qualifiers

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REGION 1..132
 note = IL-2 analog 66
 source 1..132
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 67
 PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
 ELKPLEEVLN LAQSKNFHFE PRDFVSNINV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
 ITFSQSIIST LT 132
 SEQ ID NO: 68 moltype = AA length = 132
 FEATURE Location/Qualifiers
 REGION 1..132
 note = IL-2 analog 67
 source 1..132
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 68
 PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
 ELKPLEEVLN LAQSKNFHYD PRDFVSNINV WVLDLKGSET TFMCEYADET ATIVEFLNRW 120
 ITFSQSIIST LT 132
 SEQ ID NO: 69 moltype = AA length = 132
 FEATURE Location/Qualifiers
 REGION 1..132
 note = IL-2 analog 68
 source 1..132
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 69
 PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
 ELKPLEEVLN LAQSKNFHFE PRDIVSNINE FVLELKGSET TFMCEYADET ATIVEFLNRW 120
 ITFSQSIIST LT 132
 SEQ ID NO: 70 moltype = AA length = 132
 FEATURE Location/Qualifiers
 REGION 1..132
 note = IL-2 analog 69
 source 1..132
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 70
 PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
 ELKPLEEVLN LAQSKNFHYE PRDFLSNINE WVLDLKGSET TFMCEYADET ATIVEFLNRW 120
 ITFSQSIIST LT 132
 SEQ ID NO: 71 moltype = AA length = 132
 FEATURE Location/Qualifiers
 REGION 1..132
 note = IL-2 analog 70
 source 1..132
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 71
 PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
 ELKPLEEVLN LAQSKNFHYD PRDVVSNINV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
 ITFSQSIIST LT 132
 SEQ ID NO: 72 moltype = AA length = 132
 FEATURE Location/Qualifiers
 REGION 1..132
 note = IL-2 analog 71
 source 1..132
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 72
 PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
 ELKPLEEVLN LAQSKNFHFE PRDVVSNINV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
 ITFSQSIIST LT 132
 SEQ ID NO: 73 moltype = AA length = 132
 FEATURE Location/Qualifiers
 REGION 1..132
 note = IL-2 analog 72
 source 1..132
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 73
 PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60

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ELKPLEEVLN LAQSKNFHFD PRDVGSNINV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
 ITFSQSIIST LT 132

SEQ ID NO: 74 moltype = AA length = 132
 FEATURE Location/Qualifiers
 REGION 1..132
 note = IL-2 analog 73
 source 1..132
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 74
 PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
 ELKPLEEVLN LAQSKNFHFD PRDVGSNINV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
 ITFSQSIIST LT 132

SEQ ID NO: 75 moltype = AA length = 132
 FEATURE Location/Qualifiers
 REGION 1..132
 note = IL-2 analog 74
 source 1..132
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 75
 PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
 ELKPLEEVLN LAQSKNFHFD PRDVGSNINV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
 ITFSQSIIST LT 132

SEQ ID NO: 76 moltype = AA length = 132
 FEATURE Location/Qualifiers
 REGION 1..132
 note = IL-2 analog 75
 source 1..132
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 76
 PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFAMPKKAT ELKHLQCLEE 60
 ELKPLEEVLN LAQSKNFHFD PRDVGSNINV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
 ITFSQSIIST LT 132

SEQ ID NO: 77 moltype = AA length = 132
 FEATURE Location/Qualifiers
 REGION 1..132
 note = IL-2 analog 76
 source 1..132
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 77
 PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KQFYMPKKAT ELKHLQCLEE 60
 ELKPLEEVLN LAQSKNFHFD PRDVGSNINV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
 ITFSQSIIST LT 132

SEQ ID NO: 78 moltype = AA length = 132
 FEATURE Location/Qualifiers
 REGION 1..132
 note = IL-2 analog 77
 source 1..132
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 78
 PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KEFYMPKKAT ELKHLQCLEE 60
 ELKPLEEVLN LAQSKNFHFD PRDVGSNINV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
 ITFSQSIIST LT 132

SEQ ID NO: 79 moltype = AA length = 132
 FEATURE Location/Qualifiers
 REGION 1..132
 note = IL-2 analog 78
 source 1..132
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 79
 PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAMLT KKFYMPKKAT ELKHLQCLEE 60
 ELKPLEEVLN LAQSKNFHFD PRDVGSNINV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
 ITFSQSIIST LT 132

SEQ ID NO: 80 moltype = AA length = 132
 FEATURE Location/Qualifiers
 REGION 1..132
 note = IL-2 analog 79

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source                1..132
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 80
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPELTAMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAHSKNFHF E PRDVVSININV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 81          moltype = AA length = 132
FEATURE               Location/Qualifiers
REGION                1..132
                      note = IL-2 analog 80
source                1..132
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 81
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPELTAMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAHSKNFHF E GRDVVSININV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 82          moltype = AA length = 132
FEATURE               Location/Qualifiers
REGION                1..132
                      note = IL-2 analog 81
source                1..132
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 82
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPELTAMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAHSKNFHF E VRDVVSININV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 83          moltype = AA length = 132
FEATURE               Location/Qualifiers
REGION                1..132
                      note = IL-2 analog 82
source                1..132
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 83
PTSSSTKKKTQ LQLEHLRLDL EMILNGINNY KNPKLTRMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHF D PRDEVSNINV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 84          moltype = AA length = 132
FEATURE               Location/Qualifiers
REGION                1..132
                      note = IL-2 analog 83
source                1..132
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 84
PTSSSTKKKTQ LQLEHLRLDL EMILNGINNY KNPKLTRMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHF D PRDEVSNINV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 85          moltype = AA length = 132
FEATURE               Location/Qualifiers
REGION                1..132
                      note = IL-2 analog 84
source                1..132
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 85
PTSSSTKKKTQ LQLEHLRLDL EMILNGINNY KNPKLTRMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHF D PRDEVSNINV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 86          moltype = AA length = 132
FEATURE               Location/Qualifiers
REGION                1..132
                      note = IL-2 analog 85
source                1..132
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 86
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTRMLT KKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHF E PRDVVSININV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

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SEQ ID NO: 87	moltype = AA length = 132		
FEATURE	Location/Qualifiers		
REGION	1..132		
	note = IL-2 analog 86		
source	1..132		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 87			
PTSSSTKKKTQ LQLEHLRLDL	EMILNGINNY KNPKLTRMLT	FKFYMPKKAT ELKHLQCLEE	60
ELKPLEEVLN LAQSKNPFHE	PRDVVSINIV FVLELKGSET	TFMCEYADET ATIVEFLNRW	120
ITFSQSIIST LT			132
SEQ ID NO: 88	moltype = AA length = 132		
FEATURE	Location/Qualifiers		
REGION	1..132		
	note = IL-2 analog 87		
source	1..132		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 88			
PTSSSTKKKTQ LQLEHLRLDL	EMILNGINNY KNPKLTRMLT	FKFYMPKKAT ELKHLQCLEE	60
ELKPLEEVLN LAQSKNPFHE	PRDVVSINIV FVLELKGSET	TFMCEYADET ATIVEFLNRW	120
ITFSQSIIST LT			132
SEQ ID NO: 89	moltype = AA length = 132		
FEATURE	Location/Qualifiers		
REGION	1..132		
	note = IL-2 analog 88		
source	1..132		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 89			
PTSSSTKKKTQ LQLEHLRLDL	QMILNGINNY KNPKLTRMLT	FKFYMPKKAT ELKHLQCLED	60
ELKPLEEVLN LAQSKNPFHE	PRDVVSINIV FVLELKGSET	TFMCEYADET ATIVEFLNRW	120
ITFSQSIIST LT			132
SEQ ID NO: 90	moltype = AA length = 132		
FEATURE	Location/Qualifiers		
REGION	1..132		
	note = IL-2 analog 89		
source	1..132		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 90			
PTSSSTKKKTQ LQLEHLRLDL	QMILNGINNY KNPKLTLAML	FKFYMPKKAT ELKHLQCLEE	60
ELKPLEEVLN LAQSKNPFHE	PRDVVSINIV FVLELKGSET	TFMCEYADET ATIVEFLNRW	120
ITFSQSIIST LT			132
SEQ ID NO: 91	moltype = AA length = 132		
FEATURE	Location/Qualifiers		
REGION	1..132		
	note = IL-2 analog 90		
source	1..132		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 91			
PTSSSTKKKTQ LQLEHLRLDL	QMILNGINNY KNPKLTRMLT	WKFYMPKKAT ELKHLQCLEE	60
ELKPLEEVLN LAQSKNPFHE	PRDVVSINIV FVLELKGSET	TFMCEYADET ATIVEFLNRW	120
ITFSQSIIST LT			132
SEQ ID NO: 92	moltype = AA length = 132		
FEATURE	Location/Qualifiers		
REGION	1..132		
	note = IL-2 analog 91		
source	1..132		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 92			
PTSSSTKKKTQ LQLEHLRLDL	QMILNGINNY KNPKLTRMLT	FKFYMPKKAT ELKHLQCLEE	60
ELKPLEEVLN LAQSKNPFHE	PRDVVSINIV FVLELKGSET	TFMCEYADET ATIVEFLNRW	120
ITFSQSIIST LT			132
SEQ ID NO: 93	moltype = AA length = 132		
FEATURE	Location/Qualifiers		
REGION	1..132		
	note = IL-2 analog 92		
source	1..132		
	mol_type = protein		

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                                organism = synthetic construct
SEQUENCE: 93
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTRMLT FKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHFE PRDVVSININT FVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 94      moltype = AA length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 93
source            1..132
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 94
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTRMLT FKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHFE PREVVSININT FVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 95      moltype = AA length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 94
source            1..132
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 95
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTRMLT FKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHFE PRDVVSININV FVFEFKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 96      moltype = AA length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 95
source            1..132
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 96
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTRMLT FKFYMPKKAT ELKHLQCLEE 60
ELKPLEEGLN LAQSKNFHFE PRDVVSININV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 97      moltype = AA length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 96
source            1..132
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 97
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTRMLT FKFYMPKKAT ELKHLQCLEE 60
ELKPLEEGLN LAQSKNFHFE PRDVVSININV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 98      moltype = AA length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 97
source            1..132
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 98
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAHLT FKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHFE PRDLISININV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 99      moltype = AA length = 132
FEATURE           Location/Qualifiers
REGION            1..132
                  note = IL-2 analog 98
source            1..132
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 99
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAHLT FKFYMPKKAT ELKHLQCLEE 60
ELKPLEEVLN LAQSKNFHFE PRDVISININV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

SEQ ID NO: 100     moltype = AA length = 132

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FEATURE	Location/Qualifiers	
REGION	1..132	
	note = IL-2 analog 99	
source	1..132	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 100		
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTAFLT PKFYMPKKAT ELKHLQCLEE		60
ELKPLEEVLN LAQSKNFHFE PRDLVSNINV FVLELKGSET TFMCEYADET ATIVEFLNRW		120
ITFSQSIIST LT		132
SEQ ID NO: 101	moltype = AA length = 132	
FEATURE	Location/Qualifiers	
REGION	1..132	
	note = IL-2 analog 100	
source	1..132	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 101		
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTRMLT PKFYMPKKAT ELKHLQCLEE		60
ELKPLEEVLN LAQSKNFHFE PRDYVSNINV FVLELKGSET TFMCEYADET ATIVEFLNRW		120
ITFSQSIIST LT		132
SEQ ID NO: 102	moltype = AA length = 132	
FEATURE	Location/Qualifiers	
REGION	1..132	
	note = IL-2 analog 101	
source	1..132	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 102		
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTRMLT PKFYMPKKAT ELKHLQCLEE		60
ELKPLEEVLN LAQSKNFHFE PRDLASNINV FVLELKGSET TFMCEYADET ATIVEFLNRW		120
ITFSQSIIST LT		132
SEQ ID NO: 103	moltype = AA length = 132	
FEATURE	Location/Qualifiers	
REGION	1..132	
	note = IL-2 analog 102	
source	1..132	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 103		
PTSSSTKKKTQ LQLEHLLLDL QMILNGINNY KNPKLTRMLT PKFYMPKKAT ELKHLQCLEE		60
ELKPLEEVLN LAQSKNFHFE PRDVVSNINV IVLELKGSET TFMCEYADET ATIVEFLNRW		120
ITFSQSIIST LT		132
SEQ ID NO: 104	moltype = AA length = 132	
FEATURE	Location/Qualifiers	
REGION	1..132	
	note = IL-2 analog 103	
source	1..132	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 104		
PTSSSTKKKTQ LQLEHLRLDL EMILNGINNY KNPKLTAFLT PKFYMPKKAT ELKHLQCLEE		60
ELKPLEEVLN LAQSKNFHFE PRDVVSNINV FVLELKGSET TFMCEYADET ATIVEFLNRW		120
ITFSQSIIST LT		132
SEQ ID NO: 105	moltype = AA length = 132	
FEATURE	Location/Qualifiers	
REGION	1..132	
	note = IL-2 analog 104	
source	1..132	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 105		
PTSSSTKKKTQ LQLEHLRLDL EMILNGINNY KNPKLTRMLT PKFYMPKKAT ELKHLQCLED		60
ELKPLEEVLN LAQSKNFHFE PRDVVSNINV FVLELKGSET TFMCEYADET ATIVEFLNRW		120
ITFSQSIIST LT		132
SEQ ID NO: 106	moltype = AA length = 132	
FEATURE	Location/Qualifiers	
REGION	1..132	
	note = IL-2 analog 105	
source	1..132	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 106		

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PTSSSTKTKTQ LQLEHLRLDL EMILNGINNY KNPKLTRMLT PKFYMPKKAT ELKHLQCLEE 60
ELKPLEDLN LAQSKNFHE PRDVSNINV FVLELKGSET TFMCEYADET ATIVEFLNRW 120
ITFSQSIIST LT 132

```

```

SEQ ID NO: 107      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 1
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 107
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccag gatgctcaca 120
tttaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttaaga 240
cccagggact taatcagcaa tatcaacgta atagtcttgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 108      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 2
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 108
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccag gatgctcaca 120
tttaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttaaga 240
cccagggact taatctgcaa tatcaacgta atagtcttgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 109      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 3
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 109
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac tgtaatccca aactcaccag gatgctcaca 120
tttaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttaaga 240
cccagggact taatcagcaa tatcaacgta atagtcttgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 110      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 4
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 110
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccct gtctcaccag gatgctcaca 120
tttaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttaaga 240
cccagggact taatcagcaa tatcaacgta atagtcttgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 111      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 5
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 111
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccag gatgctcaca 120
ttttgttttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180

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gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttaaga 240
cccagggact taatcagcaa tatcaacgta atagttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 112      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 6
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 112
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccag gatgctcaca 120
tttaagtttt acatgccctg taaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttaaga 240
cccagggact taatcagcaa tatcaacgta atagttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 113      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 7
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 113
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccag gatgctcaca 120
tttaagtttt acatgcccaa gtgtgccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttaaga 240
cccagggact taatcagcaa tatcaacgta atagttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 114      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 8
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 114
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccag gatgctcaca 120
tttaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gctgtaactt tcacttaaga 240
cccagggact taatcagcaa tatcaacgta atagttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 115      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 9
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 115
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
tttaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttaaga 240
cccagggact taatcagcaa tatcaacgta atagttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 116      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 10
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 116
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccag gatgctcaca 120

```

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aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttaaga 240
cccagggact taatcagcaa tatcaacgta atagttcttg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 117      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature       1..399
                   note = IL-2 analog 11
source             1..399
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 117
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccg gatgctcaca 120
gctaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttaaga 240
cccagggact taatcagcaa tatcaacgta atagttcttg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 118      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature       1..399
                   note = IL-2 analog 12
source             1..399
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 118
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccg gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttaaga 240
cccagggact taatcagcaa tatcaacgta atagttcttg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 119      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature       1..399
                   note = IL-2 analog 13
source             1..399
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 119
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccg gatgctcaca 120
gctaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttaaga 240
cccagggact taatcagcaa tatcaacgta atagttcttg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 120      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature       1..399
                   note = IL-2 analog 14
source             1..399
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 120
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gtacgattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccg gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttaaga 240
cccagggact taatcagcaa tatcaacgta atagttcttg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 121      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature       1..399
                   note = IL-2 analog 15
source             1..399
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 121
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60

```

-continued

```

cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttaaga 240
cccagggagt taatcagcaa tatcaacgta atagttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 122      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 16
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 122
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttaaga 240
cccagggagt taatcagcaa tatcaacgta atagttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 123      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 17
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 123
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttaaga 240
cccagggagt taatcagcaa tatcaacaca atagttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 124      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 18
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 124
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaacag 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttaaga 240
cccagggagt taatcagcaa tatcaacgta atagttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 125      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 19
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 125
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttagac 240
cccagggagt taatcagcaa tatcaacgta tttgttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 126      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 20
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 126

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-continued

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cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttaaga 240
cccagggacg tagtcagcaa tatcaacgta tttgttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 127      moltype = DNA  length = 399
FEATURE
misc_feature       1..399
                    note = IL-2 analog 21
source             1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 127
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcactttgac 240
cccagggacg tagtcagcaa tatcaacgta tttgttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 128      moltype = DNA  length = 399
FEATURE
misc_feature       1..399
                    note = IL-2 analog 22
source             1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 128
cctacttcaa gttctacaaa gaaaacacag gtacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttaaga 240
cccagggact taatcagcaa tatcaacgta atagtcttgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 129      moltype = DNA  length = 399
FEATURE
misc_feature       1..399
                    note = IL-2 analog 23
source             1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 129
cctacttcaa gttctacaaa gaaaacacag tttcaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttaaga 240
cccagggact taatcagcaa tatcaacgta atagtcttgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 130      moltype = DNA  length = 399
FEATURE
misc_feature       1..399
                    note = IL-2 analog 24
source             1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 130
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact ggtggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttaaga 240
cccagggact taatcagcaa tatcaacgta atagtcttgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 131      moltype = DNA  length = 399
FEATURE
misc_feature       1..399
                    note = IL-2 analog 25
source             1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

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```

SEQUENCE: 131
cctactttcaa gttctacaaa gaaaacacag ctacaactgg agcattttact gttcgattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttaaga 240
cccagggact taatcagcaa tatcaacgta atagttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 132      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 26
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 132
cctactttcaa gttctacaaa gaaaacacag ctacaactgg agcattttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttaaga 240
cccagggact taatcagcaa tttcaacgta atagttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 133      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 27
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 133
cctactttcaa gttctacaaa gaaaacacag ctacaactgg agcattttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttaaga 240
cccagggact taatcagcaa tatcaacttt atagttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 134      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 28
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 134
cctactttcaa gttctacaaa gaaaacacag ctacaactgg agcattttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttaaga 240
cccagggact taatcagcaa tatcaacgta atagttgtgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 135      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 29
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 135
cctactttcaa gttctacaaa gaaaacacag ctacaactgg agcattttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttaaga 240
cccagggact taatcagcaa tatcaacgta atagttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtacaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 136      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 30
source              1..399
                    mol_type = other DNA

```

-continued

```

                                organism = synthetic construct
SEQUENCE: 136
cctactttcaa gttctacaaa gaaaacacag ctacaactgg agcattttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
tttaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttagac 240
cccagggact taatcagcaa tatcaacgta ttgtttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 137                moltype = DNA length = 399
FEATURE                      Location/Qualifiers
misc_feature                 1..399
                             note = IL-2 analog 31
source                      1..399
                             mol_type = other DNA
                             organism = synthetic construct

SEQUENCE: 137
cctactttcaa gttctacaaa gaaaacacag ctacaactgg agcattttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
tttaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttaaga 240
cccagggagt taatcagcaa tatcaacgta atagttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 138                moltype = DNA length = 399
FEATURE                      Location/Qualifiers
misc_feature                 1..399
                             note = IL-2 analog 32
source                      1..399
                             mol_type = other DNA
                             organism = synthetic construct

SEQUENCE: 138
cctactttcaa gttctacaaa gaaaacacag ctacaactgg agcattttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
tttaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttagac 240
cccagggagt taatcagcaa tatcaacgta ttgtttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 139                moltype = DNA length = 399
FEATURE                      Location/Qualifiers
misc_feature                 1..399
                             note = IL-2 analog 33
source                      1..399
                             mol_type = other DNA
                             organism = synthetic construct

SEQUENCE: 139
cctactttcaa gttctacaaa gaaaacacag ctacaactgg agcattttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
tttaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcactttaga 240
cccagggagt taatcagcaa tatcaacgta atagttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 140                moltype = DNA length = 399
FEATURE                      Location/Qualifiers
misc_feature                 1..399
                             note = IL-2 analog 34
source                      1..399
                             mol_type = other DNA
                             organism = synthetic construct

SEQUENCE: 140
cctactttcaa gttctacaaa gaaaacacag ctacaactgg agcattttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
tttaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcactttaga 240
cccagggagt taatcagcaa tatcaacgta atagttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 141                moltype = DNA length = 399
FEATURE                      Location/Qualifiers
misc_feature                 1..399
                             note = IL-2 analog 35
source                      1..399

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```

mol_type = other DNA
organism = synthetic construct

SEQUENCE: 141
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
tttaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttaaga 240
cccagggact taatcagcaa tatcaacgta atagttttcg aatttaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 142      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 36
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 142
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
tttaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttaaga 240
cccagggact taatcagcaa tatcaacgta atagttttcg aagtaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 143      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 37
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 143
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttagac 240
cccagggact taatcagcaa tatcaacgta ttagttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 144      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 38
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 144
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttagac 240
cccagggact taatcagcaa tatcaacgta tttgttttcg aatttaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 145      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 39
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 145
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttagac 240
cccagggact taatcagcaa tatcaacgta tttgttttcg aagtaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 146      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 40

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source          1..399
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 146
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttagac 240
cccagggact taatcagcaa tatcaacgta tttgtttctg aaattaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 147      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature       1..399
                  note = IL-2 analog 41
source            1..399
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 147
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccag gatgctcaca 120
tttaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcactttgac 240
cccagggacg tagtcagcaa tatcaacgta tttgttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 148      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature       1..399
                  note = IL-2 analog 42
source            1..399
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 148
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttagaa 240
cccagggact taatcagcaa tatcaacgta tttgttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 149      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature       1..399
                  note = IL-2 analog 43
source            1..399
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 149
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttagac 240
cccagggact taatcagcaa tatcaacgta ttagttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 150      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature       1..399
                  note = IL-2 analog 44
source            1..399
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 150
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttagac 240
cccagggctc taatcagcaa tatcaacgta tttgttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 151      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature       1..399

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note = IL-2 analog 45
source      1..399
            mol_type = other DNA
            organism = synthetic construct

SEQUENCE: 151
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttagac 240
cccagggtct taatcagcaa tatcaacgta ttgtttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 152      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 46
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 152
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggtttta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttagac 240
cccagggtct taatcagcaa tatcaacgta ttgtttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 153      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 47
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 153
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggtttta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttagac 240
cccagggtct taatcagcaa tatcaacgta ttgtttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 154      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 48
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 154
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttagac 240
cccagggtct taatcagcaa tatcaacgta ttgtttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 155      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 49
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 155
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttagac 240
cccagggtct taatcagcaa tatcaacgta ttgtttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 156      moltype = DNA length = 399
FEATURE            Location/Qualifiers

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misc_feature      1..399
                  note = IL-2 analog 50
source            1..399
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 156
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccag gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcactttgac 240
cccagggacg tagtcagcaa tatcaacgta tttgttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 157      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                  note = IL-2 analog 51
source              1..399
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 157
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccag gatgctcaca 120
tttaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaacag 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttagac 240
cccagggact taatcagcaa tatcaacgta tttgttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 158      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                  note = IL-2 analog 52
source              1..399
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 158
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcactttgac 240
cccagggact taatcagcaa tatcaacgta tttgttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 159      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                  note = IL-2 analog 53
source              1..399
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 159
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcactttgac 240
cccagggagt taatcagcaa tatcaacgta tttgttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 160      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                  note = IL-2 analog 54
source              1..399
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 160
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaca gcaaaaactt tcacttagac 240
cccagggact taatcagcaa tatcaacgta tttgttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 161      moltype = DNA length = 399

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FEATURE                Location/Qualifiers
misc_feature            1..399
                        note = IL-2 analog 55
source                  1..399
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 161
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccac gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaca gcaaaaactt tcactttgac 240
cccagggact taatcagcaa tatcaacgta tttgttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 162          moltype = DNA length = 399
FEATURE                Location/Qualifiers
misc_feature            1..399
                        note = IL-2 analog 56
source                  1..399
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 162
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttg ccatgcccac gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaca gcaaaaactt tcacttagac 240
cccagggact taatcagcaa tatcaacgta tttgttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 163          moltype = DNA length = 399
FEATURE                Location/Qualifiers
misc_feature            1..399
                        note = IL-2 analog 57
source                  1..399
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 163
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttg ccatgcccac gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaca gcaaaaactt tcactttgac 240
cccagggact taatcagcaa tatcaacgta tttgttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 164          moltype = DNA length = 399
FEATURE                Location/Qualifiers
misc_feature            1..399
                        note = IL-2 analog 58
source                  1..399
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 164
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccac gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcactttgac 240
cccagggacg ctgccagcaa tatcaacgta tttgttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 165          moltype = DNA length = 399
FEATURE                Location/Qualifiers
misc_feature            1..399
                        note = IL-2 analog 59
source                  1..399
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 165
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccac gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcactttgac 240
cccagggacg ctgccagcaa tatcaacgta tatgttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

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SEQ ID NO: 166      moltype = DNA  length = 399
FEATURE            Location/Qualifiers
misc_feature       1..399
                   note = IL-2 analog 60
source             1..399
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 166
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcattttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt ccattgcccac gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcactataga 240
cccagggacg ctgccagcaa tatcaacgta tatgtttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 167      moltype = DNA  length = 399
FEATURE            Location/Qualifiers
misc_feature       1..399
                   note = IL-2 analog 61
source             1..399
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 167
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcattttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccac gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcactatgac 240
cccagggacg gtgttagcaa tatcaacgta tatgtttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 168      moltype = DNA  length = 399
FEATURE            Location/Qualifiers
misc_feature       1..399
                   note = IL-2 analog 62
source             1..399
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 168
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcattttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccac gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcactgggaa 240
cccagggacg gagccagcaa tatcaacgta tttgtttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 169      moltype = DNA  length = 399
FEATURE            Location/Qualifiers
misc_feature       1..399
                   note = IL-2 analog 63
source             1..399
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 169
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcattttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccac gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacgacgaa 240
cccagggaca caggcagcaa tatcaacgta tttgtttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 170      moltype = DNA  length = 399
FEATURE            Location/Qualifiers
misc_feature       1..399
                   note = IL-2 analog 64
source             1..399
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 170
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcattttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccac gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcactacaac 240
cccagggacg tagtcagcaa tatcaacgta tttgtttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

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```

SEQ ID NO: 171      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature       1..399
                   note = IL-2 analog 65
source            1..399
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 171
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcaactacgaa 240
cccaggggac tagtcagcaa tatcaacgta tttgttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 172      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature       1..399
                   note = IL-2 analog 66
source            1..399
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 172
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcaactcgaa 240
cccaggggact tgcgcagcaa tatcaacgta tttgttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 173      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature       1..399
                   note = IL-2 analog 67
source            1..399
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 173
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcaactatgac 240
cccaggggact tgcgcagcaa tatcaacgta tgggttctgg acctaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 174      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature       1..399
                   note = IL-2 analog 68
source            1..399
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 174
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcaactcgaa 240
cccaggggact tagtcagcaa tatcaacgaa ttcgttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 175      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature       1..399
                   note = IL-2 analog 69
source            1..399
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 175
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcaactacgaa 240
cccaggggact tctcagcaa tatcaacgaa tgggttctgg acctaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360

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attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 176 moltype = DNA length = 399
 FEATURE Location/Qualifiers
 misc_feature 1..399
 note = IL-2 analog 70
 source 1..399
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 176
 cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
 cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
 aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
 gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcactatgac 240
 cccagggacg tagtcagcaa tatcaacgta tttgttctgg aactaaaggg atctgaaaca 300
 acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
 attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 177 moltype = DNA length = 399
 FEATURE Location/Qualifiers
 misc_feature 1..399
 note = IL-2 analog 71
 source 1..399
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 177
 cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
 cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
 aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
 gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttcgaa 240
 cccagggacg tagtcagcaa tatcaacgta tttgttctgg aactaaaggg atctgaaaca 300
 acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
 attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 178 moltype = DNA length = 399
 FEATURE Location/Qualifiers
 misc_feature 1..399
 note = IL-2 analog 72
 source 1..399
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 178
 cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
 cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
 aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
 gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcactttgac 240
 cccagggacg tagtcagcaa tatcaacgta tttgttctgg aactaaaggg atctgaaaca 300
 acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
 attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 179 moltype = DNA length = 399
 FEATURE Location/Qualifiers
 misc_feature 1..399
 note = IL-2 analog 73
 source 1..399
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 179
 cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
 cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
 aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
 gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcactttgac 240
 cccagggact gggtcagcaa tatcaacgta tttgttctgg aactaaaggg atctgaaaca 300
 acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
 attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 180 moltype = DNA length = 399
 FEATURE Location/Qualifiers
 misc_feature 1..399
 note = IL-2 analog 74
 source 1..399
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 180
 cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
 cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc tatgctcaca 120
 aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
 gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcactatgac 240
 cccagggacg tagtcagcaa tatcaacgta tttgttctgg aactaaaggg atctgaaaca 300

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acattcatgt gtgaatatgc tgaatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

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SEQ ID NO: 181      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature       1..399
                   note = IL-2 analog 75
source             1..399
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 181
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagaagtttt ccatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttcgaa 240
cccagggacg tagtcagcaa tatcaacgta ttgtttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgaatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 182      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature       1..399
                   note = IL-2 analog 76
source             1..399
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 182
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aagcagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaacgg 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcactttgac 240
cccagggacg tagtcagcaa tatcaacgta ttgtttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgaatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 183      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature       1..399
                   note = IL-2 analog 77
source             1..399
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 183
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
aaggagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaacgg 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcactttgac 240
cccagggact gggtcagcaa tatcaacgta ttgtttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgaatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 184      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature       1..399
                   note = IL-2 analog 78
source             1..399
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 184
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatcccg aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttcgaa 240
cccagggacg tagtcagcaa tatcaacgta ttgtttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgaatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 185      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature       1..399
                   note = IL-2 analog 79
source             1..399
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 185
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatcccg aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaca gcaaaaactt tcacttcgaa 240

```

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cccagggacg tagtcagcaa tatcaacgta ttgtttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 186      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 80
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 186
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcattttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatcccg aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaca gcaaaaactt tcacttcgaa 240
ggcagggacg tagtcagcaa tatcaacgta ttgtttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 187      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 81
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 187
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcattttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatcccg aactcaccgc gatgctcaca 120
aagaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaca gcaaaaactt tcacttcgaa 240
gtcagggacg tagtcagcaa tatcaacgta ttgtttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 188      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 82
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 188
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcattttacg tctggattta 60
gaaatgattt tgaatggaat taataattac aagaatccca aactcaccag gatgctcaca 120
tttaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcactttgac 240
cccagggacg aagtggagcaa tatcaacgta ttgtttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 189      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 83
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 189
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcattttacg tctggattta 60
gaaatgattt tgaatggaat taataattac aagaatccca aactcaccag gatgctcaca 120
tttaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcactttgac 240
cccagggacg aagtggagcaa tatcaacgta ttgtttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 190      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 84
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 190
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcattttacg tctggattta 60
gaaatgattt tgaatggaat taataattac aagaatccca aactcaccag gatgctcaca 120
tttaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180

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gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacgtggat 240
cccagggacg aagtgaagca tatcaacgta tttgttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 191      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 85
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 191
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcattttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccag gatgctcaca 120
tttaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttcgaa 240
cccagggacg tagtcagcaa tatcaacgta tttgttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 192      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 86
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 192
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcattttacg tctggattta 60
gaaatgattt tgaatggaat taataattac aagaatccca aactcaccag gatgctcaca 120
tttaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttcgaa 240
cccagggacg tagtcagcaa tatcaacgta tttgttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 193      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 87
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 193
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcattttacg tctgtattta 60
gaaatgattt tgaatggaat taataattac aagaatccca aactcaccag gatgctcaca 120
tttaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttcgaa 240
cccagggacg tagtcagcaa tatcaacgta tttgttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 194      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 88
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 194
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcattttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccag gatgctcaca 120
tttaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagac 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttcgaa 240
cccagggacg tagtcagcaa tatcaacgta tttgttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 195      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 89
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 195
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcattttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccg gatgctcaca 120

```

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ttaaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagca ggtgctaaat ttagctcaaa gcaaaaactt tcacttcgaa 240
cccagggacg tagtcagcaa tatcaacgta ttgttctctg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 196      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 90
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 196
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccag gatgctcaca 120
tggaaagttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttcgaa 240
cccagggacg tagtcagcaa tatcaacgta ttgttctctg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 197      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 91
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 197
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccag gatgctcaca 120
ttaaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaacaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttcgaa 240
cccagggacg tagtcagcaa tatcaacgta ttgttctctg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

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SEQ ID NO: 198      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 92
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 198
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccag gatgctcaca 120
ttaaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttcgaa 240
cccagggacg tagtcagcaa tatcaacaca ttgttctctg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 199      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 93
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 199
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccag gatgctcaca 120
ttaaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttcgaa 240
cccagggagg tagtcagcaa tatcaacaca ttgttctctg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 200      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 94
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 200
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60

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cagatgattt tgaatggaat taataattac aagaatccca aactcaccag gatgctcaca 120
ttaagttttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttcgaa 240
cccagggacg tagtcagcaa tatcaacgta ttgtttctcg aattcaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 201      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 95
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 201
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccag gatgctcaca 120
ttaagttttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agggctaaat ttagctcaaa gcaaaaactt tcacttcgaa 240
cccagggacg tagtcagcaa tatcaacgta ttgtttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 202      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 96
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 202
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccag gatgctcaca 120
ttaagttttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agggctaaat ttagctgcaa gcaaaaactt tcacttcgaa 240
cccagggacg tagtcagcaa tatcaacgta ttgtttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 203      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 97
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 203
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
ttaagttttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcactttgac 240
cccagggact taatcagcaa tatcaacgta ttgtttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 204      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 98
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 204
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
ttaagttttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcactttgag 240
cccagggacg taatcagcaa tatcaacgta ttgtttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 205      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 99
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 205

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cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
tttaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcactttgag 240
cccagggact tagtcagcaa tatcaacgta ttgttcttgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 206      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 100
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 206
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccag gatgctcaca 120
tttaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttcgaa 240
cccagggact acgtcagcaa tatcaacgta ttgttcttgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 207      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 101
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 207
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccag gatgctcaca 120
tttaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttcgaa 240
cccagggacc tggccagcaa tatcaacgta ttgttcttgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 208      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 102
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 208
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttact gctggattta 60
cagatgattt tgaatggaat taataattac aagaatccca aactcaccag gatgctcaca 120
tttaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttcgaa 240
cccagggacg tagtcagcaa tatcaacgta attgttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 209      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 103
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 209
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttacg tctggattta 60
gaaatgattt tgaatggaat taataattac aagaatccca aactcaccgc gatgctcaca 120
tttaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttcgaa 240
cccagggacg tagtcagcaa tatcaacgta ttgttcttgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

```

```

SEQ ID NO: 210      moltype = DNA length = 399
FEATURE            Location/Qualifiers
misc_feature        1..399
                    note = IL-2 analog 104
source              1..399
                    mol_type = other DNA
                    organism = synthetic construct

```

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SEQUENCE: 210
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttacg tctggattta 60
gaaatgattt tgaatggaat taataattac aagaatccca aactcaccag gatgctcaca 120
tttaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagac 180
gaactcaaac ctctggagga agtgctaaat ttagctcaaa gcaaaaactt tcacttcgaa 240
cccagggacg tagtcagcaa tatcaacgta ttgttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 211          moltype = DNA length = 399
FEATURE                Location/Qualifiers
misc_feature            1..399
                        note = IL-2 analog 105
source                  1..399
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 211
cctacttcaa gttctacaaa gaaaacacag ctacaactgg agcatttacg tctggattta 60
gaaatgattt tgaatggaat taataattac aagaatccca aactcaccag gatgctcaca 120
tttaagtttt acatgcccaa gaaggccaca gaactgaaac atcttcagtg tctagaagaa 180
gaactcaaac ctctggagga cgtgctaaat ttagctcaaa gcaaaaactt tcacttcgaa 240
cccagggacg tagtcagcaa tatcaacgta ttgttctgg aactaaaggg atctgaaaca 300
acattcatgt gtgaatatgc tgatgagaca gcaaccattg tagaatttct gaacagatgg 360
attaccttta gtcaaagcat catctcaaca ctgacttga 399

SEQ ID NO: 212          moltype = AA length = 133
FEATURE                Location/Qualifiers
REGION                  1..133
                        note = General Formula 1
VARIANT                 1
                        note = Xaa is a deletion
VARIANT                 18
                        note = Xaa is leucine (L) or arginine (R)
VARIANT                 19
                        note = Xaa is leucine (L) or tyrosine (Y)
VARIANT                 22
                        note = Xaa is glutamic acid (E) or glutamine (Q)
VARIANT                 38
                        note = Xaa is alanine (A), aspartic acid (D), or arginine
                        (R)
VARIANT                 42
                        note = Xaa is alanine (A), phenylalanine (F), lysine (K),
                        or tryptophan (W)
VARIANT                 43
                        note = Xaa is glutamic acid (E), lysine (K), or glutamine
                        (Q)
VARIANT                 45
                        note = Xaa is alanine (A) or tyrosine (Y)
VARIANT                 61
                        note = Xaa is aspartic acid (D), glutamic acid (E),
                        glutamine (Q), or arginine (R)
VARIANT                 68
                        note = Xaa is aspartic acid (D) or glutamic acid (E)
VARIANT                 74
                        note = Xaa is histidine (H) or glutamine (Q)
VARIANT                 80
                        note = Xaa is phenylalanine (F), leucine (L), valine (V),
                        or tyrosine (Y)
VARIANT                 81
                        note = Xaa is aspartic acid (D), glutamic acid (E), or
                        arginine (R)
VARIANT                 84
                        note = Xaa is aspartic acid (D) or glutamic acid (E)
VARIANT                 85
                        note = Xaa is alanine (A), glutamic acid (E), glycine (G),
                        leucine (L), valine (V), tryptophan (W), or tyrosine (Y)
VARIANT                 86
                        note = Xaa is alanine (A), glycine (G), isoleucine (I), o
                        valine (V)
VARIANT                 91
                        note = Xaa is threonine (T) or valine (V)
VARIANT                 92
                        note = Xaa is phenylalanine (F), isoleucine (I), or
                        tyrosine (Y)
VARIANT                 94
                        note = Xaa is phenylalanine (F) or leucine (L)
VARIANT                 96
                        note = Xaa is phenylalanine (F) or leucine (L)
source                  1..133

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mol_type = protein
organism = synthetic construct

SEQUENCE: 212
XPTSSSTKKT QLQLEHLXXD LXMILNGINN YKNPKLTXML TXXFXMPKKA TELKHLQCLE 60
XELKPLEXVL NLAXSKNFHX XPRXXSNIN XXVXEXKGSE TTFMCEYADE TATIVEFLNR 120
WITFSQSIIS TLT 133

SEQ ID NO: 213      moltype = AA  length = 133
FEATURE            Location/Qualifiers
REGION            1..133
                  note = General Formula 2
VARIANT           1
                  note = Xaa is a deletion
VARIANT           18
                  note = Xaa is leucine (L) or arginine (R)
VARIANT           22
                  note = Xaa is glutamic acid (E) or glutamine (Q)
VARIANT           38
                  note = Xaa is alanine (A) or arginine (R)
VARIANT           42
                  note = Xaa is phenylalanine (F) or lysine (K)
VARIANT           61
                  note = Xaa is aspartic acid (D) or glutamic acid (E)
VARIANT           68
                  note = Xaa is aspartic acid (D) or glutamic acid (E)
VARIANT           81
                  note = Xaa is aspartic acid (D) or glutamic acid (E)
VARIANT           85
                  note = Xaa is leucine (L) or valine (V)
VARIANT           86
                  note = Xaa is isoleucine (I) or valine (V)
source            1..133
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 213
XPTSSSTKKT QLQLEHLXLD LXMILNGINN YKNPKLTXML TXKFYMPKKA TELKHLQCLE 60
XELKPLEXVL NLAQSKNFHF XPRDXXSNIN VFLVLELKGSE TTFMCEYADE TATIVEFLNR 120
WITFSQSIIS TLT 133

SEQ ID NO: 214      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature       1..33
                  note = primer #1 (IL2_L12F_F)
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 214
caaagaaaac acagtttcaa ctggagcatt tac 33

SEQ ID NO: 215      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature       1..33
                  note = primer #2 (IL2_L12F_R)
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 215
gtaaatgctc cagttgaaac tgtgttttct ttg 33

SEQ ID NO: 216      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature       1..33
                  note = primer #3 (IL2_L12V_F)
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 216
caaagaaaac acaggtacaa ctggagcatt tac 33

SEQ ID NO: 217      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature       1..33
                  note = primer #4 (IL2_L12V_R)
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 217
gtaaatgctc cagttgtacc tgtgttttct ttg 33

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SEQ ID NO: 218      moltype = DNA  length = 39
FEATURE            Location/Qualifiers
misc_feature        1..39
                    note = primer #5 (IL2_L18R_L19R_Q22E_F)
source              1..39
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 218
ctggagcatt tacgtcgtga tttagaaatg attttgaat      39

SEQ ID NO: 219      moltype = DNA  length = 39
FEATURE            Location/Qualifiers
misc_feature        1..39
                    note = primer #6 (IL2_L18R_L19R_Q22E_R)
source              1..39
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 219
attcaaaatc atttctaaat cacgacgtaa atgctccag      39

SEQ ID NO: 220      moltype = DNA  length = 39
FEATURE            Location/Qualifiers
misc_feature        1..39
                    note = primer #7 (IL2_L18R_Q22E_F)
source              1..39
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 220
ctggagcatt tacgtctgga tttagaaatg attttgaat      39

SEQ ID NO: 221      moltype = DNA  length = 39
FEATURE            Location/Qualifiers
misc_feature        1..39
                    note = primer #8 (IL2_L18R_Q22E_R)
source              1..39
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 221
attcaaaatc atttctaaat ccagacgtaa atgctccag      39

SEQ ID NO: 222      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = primer #9 (IL2_L19F_F)
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 222
gagcatttac tgttcgattt acagatgatt ttg      33

SEQ ID NO: 223      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = primer #10 (IL2_L19F_R)
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 223
caaaatcatc tgtaaatcga acagtaaatg ctc      33

SEQ ID NO: 224      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = primer #11 (IL2_L19V_F)
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 224
gagcatttac tgggtggattt acagatgatt ttg      33

SEQ ID NO: 225      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = primer #12 (IL2_L19V_R)
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 225
caaaatcatc tgtaaatcca ccagtaaatg ctc      33

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SEQ ID NO: 226      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = primer #13 (IL2_L19Y_F)
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 226
caactggagc atttactgta cgatttacag atg                               33

SEQ ID NO: 227      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = primer #14 (IL2_L19Y_R)
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 227
catctgtaaa tcgtacagta aatgctccag ttg                               33

SEQ ID NO: 228      moltype = DNA  length = 34
FEATURE            Location/Qualifiers
misc_feature        1..34
                    note = primer #15 (IL2_D20F_F)
source              1..34
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 228
ggagcattta ctgctgtttt tacagatgat ttg                               34

SEQ ID NO: 229      moltype = DNA  length = 34
FEATURE            Location/Qualifiers
misc_feature        1..34
                    note = primer #16 (IL2_D20F_R)
source              1..34
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 229
caaaatcatc tgtaaaaaca gcagtaaatg ctcc                               34

SEQ ID NO: 230      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = primer #17 (IL2_D20V_F)
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 230
gagcatttac tgctggtttt acagatgatt ttg                               33

SEQ ID NO: 231      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = primer #18 (IL2_D20V_R)
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 231
caaaatcatc tgtaaaacca gcagtaaatg ctc                               33

SEQ ID NO: 232      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = primer #19 (IL2_K32C_F)
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 232
ggaattaata attactgtaa tcccaaactc acc                               33

SEQ ID NO: 233      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = primer #20 (IL2_K32C_R)
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 233

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ggtgagtttg ggattacagt aattattaat tcc                                     33

SEQ ID NO: 234      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = primer #21 (IL2_K35C_F)
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 234
aattacaaga atccctgtct caccaggatg ctc                                     33

SEQ ID NO: 235      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = primer #22 (IL2_K35C_R)
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 235
gagcatcctg gtgagacagg gattcttgta att                                     33

SEQ ID NO: 236      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = primer #23 (IL2_K35E_F)
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 236
taattacaag aatcccgaac tcaccgcgat gct                                     33

SEQ ID NO: 237      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = primer #24 (IL2_K35E_R)
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 237
agcatcgcgg tgagttcggg attcttgtaa tta                                     33

SEQ ID NO: 238      moltype = DNA  length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                    note = primer #25 (IL2_R38A_F)
source              1..25
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 238
ccaaactcac cgcgatgctc acatt                                             25

SEQ ID NO: 239      moltype = DNA  length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                    note = primer #26 (IL2_R38A_R)
source              1..25
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 239
aatgtgagca tcgcggtgag tttag                                             25

SEQ ID NO: 240      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = primer #27 (IL2_R38A_F42F_F)
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 240
caccgcgatg ctcacattta agttttacat gcc                                     33

SEQ ID NO: 241      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = primer #28 (IL2_R38A_F42F_R)
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 241
ggcatgtaaa acttaaatgt gagcatcgcg gtg                                     33

SEQ ID NO: 242      moltype = DNA length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                    note = primer #29 (IL2_A38R_F)
source              1..25
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 242
ccaaactcac caggatgctc acatt                                             25

SEQ ID NO: 243      moltype = DNA length = 25
FEATURE            Location/Qualifiers
misc_feature        1..25
                    note = primer #30 (IL2_A38R_R)
source              1..25
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 243
aatgtgagca tcctgggtgag ttggg                                           25

SEQ ID NO: 244      moltype = DNA length = 30
FEATURE            Location/Qualifiers
misc_feature        1..30
                    note = primer #31 (IL2_R38D_F)
source              1..30
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 244
aagaatccca aactcacgga tatgctcaca                                       30

SEQ ID NO: 245      moltype = DNA length = 30
FEATURE            Location/Qualifiers
misc_feature        1..30
                    note = primer #32 (IL2_R38D_R)
source              1..30
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 245
tgtgagcata tcggtgagtt tgggattcct                                       30

SEQ ID NO: 246      moltype = DNA length = 30
FEATURE            Location/Qualifiers
misc_feature        1..30
                    note = primer #33 (IL2_F42A_F)
source              1..30
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 246
ccaggatgct cacagctaag ttttacatgc                                       30

SEQ ID NO: 247      moltype = DNA length = 30
FEATURE            Location/Qualifiers
misc_feature        1..30
                    note = primer #34 (IL2_F42A_R)
source              1..30
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 247
gcatgtaaaa cttagctgtg agcatcctgg                                       30

SEQ ID NO: 248      moltype = DNA length = 30
FEATURE            Location/Qualifiers
misc_feature        1..30
                    note = primer #35 (IL2_F42K_F)
source              1..30
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 248
ccaggatgct cacaagaag ttttacatgc                                       30

SEQ ID NO: 249      moltype = DNA length = 30
FEATURE            Location/Qualifiers
misc_feature        1..30
                    note = primer #36 (IL2_F42K_R)
source              1..30
                    mol_type = other DNA

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                organism = synthetic construct
SEQUENCE: 249
gcatgtaaaa cttctttgtg agcatcctgg                30

SEQ ID NO: 250      moltype = DNA  length = 30
FEATURE            Location/Qualifiers
misc_feature        1..30
                    note = primer #37 (IL2_F42K_K43E_F)
source              1..30
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 250
atgctcacia aggagtttta catgcccaag                30

SEQ ID NO: 251      moltype = DNA  length = 30
FEATURE            Location/Qualifiers
misc_feature        1..30
                    note = primer #38 (IL2_F42K_K43E_R)
source              1..30
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 251
cttgggcatg taaaactcct ttgtgagcat                30

SEQ ID NO: 252      moltype = DNA  length = 30
FEATURE            Location/Qualifiers
misc_feature        1..30
                    note = primer #39 (IL2_F42K_K43Q_F)
source              1..30
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 252
atgctcacia agcagtttta catgcccaag                30

SEQ ID NO: 253      moltype = DNA  length = 30
FEATURE            Location/Qualifiers
misc_feature        1..30
                    note = primer #40 (IL2_F42K_K43Q_R)
source              1..30
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 253
cttgggcatg taaaactgct ttgtgagcat                30

SEQ ID NO: 254      moltype = DNA  length = 30
FEATURE            Location/Qualifiers
misc_feature        1..30
                    note = primer #41 (IL2_K42F_F)
source              1..30
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 254
ccaggatgct cacatttaag tttacatgc                30

SEQ ID NO: 255      moltype = DNA  length = 30
FEATURE            Location/Qualifiers
misc_feature        1..30
                    note = primer #42 (IL2_K42F_R)
source              1..30
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 255
gcatgtaaaa cttaaatgtg agcatcctgg                30

SEQ ID NO: 256      moltype = DNA  length = 39
FEATURE            Location/Qualifiers
misc_feature        1..39
                    note = primer #43 (IL2_K42F_F_long)
source              1..39
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 256
ctcaccagga tgctcacatt taagtgttac atgcccaag                39

SEQ ID NO: 257      moltype = DNA  length = 39
FEATURE            Location/Qualifiers
misc_feature        1..39
                    note = primer #44 (IL2_K42F_R_long)
source              1..39

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	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 257		
cttgggcatg taaaacttaa atgtgagcat cctgggtgag		39
SEQ ID NO: 258	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = primer #45 (IL2_F42W_F)	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 258		
accaggatgc tcacatggaa gttttacatg ccc		33
SEQ ID NO: 259	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = primer #46 (IL2_F42W_R)	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 259		
gggcatgtaa aacttccatg tgagcatcct ggt		33
SEQ ID NO: 260	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = primer #47 (IL2_K43C_F)	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 260		
aggatgctca cattttgttt ttacatgccc aag		33
SEQ ID NO: 261	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = primer #48 (IL2_K43C_R)	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 261		
cttgggcatg taaaacaaaa atgtgagcat cct		33
SEQ ID NO: 262	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
misc_feature	1..19	
	note = primer #49 (IL2_Y45A_F)	
source	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 262		
gaagtttgcc atgcccaga		19
SEQ ID NO: 263	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
misc_feature	1..19	
	note = primer #50 (IL2_Y45A_R)	
source	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 263		
cttgggcatg gcaaacttc		19
SEQ ID NO: 264	moltype = DNA length = 34	
FEATURE	Location/Qualifiers	
misc_feature	1..34	
	note = primer #51 (IL2_Y45A_F_long)	
source	1..34	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 264		
gctcacaaaag aagtttgcca tgcccagaag ggcc		34
SEQ ID NO: 265	moltype = DNA length = 34	
FEATURE	Location/Qualifiers	
misc_feature	1..34	
	note = primer #52 (IL2_Y45A_R_long)	

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source                1..34
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 265
ggccttcttg ggcattggcaa acttctttgt gagc                34

SEQ ID NO: 266        moltype = DNA length = 33
FEATURE              Location/Qualifiers
misc_feature          1..33
                      note = primer #53 (IL2_K48C_F)
source                1..33
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 266
aagttttaca tgccttgtaa ggccacagaa ctg                33

SEQ ID NO: 267        moltype = DNA length = 33
FEATURE              Location/Qualifiers
misc_feature          1..33
                      note = primer #54 (IL2_K48C_R)
source                1..33
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 267
cagttctgtg gccttacagg gcatgtaaaa ctt                33

SEQ ID NO: 268        moltype = DNA length = 33
FEATURE              Location/Qualifiers
misc_feature          1..33
                      note = primer #55 (IL2_K49C_F)
source                1..33
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 268
ttttacatgc ccaagtgatc cacagaactg aaa                33

SEQ ID NO: 269        moltype = DNA length = 33
FEATURE              Location/Qualifiers
misc_feature          1..33
                      note = primer #56 (IL2_K49C_R)
source                1..33
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 269
tttcagttct gtggcacact tgggcatgta aaa                33

SEQ ID NO: 270        moltype = DNA length = 33
FEATURE              Location/Qualifiers
misc_feature          1..33
                      note = primer #57 (IL2_E61D_F)
source                1..33
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 270
cttcagtgtc tagaagacga actcaaacct ctg                33

SEQ ID NO: 271        moltype = DNA length = 33
FEATURE              Location/Qualifiers
misc_feature          1..33
                      note = primer #58 (IL2_E61D_R)
source                1..33
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 271
cagagggttg agttcgtctt ctagacactg aag                33

SEQ ID NO: 272        moltype = DNA length = 33
FEATURE              Location/Qualifiers
misc_feature          1..33
                      note = primer #59 (IL2_E61Q_F)
source                1..33
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 272
cttcagtgtc tagaacaaga actcaaacct ctg                33

SEQ ID NO: 273        moltype = DNA length = 33
FEATURE              Location/Qualifiers
misc_feature          1..33

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source	note = primer #60 (IL2_E61Q_R) 1..33 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 273		
cagagggttg agttcttggt ctagacactg aag		33
SEQ ID NO: 274	moltype = DNA length = 30	
FEATURE	Location/Qualifiers	
misc_feature	1..30	
	note = primer #61 (IL2_E61R_F)	
source	1..30 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 274		
cttcagtgtc tagaacggga actcaaacct		30
SEQ ID NO: 275	moltype = DNA length = 30	
FEATURE	Location/Qualifiers	
misc_feature	1..30	
	note = primer #62 (IL2_E61R_R)	
source	1..30 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 275		
agggttgagt tccggttcta gacactgaag		30
SEQ ID NO: 276	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = primer #63 (IL2_E68D_F)	
source	1..33 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 276		
ctcaaacctc tggaggacgt gctaaattta gct		33
SEQ ID NO: 277	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = primer #64 (IL2_E68D_R)	
source	1..33 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 277		
agctaaattt agcacgtcct ccagagggtt gag		33
SEQ ID NO: 278	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = primer #65 (IL2_E68Q_F)	
source	1..33 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 278		
ctcaaacctc tggagcaggt gctaaattta gct		33
SEQ ID NO: 279	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = primer #66 (IL2_E68Q_R)	
source	1..33 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 279		
agctaaattt agcacctgct ccagagggtt gag		33
SEQ ID NO: 280	moltype = DNA length = 32	
FEATURE	Location/Qualifiers	
misc_feature	1..32	
	note = primer #67 (IL2_V69G_F)	
source	1..32 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 280		
caaacctctg gaggaagggc taaatttagc tc		32
SEQ ID NO: 281	moltype = DNA length = 32	
FEATURE	Location/Qualifiers	

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misc_feature      1..32
                  note = primer #68 (IL2_V69G_R)
source            1..32
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 281
gagctaaatt tagcccttcc tccagaggtt tg                               32

SEQ ID NO: 282      moltype = DNA length = 40
FEATURE            Location/Qualifiers
misc_feature        1..40
                  note = primer #69 (IL2_V69G_Q74A_F)
source             1..40
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 282
ctggaggaag ggctaaattt agctgcaagc aaaaactttc                       40

SEQ ID NO: 283      moltype = DNA length = 40
FEATURE            Location/Qualifiers
misc_feature        1..40
                  note = primer #70 (IL2_V69G_Q74A_R)
source             1..40
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 283
gaaagttttt gcttgcaagct aaatttagcc cttcctccag                       40

SEQ ID NO: 284      moltype = DNA length = 24
FEATURE            Location/Qualifiers
misc_feature        1..24
                  note = primer #71 (IL2_Q74H_F)
source             1..24
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 284
ctaaatttag ctcacagcaa aaac                                          24

SEQ ID NO: 285      moltype = DNA length = 24
FEATURE            Location/Qualifiers
misc_feature        1..24
                  note = primer #72 (IL2_Q74H_R)
source             1..24
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 285
gtttttgctg tgagctaaat ttag                                          24

SEQ ID NO: 286      moltype = DNA length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                  note = primer #73 (IL2_K76C_F)
source             1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 286
aatttagctc aaagctgtaa ctttcactta aga                                33

SEQ ID NO: 287      moltype = DNA length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                  note = primer #74 (IL2_K76C_R)
source             1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 287
tcttaagtga aagttacagc tttagactaa att                                33

SEQ ID NO: 288      moltype = DNA length = 32
FEATURE            Location/Qualifiers
misc_feature        1..32
                  note = primer #75 (IL2_L80D_R81E_F)
source             1..32
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 288
aaagcaaaaa ctttcacgac gaaccaggg ac                                  32

SEQ ID NO: 289      moltype = DNA length = 32

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FEATURE	Location/Qualifiers	
misc_feature	1..32	
	note = primer #76 (IL2_L80D_R81E_R)	
source	1..32	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 289		
gtccctgggt tcgtcgtgaa agtttttgc	tt	32
SEQ ID NO: 290	moltype = DNA length = 29	
FEATURE	Location/Qualifiers	
misc_feature	1..29	
	note = primer #77 (IL2_L80F_F)	
source	1..29	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 290		
gcaaaaactt tcactttgaa cccagggac		29
SEQ ID NO: 291	moltype = DNA length = 29	
FEATURE	Location/Qualifiers	
misc_feature	1..29	
	note = primer #78 (IL2_L80F_R)	
source	1..29	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 291		
gtccctgggt ctaaagtga agtttttgc		29
SEQ ID NO: 292	moltype = DNA length = 29	
FEATURE	Location/Qualifiers	
misc_feature	1..29	
	note = primer #79 (IL2_L80F_R81D_F)	
source	1..29	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 292		
gcaaaaactt tcactttgac cccagggac		29
SEQ ID NO: 293	moltype = DNA length = 29	
FEATURE	Location/Qualifiers	
misc_feature	1..29	
	note = primer #80 (IL2_L80F_R81D_R)	
source	1..29	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 293		
gtccctgggg tcaaagtga agtttttgc		29
SEQ ID NO: 294	moltype = DNA length = 32	
FEATURE	Location/Qualifiers	
misc_feature	1..32	
	note = primer #81 (IL2_L80F_R81E_F)	
source	1..32	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 294		
aaagcaaaaa ctttcacttc gaaccaggg ac		32
SEQ ID NO: 295	moltype = DNA length = 32	
FEATURE	Location/Qualifiers	
misc_feature	1..32	
	note = primer #82 (IL2_L80F_R81E_R)	
source	1..32	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 295		
gtccctgggt tcgaagtga agtttttgc	tt	32
SEQ ID NO: 296	moltype = DNA length = 32	
FEATURE	Location/Qualifiers	
misc_feature	1..32	
	note = primer #83 (IL2_L80V_R81D_F)	
source	1..32	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 296		
aaagcaaaaa ctttcacgtg gatccaggg ac		32

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SEQ ID NO: 297	moltype = DNA length = 32	
FEATURE	Location/Qualifiers	
misc_feature	1..32	
	note = primer #84 (IL2_L80V_R81D_R)	
source	1..32	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 297		
gtccctggga tccacgtgaa agtttttgct tt		32
SEQ ID NO: 298	moltype = DNA length = 32	
FEATURE	Location/Qualifiers	
misc_feature	1..32	
	note = primer #85 (IL2_L80W_R81E_F)	
source	1..32	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 298		
aaagcaaaaa ctttcactgg gaaccaggg ac		32
SEQ ID NO: 299	moltype = DNA length = 32	
FEATURE	Location/Qualifiers	
misc_feature	1..32	
	note = primer #86 (IL2_L80W_R81E_R)	
source	1..32	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 299		
gtccctgggt tcccagtgaa agtttttgct tt		32
SEQ ID NO: 300	moltype = DNA length = 29	
FEATURE	Location/Qualifiers	
misc_feature	1..29	
	note = primer #87 (IL2_L80Y_R81D_F)	
source	1..29	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 300		
gcaaaaactt tcactatgac cccagggac		29
SEQ ID NO: 301	moltype = DNA length = 29	
FEATURE	Location/Qualifiers	
misc_feature	1..29	
	note = primer #88 (IL2_L80Y_R81D_R)	
source	1..29	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 301		
gtccctgggg tcatagtgaa agtttttgct		29
SEQ ID NO: 302	moltype = DNA length = 32	
FEATURE	Location/Qualifiers	
misc_feature	1..32	
	note = primer #89 (IL2_L80Y_R81E_F)	
source	1..32	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 302		
aaagcaaaaa ctttcactac gaaccaggg ac		32
SEQ ID NO: 303	moltype = DNA length = 32	
FEATURE	Location/Qualifiers	
misc_feature	1..32	
	note = primer #90 (IL2_L80Y_R81E_R)	
source	1..32	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 303		
gtccctgggt tcgtagtgaa agtttttgct tt		32
SEQ ID NO: 304	moltype = DNA length = 32	
FEATURE	Location/Qualifiers	
misc_feature	1..32	
	note = primer #91 (IL2_L80Y_R81N_F)	
source	1..32	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 304		
aaagcaaaaa ctttcactac aacccaggg ac		32

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SEQ ID NO: 305      moltype = DNA  length = 32
FEATURE            Location/Qualifiers
misc_feature        1..32
                    note = primer #92 (IL2_L80Y_R81N_R)
source              1..32
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 305
gtccctgggg ttgtagtgaa agtttttgc tt                               32

SEQ ID NO: 306      moltype = DNA  length = 29
FEATURE            Location/Qualifiers
misc_feature        1..29
                    note = primer #93 (IL2_L80Y_R81R_F)
source              1..29
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 306
gcaaaaactt tcactataga cccagggac                                   29

SEQ ID NO: 307      moltype = DNA  length = 29
FEATURE            Location/Qualifiers
misc_feature        1..29
                    note = primer #94 (IL2_L80Y_R81R_R)
source              1..29
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 307
gtccctgggt ctatagtgaa agtttttgc                                   29

SEQ ID NO: 308      moltype = DNA  length = 31
FEATURE            Location/Qualifiers
misc_feature        1..31
                    note = primer #95 (IL2_R81D_D84E_F)
source              1..31
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 308
ctttcactta gaccccaggg agttaatcag c                               31

SEQ ID NO: 309      moltype = DNA  length = 31
FEATURE            Location/Qualifiers
misc_feature        1..31
                    note = primer #96 (IL2_R81D_D84E_R)
source              1..31
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 309
gctgattaac tcctgggggt ctaagtgaag g                               31

SEQ ID NO: 310      moltype = DNA  length = 34
FEATURE            Location/Qualifiers
misc_feature        1..34
                    note = primer #97 (IL2_R81D_F)
source              1..34
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 310
caaaaacttt cacttagacc ccagggactt aatc                             34

SEQ ID NO: 311      moltype = DNA  length = 34
FEATURE            Location/Qualifiers
misc_feature        1..34
                    note = primer #98 (IL2_R81D_R)
source              1..34
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 311
gattaagtcc ctggggtcta agtgaaagtt ttg                               34

SEQ ID NO: 312      moltype = DNA  length = 51
FEATURE            Location/Qualifiers
misc_feature        1..51
                    note = primer #99 (IL2_R81D_F_long)
source              1..51
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 312

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ctcaaagcaa aaactttcac ttagacccca gggacttaat cagcaaatat c          51

SEQ ID NO: 313      moltype = DNA  length = 51
FEATURE            Location/Qualifiers
misc_feature        1..51
                    note = primer #100 (IL2_R81D_R_long)
source              1..51
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 313
gatatttgct gattaagtcc ctgggggtcta agtgaaagtt tttgctttga g          51

SEQ ID NO: 314      moltype = DNA  length = 43
FEATURE            Location/Qualifiers
misc_feature        1..43
                    note = primer #101 (IL2_R81E_D84E_L85V_I86V_V91T_I92F_F)
source              1..43
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 314
gaaccaggagg aggtagtcag caatatcaac acatttgttc tgg          43

SEQ ID NO: 315      moltype = DNA  length = 43
FEATURE            Location/Qualifiers
misc_feature        1..43
                    note = primer #102 (IL2_R81E_D84E_L85V_I86V_V91T_I92F_R)
source              1..43
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 315
ccagaacaaaa tgtgttgata ttgctgacta cctccctggg ttc          43

SEQ ID NO: 316      moltype = DNA  length = 31
FEATURE            Location/Qualifiers
misc_feature        1..31
                    note = primer #103 (IL2_R81E_F)
source              1..31
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 316
aaactttcac ttagaaccca gggacttaat c          31

SEQ ID NO: 317      moltype = DNA  length = 31
FEATURE            Location/Qualifiers
misc_feature        1..31
                    note = primer #104 (IL2_R81E_R)
source              1..31
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 317
gattaagtcc ctgggttcta agtgaaagtt t          31

SEQ ID NO: 318      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = primer #105 (IL2_P82G_F)
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 318
aactttcact tcgaaggcag ggacgtagtc agc          33

SEQ ID NO: 319      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = primer #106 (IL2_P82G_R)
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 319
gctgactacg tccctgcctt cgaagtgaaa gtt          33

SEQ ID NO: 320      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = primer #107 (IL2_P82V_F)
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 320
aaccttcact tcgaagtcag ggacgtagtc agc                                     33

SEQ ID NO: 321      moltype = DNA length = 33
FEATURE            Location/Qualifiers
misc_feature       1..33
                   note = primer #108 (IL2_P82V_R)
source            1..33
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 321
gctgactacg tccctgactt cgaagtgaac gtt                                     33

SEQ ID NO: 322      moltype = DNA length = 34
FEATURE            Location/Qualifiers
misc_feature       1..34
                   note = primer #109 (IL2_D84E_F)
source            1..34
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 322
cttaagaccc agggagttaa tcagcaatat caac                                     34

SEQ ID NO: 323      moltype = DNA length = 34
FEATURE            Location/Qualifiers
misc_feature       1..34
                   note = primer #110 (IL2_D84E_R)
source            1..34
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 323
gttgatattg ctgattaact cccctgggtct taag                                     34

SEQ ID NO: 324      moltype = DNA length = 32
FEATURE            Location/Qualifiers
misc_feature       1..32
                   note = primer #111 (IL2_D84F_F)
source            1..32
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 324
cacttagacc ccagggtctt aatcagcaat at                                       32

SEQ ID NO: 325      moltype = DNA length = 32
FEATURE            Location/Qualifiers
misc_feature       1..32
                   note = primer #112 (IL2_D84F_R)
source            1..32
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 325
atattgctga ttaagaacct ggggtctaag tg                                       32

SEQ ID NO: 326      moltype = DNA length = 31
FEATURE            Location/Qualifiers
misc_feature       1..31
                   note = primer #113 (IL2_D84V_F)
source            1..31
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 326
acttagaccc cagggtctta atcagcaata t                                       31

SEQ ID NO: 327      moltype = DNA length = 31
FEATURE            Location/Qualifiers
misc_feature       1..31
                   note = primer #114 (IL2_D84V_R)
source            1..31
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 327
atattgctga ttaagaccct ggggtctaag t                                       31

SEQ ID NO: 328      moltype = DNA length = 35
FEATURE            Location/Qualifiers
misc_feature       1..35
                   note = primer #115 (IL2_L85A_I86A_F)
source            1..35
                   mol_type = other DNA

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                organism = synthetic construct
SEQUENCE: 328
ctttgacccc agggacgctg ccagcaatat caacg                               35

SEQ ID NO: 329      moltype = DNA  length = 35
FEATURE            Location/Qualifiers
misc_feature        1..35
                    note = primer #116 (IL2_L85A_I86A_R)
source              1..35
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 329
cgttgatatt gctggcagcg tccctgggggt caaag                               35

SEQ ID NO: 330      moltype = DNA  length = 30
FEATURE            Location/Qualifiers
misc_feature        1..30
                    note = primer #117 (IL2_L85E_I86V_F)
source              1..30
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 330
cccagggacg aagtgcagaa tatcaacgta                                     30

SEQ ID NO: 331      moltype = DNA  length = 30
FEATURE            Location/Qualifiers
misc_feature        1..30
                    note = primer #118 (IL2_L85E_I86V_R)
source              1..30
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 331
tacgttgata ttgctcactt cgtccctggg                                     30

SEQ ID NO: 332      moltype = DNA  length = 27
FEATURE            Location/Qualifiers
misc_feature        1..27
                    note = primer #119 (IL2_L85F_I86L_F)
source              1..27
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 332
cccagggact tcctcagcaa tatcaac                                       27

SEQ ID NO: 333      moltype = DNA  length = 27
FEATURE            Location/Qualifiers
misc_feature        1..27
                    note = primer #120 (IL2_L85F_I86L_R)
source              1..27
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 333
gttgatattg ctgaggaagt ccctggg                                       27

SEQ ID NO: 334      moltype = DNA  length = 27
FEATURE            Location/Qualifiers
misc_feature        1..27
                    note = primer #121 (IL2_L85F_I86V_F)
source              1..27
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 334
cccagggact tcgtcagcaa tatcaac                                       27

SEQ ID NO: 335      moltype = DNA  length = 27
FEATURE            Location/Qualifiers
misc_feature        1..27
                    note = primer #122 (IL2_L85F_I86V_R)
source              1..27
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 335
gttgatattg ctgacgaagt ccctggg                                       27

SEQ ID NO: 336      moltype = DNA  length = 27
FEATURE            Location/Qualifiers
misc_feature        1..27
                    note = primer #123 (IL2_L85G_I86A_F)
source              1..27

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	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 336		
cccagggacg gagccagcaa tatcaac		27
SEQ ID NO: 337	moltype = DNA length = 27	
FEATURE	Location/Qualifiers	
misc_feature	1..27	
	note = primer #124 (IL2_L85G_I86A_R)	
source	1..27	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 337		
gttgatattg ctggctccgt cctggg		27
SEQ ID NO: 338	moltype = DNA length = 35	
FEATURE	Location/Qualifiers	
misc_feature	1..35	
	note = primer #125 (IL2_L85G_I86V_F)	
source	1..35	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 338		
ctttgacccc agggacgggtg ttagcaatat caacg		35
SEQ ID NO: 339	moltype = DNA length = 35	
FEATURE	Location/Qualifiers	
misc_feature	1..35	
	note = primer #126 (IL2_L85G_I86V_R)	
source	1..35	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 339		
cgttgatatt gctaaccag tccctggggt caaag		35
SEQ ID NO: 340	moltype = DNA length = 27	
FEATURE	Location/Qualifiers	
misc_feature	1..27	
	note = primer #127 (IL2_L85I_I86V_F)	
source	1..27	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 340		
cccagggaca tagtcagcaa tatcaac		27
SEQ ID NO: 341	moltype = DNA length = 27	
FEATURE	Location/Qualifiers	
misc_feature	1..27	
	note = primer #128 (IL2_L85I_I86V_R)	
source	1..27	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 341		
gttgatattg ctgactatgt cctggg		27
SEQ ID NO: 342	moltype = DNA length = 34	
FEATURE	Location/Qualifiers	
misc_feature	1..34	
	note = primer #129 (IL2_L85L_I86A_F)	
source	1..34	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 342		
cttcgaaccc agggacctgg ccagcaatat caac		34
SEQ ID NO: 343	moltype = DNA length = 34	
FEATURE	Location/Qualifiers	
misc_feature	1..34	
	note = primer #130 (IL2_L85L_I86A_R)	
source	1..34	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 343		
gttgatattg ctggccaggt cctggggttc gaag		34
SEQ ID NO: 344	moltype = DNA length = 34	
FEATURE	Location/Qualifiers	
misc_feature	1..34	
	note = primer #131 (IL2_L85L_I86V_F)	

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source                1..34
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 344
ctttgagccc agggaacttag tcagcaatat caac                               34

SEQ ID NO: 345         moltype = DNA length = 34
FEATURE               Location/Qualifiers
misc_feature          1..34
                      note = primer #132 (IL2_L85L_I86V_R)
source                1..34
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 345
gttgatattg ctgactaagt cccctgggctc aaag                               34

SEQ ID NO: 346         moltype = DNA length = 27
FEATURE               Location/Qualifiers
misc_feature          1..27
                      note = primer #133 (IL2_L85T_I86G_F)
source                1..27
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 346
cccagggaca caggcagcaa tatcaac                                         27

SEQ ID NO: 347         moltype = DNA length = 27
FEATURE               Location/Qualifiers
misc_feature          1..27
                      note = primer #134 (IL2_L85T_I86G_R)
source                1..27
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 347
gttgatattg ctgcctgtgt cccctggg                                         27

SEQ ID NO: 348         moltype = DNA length = 29
FEATURE               Location/Qualifiers
misc_feature          1..29
                      note = primer #135 (IL2_L85V_I86G_F)
source                1..29
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 348
cccaggggacg taggcagcaa tatcaacgt                                       29

SEQ ID NO: 349         moltype = DNA length = 29
FEATURE               Location/Qualifiers
misc_feature          1..29
                      note = primer #136 (IL2_L85V_I86G_R)
source                1..29
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 349
acgttgatat tgctgcctac gtcctctggg                                       29

SEQ ID NO: 350         moltype = DNA length = 31
FEATURE               Location/Qualifiers
misc_feature          1..31
                      note = primer #137 (IL2_L85V_I86I_codon_F)
source                1..31
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 350
gaacccaggg acgtaatcag caatatcaac g                                     31

SEQ ID NO: 351         moltype = DNA length = 31
FEATURE               Location/Qualifiers
misc_feature          1..31
                      note = primer #138 (IL2_L85V_I86I_codon_R)
source                1..31
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 351
cgttgatatt gctgattacg tccctggggtt c                                     31

SEQ ID NO: 352         moltype = DNA length = 30
FEATURE               Location/Qualifiers
misc_feature          1..30

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source	note = primer #139 (IL2_L85V_I86V_F) 1..30 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 352		
gaccagggga cgtagtcagc aatatcaacg		30
SEQ ID NO: 353	moltype = DNA length = 30	
FEATURE	Location/Qualifiers	
misc_feature	1..30	
source	note = primer #140 (IL2_L85V_I86V_R) 1..30 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 353		
cgttgatatt gctgactacg tccctggggtc		30
SEQ ID NO: 354	moltype = DNA length = 29	
FEATURE	Location/Qualifiers	
misc_feature	1..29	
source	note = primer #141 (IL2_L85W_I86V_F) 1..29 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 354		
cccagggact gggtcagcaa tatcaacgt		29
SEQ ID NO: 355	moltype = DNA length = 29	
FEATURE	Location/Qualifiers	
misc_feature	1..29	
source	note = primer #142 (IL2_L85W_I86V_R) 1..29 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 355		
acgttgatat tgctgaccca gtcctgggg		29
SEQ ID NO: 356	moltype = DNA length = 30	
FEATURE	Location/Qualifiers	
misc_feature	1..30	
source	note = primer #143 (IL2_L85W_I86V_Long_F) 1..30 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 356		
tttgaccca gggactgggt cagcaatatc		30
SEQ ID NO: 357	moltype = DNA length = 30	
FEATURE	Location/Qualifiers	
misc_feature	1..30	
source	note = primer #144 (IL2_L85W_I86V_Long_R) 1..30 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 357		
gatattgctg acccagtgccc tgggggtcaaa		30
SEQ ID NO: 358	moltype = DNA length = 34	
FEATURE	Location/Qualifiers	
misc_feature	1..34	
source	note = primer #145 (IL2_L85Y_I86V_F) 1..34 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 358		
cttcgaaccc agggactacg tcagcaatat caac		34
SEQ ID NO: 359	moltype = DNA length = 34	
FEATURE	Location/Qualifiers	
misc_feature	1..34	
source	note = primer #146 (IL2_L85Y_I86V_R) 1..34 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 359		
gttgatattg ctgacgtagt ccttgggttc gaag		34
SEQ ID NO: 360	moltype = DNA length = 31	
FEATURE	Location/Qualifiers	

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misc_feature      1..31
                  note = primer #147 (IL2_I86V_V91T_I92F_F)
source            1..31
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 360
gtcagcaata tcaacacatt tgttctggaa c                               31

SEQ ID NO: 361      moltype = DNA length = 31
FEATURE            Location/Qualifiers
misc_feature       1..31
                  note = primer #148 (IL2_I86V_V91T_I92F_R)
source            1..31
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 361
gttcagaaac aaatgtgttg atattgctga c                               31

SEQ ID NO: 362      moltype = DNA length = 33
FEATURE            Location/Qualifiers
misc_feature       1..33
                  note = primer #149 (IL2_S87C_F)
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 362
cccagggact taatctgcaa tatcaacgta ata                             33

SEQ ID NO: 363      moltype = DNA length = 33
FEATURE            Location/Qualifiers
misc_feature       1..33
                  note = primer #150 (IL2_S87C_R)
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 363
tattacgttg atattgcaga ttaagtcctt ggg                             33

SEQ ID NO: 364      moltype = DNA length = 34
FEATURE            Location/Qualifiers
misc_feature       1..34
                  note = primer #151 (IL2_N88F_F)
source            1..34
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 364
cagggaactta atcagcttta tcaacgtatt tggt                             34

SEQ ID NO: 365      moltype = DNA length = 34
FEATURE            Location/Qualifiers
misc_feature       1..34
                  note = primer #152 (IL2_N88F_R)
source            1..34
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 365
aacaaatacgt ttgataaagc tgattaagtc cctg                             34

SEQ ID NO: 366      moltype = DNA length = 33
FEATURE            Location/Qualifiers
misc_feature       1..33
                  note = primer #153 (IL2_N88Q_F)
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 366
gggacttaat cagccaaatc aacgtaatag ttc                             33

SEQ ID NO: 367      moltype = DNA length = 33
FEATURE            Location/Qualifiers
misc_feature       1..33
                  note = primer #154 (IL2_N88Q_R)
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 367
gaactattac gttgatttgg ctgattaagt ccc                             33

SEQ ID NO: 368      moltype = DNA length = 34

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FEATURE	Location/Qualifiers	
misc_feature	1..34	
	note = primer #155 (IL2_N88V_F)	
source	1..34	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 368		
cagggactta atcagcgtta tcaacgtatt tggt		34
SEQ ID NO: 369	moltype = DNA length = 34	
FEATURE	Location/Qualifiers	
misc_feature	1..34	
	note = primer #156 (IL2_N88V_R)	
source	1..34	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 369		
aacaaatcgt ttgataacgc tgattaagtc cctg		34
SEQ ID NO: 370	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = primer #157 (IL2_I89F_F)	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 370		
gacttaatca gcaatttcaa cgtaatagtt ctg		33
SEQ ID NO: 371	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = primer #158 (IL2_I89F_R)	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 371		
cagaactatt acgttgaaat tgcgtgattaa gtc		33
SEQ ID NO: 372	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = primer #159 (IL2_V91E_I92F_F)	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 372		
agcaatatca acgaattcgt tctggaacta aag		33
SEQ ID NO: 373	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = primer #160 (IL2_V91E_I92F_R)	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 373		
ctttagttcc agaacgaatt cgttgatatt gct		33
SEQ ID NO: 374	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = primer #161 (IL2_V91E_I92W_F)	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 374		
agcaatatca acgaatgggt tctggaacta aag		33
SEQ ID NO: 375	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = primer #162 (IL2_V91E_I92W_R)	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 375		
ctttagttcc agaaccatt cgttgatatt gct		33

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SEQ ID NO: 376	moltype = DNA length = 34	
FEATURE	Location/Qualifiers	
misc_feature	1..34	
	note = primer #163 (IL2_V91F_F)	
source	1..34	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 376		
cagcaatatc aactttatag ttctggaact aaag		34
SEQ ID NO: 377	moltype = DNA length = 34	
FEATURE	Location/Qualifiers	
misc_feature	1..34	
	note = primer #164 (IL2_V91F_R)	
source	1..34	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 377		
ctttagttcc agaactataa agttgatatt gctg		34
SEQ ID NO: 378	moltype = DNA length = 34	
FEATURE	Location/Qualifiers	
misc_feature	1..34	
	note = primer #165 (IL2_V91T_F)	
source	1..34	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 378		
cagcaatatc aacacaatag ttctggaact aaag		34
SEQ ID NO: 379	moltype = DNA length = 34	
FEATURE	Location/Qualifiers	
misc_feature	1..34	
	note = primer #166 (IL2_V91T_R)	
source	1..34	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 379		
ctttagttcc agaactattg tgttgatatt gctg		34
SEQ ID NO: 380	moltype = DNA length = 32	
FEATURE	Location/Qualifiers	
misc_feature	1..32	
	note = primer #167 (IL2_I92F_F)	
source	1..32	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 380		
gcaatatcaa cgtatttggt ctggaactaa ag		32
SEQ ID NO: 381	moltype = DNA length = 32	
FEATURE	Location/Qualifiers	
misc_feature	1..32	
	note = primer #168 (IL2_I92F_R)	
source	1..32	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 381		
ctttagttcc agaacaaata cgttgatatt gc		32
SEQ ID NO: 382	moltype = DNA length = 49	
FEATURE	Location/Qualifiers	
misc_feature	1..49	
	note = primer #169 (IL2_I92F_F_long)	
source	1..49	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 382		
gacttaatca gcaatatcaa cgtatttggt ctggaactaa agggatctg		49
SEQ ID NO: 383	moltype = DNA length = 49	
FEATURE	Location/Qualifiers	
misc_feature	1..49	
	note = primer #170 (IL2_I92F_R_long)	
source	1..49	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 383		
cagatccctt tagttccaga acaaatcgt tgatattgct gattaagtc		49

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SEQ ID NO: 384      moltype = DNA length = 45
FEATURE            Location/Qualifiers
misc_feature        1..45
                    note = primer #171 (IL2_I92F_F_long2)
source              1..45
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 384
cttaatcagc aatatcaacg tatttgttct ggaactaaag ggatc      45

SEQ ID NO: 385      moltype = DNA length = 45
FEATURE            Location/Qualifiers
misc_feature        1..45
                    note = primer #172 (IL2_I92F_R_long2)
source              1..45
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 385
gatcccttta gttccagaac aaatacgttg atattgctga ttaag      45

SEQ ID NO: 386      moltype = DNA length = 32
FEATURE            Location/Qualifiers
misc_feature        1..32
                    note = primer #173 (IL2_I92F_L94F_L96F_F)
source              1..32
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 386
caacgtattt gttttcgaat tcaagggatc tg      32

SEQ ID NO: 387      moltype = DNA length = 32
FEATURE            Location/Qualifiers
misc_feature        1..32
                    note = primer #174 (IL2_I92F_L94F_L96F_R)
source              1..32
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 387
cagatccctt gaattcgaaa acaaatacgt tg      32

SEQ ID NO: 388      moltype = DNA length = 39
FEATURE            Location/Qualifiers
misc_feature        1..39
                    note = primer #175 (IL2_I92F_L94F_L96F_F)
source              1..39
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 388
gcaatatcaa cgtatttggt ttcgaaatta agggatctg      39

SEQ ID NO: 389      moltype = DNA length = 39
FEATURE            Location/Qualifiers
misc_feature        1..39
                    note = primer #176 (IL2_I92F_L94F_L96F_R)
source              1..39
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 389
cagatccctt aaattcgaaa acaaatacgt tgatattgc      39

SEQ ID NO: 390      moltype = DNA length = 39
FEATURE            Location/Qualifiers
misc_feature        1..39
                    note = primer #177 (IL2_I92F_L94F_L96I_F)
source              1..39
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 390
gcaatatcaa cgtatttggt ttcgaaatta agggatctg      39

SEQ ID NO: 391      moltype = DNA length = 39
FEATURE            Location/Qualifiers
misc_feature        1..39
                    note = primer #178 (IL2_I92F_L94F_L96I_R)
source              1..39
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 391

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cagatccctt aatttcgaaa acaaatacgt tgatattgc	39
SEQ ID NO: 392	moltype = DNA length = 39
FEATURE	Location/Qualifiers
misc_feature	1..39
	note = primer #179 (IL2_I92F_L94F_L96V_F)
source	1..39
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 392	
gcaatatcaa cgtatttggt ttcgaagtaa agggatctg	39
SEQ ID NO: 393	moltype = DNA length = 39
FEATURE	Location/Qualifiers
misc_feature	1..39
	note = primer #180 (IL2_I92F_L94F_L96V_R)
source	1..39
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 393	
cagatccctt tacttcgaaa acaaatacgt tgatattgc	39
SEQ ID NO: 394	moltype = DNA length = 33
FEATURE	Location/Qualifiers
misc_feature	1..33
	note = primer #181 (IL2_I92I_F)
source	1..33
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 394	
gcaatatcaa cgtaattggt ctggaactaa agg	33
SEQ ID NO: 395	moltype = DNA length = 33
FEATURE	Location/Qualifiers
misc_feature	1..33
	note = primer #182 (IL2_I92I_R)
source	1..33
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 395	
cctttagttc cagaacaatt acgttgatat tgc	33
SEQ ID NO: 396	moltype = DNA length = 33
FEATURE	Location/Qualifiers
misc_feature	1..33
	note = primer #183 (IL2_I92L_F)
source	1..33
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 396	
gcaatatcaa cgtattagtt ctggaactaa agg	33
SEQ ID NO: 397	moltype = DNA length = 33
FEATURE	Location/Qualifiers
misc_feature	1..33
	note = primer #184 (IL2_I92L_R)
source	1..33
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 397	
cctttagttc cagaactaat acgttgatat tgc	33
SEQ ID NO: 398	moltype = DNA length = 33
FEATURE	Location/Qualifiers
misc_feature	1..33
	note = primer #185 (IL2_I92W_F)
source	1..33
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 398	
agcaatatca acgtatgggt tctggaacta aag	33
SEQ ID NO: 399	moltype = DNA length = 33
FEATURE	Location/Qualifiers
misc_feature	1..33
	note = primer #186 (IL2_I92W_R)
source	1..33
	mol_type = other DNA
	organism = synthetic construct

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SEQUENCE: 399
ctttagttcc agaaccata cgttgatatt gct 33

SEQ ID NO: 400 moltype = DNA length = 32
FEATURE Location/Qualifiers
misc_feature 1..32
 note = primer #187 (IL2_I92Y_F)
source 1..32
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 400
gcaatatcaa cgtatatggt ctggaactaa ag 32

SEQ ID NO: 401 moltype = DNA length = 32
FEATURE Location/Qualifiers
misc_feature 1..32
 note = primer #188 (IL2_I92Y_R)
source 1..32
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 401
ctttagttcc agaacatata cgttgatatt gc 32

SEQ ID NO: 402 moltype = DNA length = 33
FEATURE Location/Qualifiers
misc_feature 1..33
 note = primer #189 (IL2_L94F_L96F_F)
source 1..33
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 402
cgtaatagtt ttcgaattta agggatctga aac 33

SEQ ID NO: 403 moltype = DNA length = 33
FEATURE Location/Qualifiers
misc_feature 1..33
 note = primer #190 (IL2_L94F_L96F_R)
source 1..33
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 403
gtttcagatc ccttaaattc gaaaactatt acg 33

SEQ ID NO: 404 moltype = DNA length = 33
FEATURE Location/Qualifiers
misc_feature 1..33
 note = primer #191 (IL2_L94F_L96I_F)
source 1..33
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 404
cgtaatagtt ttcgaaatta agggatctga aac 33

SEQ ID NO: 405 moltype = DNA length = 33
FEATURE Location/Qualifiers
misc_feature 1..33
 note = primer #192 (IL2_L94F_L96I_R)
source 1..33
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 405
gtttcagatc ccttaatttc gaaaactatt acg 33

SEQ ID NO: 406 moltype = DNA length = 33
FEATURE Location/Qualifiers
misc_feature 1..33
 note = primer #193 (IL2_L94F_L96V_F)
source 1..33
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 406
cgtaatagtt ttcgaagtaa agggatctga aac 33

SEQ ID NO: 407 moltype = DNA length = 33
FEATURE Location/Qualifiers
misc_feature 1..33
 note = primer #194 (IL2_L94F_L96V_R)
source 1..33
 mol_type = other DNA

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                organism = synthetic construct
SEQUENCE: 407
gtttcagatc cctttacttc gaaaactatt acg                               33

SEQ ID NO: 408      moltype = DNA  length = 30
FEATURE            Location/Qualifiers
misc_feature        1..30
                    note = primer #195 (IL2_E95D_F)
source              1..30
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 408
gttctggacc taaagggatc tgaacaaca                                     30

SEQ ID NO: 409      moltype = DNA  length = 30
FEATURE            Location/Qualifiers
misc_feature        1..30
                    note = primer #196 (IL2_E95D_R)
source              1..30
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 409
tgttgtttca gatcccttta ggtccagaac                                   30

SEQ ID NO: 410      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = primer #197 (IL2_C125S_F)
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 410
agatggatta cctttagtca aagcatcatc tca                               33

SEQ ID NO: 411      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = primer #198 (IL2_C125S_R)
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 411
tgagatgatg ctttgactaa aggtaatcca tct                               33

SEQ ID NO: 412      moltype = DNA  length = 34
FEATURE            Location/Qualifiers
misc_feature        1..34
                    note = primer #199 (IL2_Q126T_F)
source              1..34
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 412
gatggattac ctttagtaca agcatcatct caac                             34

SEQ ID NO: 413      moltype = DNA  length = 34
FEATURE            Location/Qualifiers
misc_feature        1..34
                    note = primer #200 (IL2_Q126T_R)
source              1..34
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 413
gttgagatga tgcttg tact aaaggtaatc catc                             34

SEQ ID NO: 414      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = primer #201 (IL2_desA1_F)
source              1..33
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 414
taagaatata catatgccta cttcaagttc tac                               33

SEQ ID NO: 415      moltype = DNA  length = 33
FEATURE            Location/Qualifiers
misc_feature        1..33
                    note = primer #202 (IL2_desA1_R)
source              1..33

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mol_type = other DNA
organism = synthetic construct

SEQUENCE: 415
gtagaacttg aagtaggcat atgtatattc tta                               33

SEQ ID NO: 416      moltype = DNA length = 34
FEATURE            Location/Qualifiers
misc_feature        1..34
                    note = primer #203 (IL2_NdeI_desA1_N-term)
source              1..34
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 416
cgccatatgc ctacttcaag ttctacaaag aaaa                               34

SEQ ID NO: 417      moltype = DNA length = 34
FEATURE            Location/Qualifiers
misc_feature        1..34
                    note = primer #204 (IL2_BHI_C-term)
source              1..34
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 417
cggggtcctc aagtcagtgt tgagatgatg cttt                               34

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The invention claimed is:

1. An interleukin-2 analog comprising an amino acid sequence represented by General Formula 1 below:

[General Formula 1]

X1-P-T-S-S-S-T-K-K-T-Q-L-Q-L-E-H-L-X18-X19-D-L-X22-M-I-L-N-G-I-N-N-Y-K-N-P-K-L-T-X38-M-L-T-X42-X43-F-X45-M-P-K-K-A-T-E-L-K-H-L-Q-C-L-E-X61-E-L-K-P-L-E-X68-V-L-N-L-A-X74-S-K-N-F-H-X80-X81-P-R-X84-X85-X86-S-N-I-N-X91-X92-V-X94-E-X96-K-G-S-E-T-T-F-M-C-E-Y-A-D-E-T-A-T-I-V-E-F-L-N-R-W-I-T-F-S-Q-S-I-I-S-T-L-T (General Formula 1, SEQ ID NO: 212)

wherein in General Formula 1 above,

X1 is a deletion;

X18 is leucine (L) or arginine (R);

X19 is leucine (L) or tyrosine (Y);

X22 is glutamic acid (E) or glutamine (Q);

X38 is alanine (A);

X42 is alanine (A), phenylalanine (F), lysine (K), or tryptophan (W);

X43 is glutamic acid (E), lysine (K), or glutamine (Q);

X45 is alanine (A) or tyrosine (Y);

X61 is aspartic acid (D), glutamic acid (E), glutamine (Q), or arginine (R);

X68 is aspartic acid (D) or glutamic acid (E);

X74 is histidine (H) or glutamine (Q);

X80 is phenylalanine (F);

X81 is aspartic acid (D), glutamic acid (E), or arginine (R);

X84 is aspartic acid (D) or glutamic acid (E);

X85 is alanine (A), glutamic acid (E), glycine (G), leucine (L), valine (V), tryptophan (W), or tyrosine (Y);

X86 is alanine (A), glycine (G), isoleucine (I), or valine (V);

X91 is threonine (T) or valine (V);

X92 is phenylalanine (F);

X94 is phenylalanine (F) or leucine (L); and

X96 is phenylalanine (F) or leucine (L).

2. The interleukin-2 analog of claim 1, wherein the interleukin-2 analog comprises any one sequence selected from the group consisting of amino acid sequences of SEQ ID NOS: 54, 56, 58, 59, 72, 74, 76, 77, 78, 98, and 100.

3. The interleukin-2 analog of claim 1, wherein in General Formula 1 above,

X43 is lysine (K);

X45 is tyrosine (Y);

X61 is aspartic acid (D), glutamic acid (E), or glutamine (Q);

X68 is glutamic acid (E);

X74 is glutamine (Q);

X80 is phenylalanine (F);

X85 is leucine (L), valine (V), or tyrosine (Y);

X86 is isoleucine (I) or valine (V); and

X92 is phenylalanine (F).

4. The interleukin-2 analog of claim 3, wherein the interleukin-2 analog comprises any one sequence selected from the group consisting of amino acid sequences of SEQ ID NOS: 54, 72, 98, and 100.

5. The interleukin-2 analog of claim 1, wherein the interleukin-2 analog further comprises one or more amino acids at the C-terminus thereof.

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